भारत सरकार
Government of India
जल शक्ति मंत्रालय,
Ministry of Jal Shakti,
जल संसाधन, नदी विकास और गंगा संरक्षण विभाग,
Department of Water Resources,
River Development and
Ganga Rejuvenation

केंद्रीय भूमि जल बोर्ड
Central Ground Water Board

वार्षिक रिपोर्ट

ANNUAL REPORT

2024-25

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FARIDABAD
JANUARY 2026

डॉ. सुनील कुमार अम्बष्ट
अध्यक्ष
Dr. Sunil Kumar Ambast
Chairman

सत्यमेव जयते

भारत सरकार
जल शक्ति मंत्रालय
जल संसाधन,
नदी विकास और गंगा संरक्षण विभाग
केन्द्रीय भूमि जल बोर्ड
Government of India
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Department of Water Resources,
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Central Ground Water Board

MESSAGE

It gives me great pleasure to present the Annual Report of the Central Ground Water Board for the Year 2024-25. The report encapsulates our collective efforts, achievements, and future priorities in the vital domain of Ground water development, Management, and Conservation.

Ground water continues to be the lifeline of our country, sustaining drinking water needs, agriculture, industries, and ecosystems. However, the increasing demand, coupled with challenges of over-extraction, contamination, and climate variability, calls for prudent management, scientific monitoring, and innovative interventions. CGWB, as the national apex organization, remains committed to addressing these challenges with a vision of ensuring sustainability and equity in ground water resources.

During the year, the Board has undertaken comprehensive hydrogeological surveys, aquifer mapping, monitoring of ground water levels and quality, and assessments for resource planning. Special emphasis has been laid on participatory ground water management, capacity building of stakeholders, and supporting State Governments in implementation of sustainable water management practices. Our technical expertise has also contributed significantly to policy inputs, awareness programmes, and research-based initiatives.

This Annual Report reflects not only our activities but also our resolve to move forward with renewed dedication. The journey ahead requires integration of science, community participation, and governance. I take this opportunity to acknowledge the dedicated efforts of CGWB officers and staff, as well as the guidance received from the Ministry of Jal Shakti, Government of India.

I am confident that with collective will and scientific stewardship, we shall continue to safeguard our precious ground water resources for present and future generations.

(Dr.Sunil Kumar Ambast)

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वार्षिक रिपोर्ट, 2024-25

सारांश

एनएचएस अनुभाग

- केंद्रीय भूमि जल बोर्ड (CGWB) में वर्तमान में (डग वेल : 16,431, पीज़ोमीटर: 8,680, हैंड पंप: 966, स्प्रिंग: 274 सहित) पूरे देश में 26,351 भूजल प्रेक्षण कूपों का एक नेटवर्क है ।
- एएपी (AAP) 2024-25 में, सीजीडब्ल्यूबी द्वारा कुल 64,696 भूजल नमूनों का विश्लेषण किया गया, जिनमें मूल पैरामीटर के लिए 32,853 और भारी धातुओं के लिए 31,843 नमूने शामिल किए गए थे ।

भूभौतिकीय अनुभाग

- एएपी 2024-25 के दौरान, सीजीडब्ल्यूबी द्वारा फील्ड अधिकारियों के माध्यम से नैक्यूम (NAQUIM) 2.0, अल्पकालिक जल आपूर्ति अन्वेषण और विशेष अध्ययनों के लिए 2,230 वर्टिकल इलेक्ट्रिकल साउंडिंग (VES) और 1,633 ट्रांजिएंन्ट इलेक्ट्रो-मैग्नेटिक (TEM) सर्वेक्षण किए गए ।
- इन-हाउस और आउटसोर्सिंग के माध्यम से कुल 1,593 बोरहोल की लॉगिंग की गई । हरियाणा में यमुना नदी के फरीदाबाद बाढ़ के मैदानों में एफएमडीए (FMDA) अध्ययन के लिए उन्नत तकनीक (2D इमेजिंग) का उपयोग करके 14 स्थलों पर एक विशेष अध्ययन भी किया गया ।
- केंद्रीय भूमि जल बोर्ड (CGWB) द्वारा राजस्थान, गुजरात और हरियाणा राज्यों के शुष्क क्षेत्रों के कुछ हिस्सों में हेलीबोर्न भूभौतिकीय सर्वेक्षण की अत्याधुनिक तकनीक का उपयोग करके एक्विफर (जलभृत) की उच्च-रिज़ॉल्यूशन मैपिंग शुरू की गई है ।
- राजस्थान, गुजरात और हरियाणा राज्यों के 92 ब्लॉकों को शामिल करते हुए 97,165 वर्ग किमी क्षेत्र और 40,313 लाइन किमी फ्लाइट लाइन की चरण-I हेलीबोर्न सर्वेक्षण परियोजना पूरी हो गई है । राजस्थान में 8 जिलों के 47 ब्लॉकों को शामिल किया गया जिसमे बीकानेर, सीकर, जैसलमेर जिले में 1-1 ब्लॉक; चूरू जिले में 7, गंगानगर जिले में 9, जालोर जिले में 8, पाली जिले में 10 और जोधपुर जिले में 10 ब्लॉक हैं। गुजरात के राजकोट में 11 ब्लॉक, जामनगर में 6, मोरबी में 5, सुरेंद्रनगर में 10 और देवभूमि द्वारका जिले में 4 ब्लॉक शामिल किए गए जबकि हरियाणा के कुरुक्षेत्र में 5 ब्लॉक और यमुनानगर जिले में 4 ब्लॉकों का सर्वेक्षण किया गया ।

भूजल अन्वेषण

- बोर्ड द्वारा भूजल अन्वेषण कार्य 73 ड्रिलिंग रिग्स (22 डायरेक्ट रोटरी, 49 डाउन द होल और 02 डुअल रोटरी) के बेड़े के माध्यम से किया जा रहा है ।
- केंद्रीय भूमि जल बोर्ड द्वारा वर्ष 2024-25 के दौरान अपने इन-हाउस भूजल अन्वेषण कार्यक्रम के तहत 750 कुओं (EW- 199, OW- 68, PZ- 477, RW-6) का निर्माण किया गया । 199 अन्वेषण कूपों (Exploratory wells) में से, 168 कूपों का निर्माण कठोर चट्टानी क्षेत्रों में, 31 कुओं का जलोढ़ संरचना क्षेत्रों में, और 15 अन्वेषण कूपों का निर्माण देश के जनजातीय क्षेत्रों में किया गया ।
- बोर्ड द्वारा दिनांक 31.03.2024 तक, देश में भूजल विकास के लिए उपयुक्त क्षेत्रों की पहचान करने के लिए 48,638 बोरहोल (आउटसोर्सिंग के माध्यम से 10,349 बोरहोल सहित) ड्रिल किए गए हैं ।

भूजल संसाधन आकलन

- सीजीडब्ल्यूबी और राज्य भूजल विभागों द्वारा संयुक्त रूप से राज्य स्तरीय समितियों के पर्यवेक्षण में पूरे देश के डॉयनेमिक भूजल संसाधनों (वर्ष 2024 तक) का आकलन किया गया है।
- वर्ष 2024 के दौरान भूजल संसाधनों का आकलन किया गया, जिसके परिणामस्वरूप **'भारत के डॉयनेमिक भूजल संसाधनों का राष्ट्रीय संकलन, 2024' (National Compilation of Dynamic Ground Water Resources of India, 2024)** विषयक रिपोर्ट जारी की गई । उक्त रिपोर्ट के अनुसार देश में कुल वार्षिक भूजल पुनर्भरण (recharge) का आकलन 446.91 बिलियन क्यूबिक मीटर (bcm) किया गया है । वार्षिक निष्कर्षण योग्य भूजल संसाधन का आकलन 406.19 बिलियन क्यूबिक मीटर किया गया है, जबकि वार्षिक भूजल निष्कर्षण 245.64 बिलियन क्यूबिक मीटर है । भूजल निष्कर्षण का स्तर 60.47% है ।
- देश की कुल 6746 आकलन इकाइयों (ब्लॉक / तालुका / मण्डल) में से 751 (11.13%) इकाइयों को 'अति-दोहित' (Over-exploited) के रूप में वर्गीकृत किया गया है । यह वार्षिक पुनर्भरणीय भूजल संसाधनों से अधिक भूजल निष्कर्षण की स्थिति को दर्शाता है। 206 (3.05%) आकलन इकाइयों में भूजल निष्कर्षण का स्तर 90-100% के मध्य है और उन्हें 'गंभीर' (Critical) के रूप में वर्गीकृत किया गया है । 711 (10.54%) इकाइयाँ "अर्ध-गंभीर" (Semi-critical) हैं, जहाँ भूजल निष्कर्षण का चरण 70-90% के मध्य है, और 4951 (73.39%) इकाइयों को "सुरक्षित" (Safe) के रूप में वर्गीकृत किया गया है। 127 (1.88%) आकलन इकाइयों को इकाइयों को "लवणीय" (Saline) के रूप में वर्गीकृत किया गया है।

नैक्यूम

- नैक्यूम 2.0 अध्ययन विभिन्न प्राथमिकता वाले क्षेत्रों पर ध्यान केंद्रित करने के लिए डिज़ाइन किए गए हैं । वर्ष 2024-25 में, नैक्यूम 2.0 के तहत देश भर में 23,724 वर्ग किमी क्षेत्र में कुल 39 अध्ययन किए गए हैं।
- एएपी 2024-25 में नैक्यूम के निष्कर्षों एवं अन्य संबंधित अध्ययनों का प्रसार करने के लिए तथा जागरूकता सृजन हेतु 63 जन संपर्क कार्यक्रम (PIP) आयोजित किए गए । इनमें भूजल के मुद्दों और संभावित समाधानों पर क्षेत्र-विशिष्ट चर्चाएँ भी शामिल हैं ।

जल संरक्षण / कृत्रिम पुनर्भरण

- सीजीडब्ल्यूबी द्वारा एक पायलट अध्ययन के रूप में 08 राज्यों के चयनित 09 ब्लॉकों में कृत्रिम पुनर्भरण संरचनाओं के स्थल चयन और कृत्रिम पुनर्भरण संचारनाओं की डिजाइन तथा मनरेगा (MGNREGS) अधिकारियों के क्षमता निर्माण में तकनीकी मार्गदर्शन प्रदान किया जा रहा है । अब तक, आंध्र प्रदेश, तेलंगाना और तमिलनाडु (प्रत्येक में एक ब्लॉक) में कार्य पूरा किया जा चुका है। सीजीडब्ल्यूबी द्वारा हिमाचल प्रदेश के सिरमौर जिले की काला अंब घाटी में 07 चेक डैम और 12 पुनर्भरण कूप के निर्माण का प्रस्ताव मंत्रालय के समक्ष प्रस्तुत किया गया है।
- जिला पुनर्भरण योजना (DRP) 1:50,000 के पैमाने पर तैयार की जाएगी । वर्ष 2023 के दौरान जल शक्ति मंत्रालय द्वारा यह परामर्श दिया गया है कि जल शक्ति अभियान 2023 के दौरान चिह्नित 150 जल की कमी वाले जिलों का जिला पुनर्भरण योजना तैयार किया जाए जिसमे से जेएसए 2019 और जेएसए 2023 में 50 जिले पहले से ही चिह्नित थे। अतः जेएसए 2019 और जेएसए 2023 दोनों में चिह्नित कुल 350 जिला पुनर्भरण योजनाओं में से 343 जिला पुनर्भरण योजनाओं को अंतिम रूप दिया जा चुका है और शेष को अंतिम रूप दिया जा रहा है ।

केंद्रीय भूमि जल प्राधिकरण

- सीजीडब्ल्यूए द्वारा दिनांक 24.09.2020 के दिशानिर्देशों और यथासंशोधित दिनांक 29.03.2023 के दिशानिर्देशों के अनुसार भूजल निकासी के लिए एनओसी (NOC) प्रदान करने हेतु उद्योगों / अवसंरचनात्मक इकाइयों/खनन परियोजनाओं से प्राप्त आवेदनों का मूल्यांकन किया गया। वर्ष 2024-25 के दौरान कुल 1,334 नई एनओसी जारी की गईं और 1,954 एनओसी का नवीनीकरण किया गया। इसके अतिरिक्त, 4,658 नए आवेदनों और 90 नवीनीकरण मामलों में छूट दी गई। अतः वर्ष 2024-25 के दौरान कुल 8,036 एनओसी/छूट जारी किए गए।

- सीजीडब्ल्यूए द्वारा दिनांक 19 सितंबर 2024 को 8वें भारत जल सप्ताह आयोजन के दौरान भूजल उपयोगकर्ताओं को एनओसी जारी करने के लिए नया पोर्टल 'भूनीर ऐप' (BhuNeer APP) लॉन्च किया गया । यह पोर्टल देश में बड़े पैमाने पर उपयोक्ताओं को सेवा उपलब्ध करवाने के लिए विकसित किया गया है तथा इसमे उपयोगकर्ता पंजीकरण, प्रमाणीकरण, रीयल-टाइम ट्रांजेक्शन प्रोसेसिंग, रिपोर्टिंग तथा भुगतान गेटवे और डिजिटल रिकॉर्ड्स रेपोजेटीरी जैसी सुविधाओं सहित तृतीय पक्ष प्रणाली के साथ संपर्क आदि सुविधाएं भी उपलब्ध है।

सार्वजनिक निवेश बोर्ड

- एएपी 2024-25 के अनुसार, विभिन्न राज्यों में कुल 2,826 पीज़ोमीटर का निर्माण किया गया है और 298 डीडब्ल्यूएलआर (DWLRs) संस्थापित किए गए हैं ।
- विभिन्न पैकेजों के तहत राज्यों में कुल 567 अन्वेषण कूपों (EWs) और प्रेक्षण कूपों (OWs) का निर्माण किया गया है ।

राजीव गांधी राष्ट्रीय भूमि जल प्रशिक्षण एवं अनुसंधान संस्थान

- एएपी 2024-2025 के दौरान, "स्थानीय भूजल-संबंधी मुद्दों और सहभागी भूजल प्रबंधन" विषय पर ग्राम एवं ब्लॉक स्तर पर क्षमता निर्माण हेतु 52 ब्लॉक स्तरीय (टीयर-III) प्रशिक्षण आयोजित किए गए । इन सभी 52 प्रशिक्षणों का आयोजन भौतिक / ऑफलाइन मोड में किया गया जिसमे 2,315 महिला प्रतिभागियों सहित कुल 6,420 प्रशिक्षार्थियों ने भाग लिया ।
- वर्ष 2024 में आरजीएनजीडब्ल्यूटीआरआई, रायपुर ने 'भूजल न्यूज' खंड 33 (2023-24 अंक 1-4) में नैक्यूम (NAQUIM) 1.0 के परिणामों को शामिल करते हुए नैक्यूम विशेष खंड प्रकाशित किया।
- आरजीएनजीडब्ल्यूटीआरआई, रायपुर द्वारा जल ऑडिटर सर्टिफिकेशन के एक बैच का प्रशिक्षण पूरा किया गया जिसमे 3 जल ऑडिट सर्टिफिकेट पाठ्यक्रम शामिल थे। वर्ष 2024-25 के दौरान कुल 13 जल ऑडिट सर्टिफिकेशन (Water Auditor) प्रमाणपत्र प्रदान किए जाने से कुल प्रमाणित जल लेखा परीक्षकों की संख्या 46 हो गई है ।
- आरजीएनजीडब्ल्यूटीआरआई, रायपुर द्वारा राजभाषा गतिविधियों के अंतर्गत केंद्र सरकार के अन्य संगठनों के सहयोग से संयुक्त तिमाही राजभाषा कार्यशाला का आयोजन नियमित रूप से किया जाता है। आरजीएनजीडब्ल्यूटीआरआई, रायपुर को नराकास की दूसरी अर्द्धवार्षिक बैठक में अपने दैनिक सरकारी कार्यों में राजभाषा के प्रचार-प्रसार हेतु प्रथम पुरस्कार से सम्मानित किया गया ।
- एएपी 2024-25 में आरजीएनजीडब्ल्यूटीआरआई द्वारा राजभाषा हिन्दी में 'प्रशिक्षण भारती (खंड-1)' नामक प्रथम पत्रिका का प्रकाशन किया गया ।

Annual General Report, 2024-25

Summary

NHS Section

- CGWB currently has a network of 26351 (Dug Wells: 16431, Piezometers: 8680, Hand Pump: 966, spring: 274) groundwater observation wells throughout the Country.
- In the AAP 2024-25, a total of 64696 groundwater samples were analyzed by CGWB which included 32853 for Basic parameters, 31843 for Heavy Metals.

Geophysical Section

- During the AAP 2024-25, CGWB through its field officers carried out 2230 Vertical Electrical Sounding (VES), 1633 Transient Electro-magnetic (TEM) survey for NAQUIM 2.0, short term water supply investigations, special studies.
- A total 1593 boreholes logging conducted in in-house and outsourcing. A special study was also carried out at 14 locations using advanced technology (2D Imaging) for FMDA study in Faridabad flood plains of the Yamuna River in Haryana.
- Central Ground Water Board (CGWB) has initiated high resolution mapping of aquifers using state of art technology heliborne geophysical survey in parts of the arid areas spread over the states of Rajasthan, Gujarat and Haryana.
- The phase-I heliborne survey project of an area of 97165 sq.km and 40313line km flight line has been completed covering 92 Blocks in Rajasthan, Gujarat and Haryana states. In Rajasthan, 47 Blocks spanning in parts of 8 districts were addressed, with 1 Block each in Bikaner, Sikar, and Jaisalmer districts, 7 Blocks in Churu district, 9 blocks in Ganganagar district, 8 blocks in Jalore district, 10 blocks in Pali district, and 10 blocks in Jodhpur district. In Gujarat, the areas covered included 11 blocks in Rajkot, 6 blocks in Jamnagar, 5 blocks in Morbi, 10 blocks in Surendranagar, and 4 blocks in Devbhumi Dwarka districts, while in Haryana, 5 blocks in Kurukshetra and 4 blocks in Yamunanagar districts were surveyed.

Ground Water Exploration

- Groundwater exploration is being carried out by the Board through its fleet of 73 drilling rigs (22 Direct Rotary, 49 Down the Hole and 02 Dual Rotary).
- During the year 2024-25, Central Ground Water Board under its in-house Ground Water Exploration programme constructed 750 wells (EW- 199, OW- 68, PZ- 477, RW-6). Out of 199 exploratory wells, 168 wells were constructed in hard rock, 31 wells in alluvium and formations 15 exploratory wells constructed in Tribal areas of the country.
- The Board since inception has drilled 48638 bore holes (including 10349 bore holes through outsourcing), as on 31.03.2024, to identify worthy areas for ground water development in the country.

Ground Water Resource Assessment

- The Dynamic ground water resources (as in 2024) of the entire country have been assessed jointly by CGWB and State Ground Water Departments under the supervision of the State level Committees.
- Ground Water Resource Assessment were carried out during the year 2024, culminating in the release of the report **'National Compilation of Dynamic Ground Water Resources of India, 2024'**. As per the report, the total annual groundwater recharge in the country has been assessed as 446.91 billion cubic meter (bcm). The annual extractable ground water resource has been assessed as 406.19bcm. The annual groundwater extraction is 245.64bcm. The stage of groundwater extraction is 60.47 %.
- Out of the total 6746 assessment units (Blocks/Taluks/Mandals) in the country, 751(11.13%) units have been categorized as 'Over-exploited' indicating ground water extraction exceeding the annually replenishable ground water recharge. In, 206 (3.05 %) assessment units the stage of groundwater extraction is between 90-100% and have been categorized as 'Critical'. There are 711 (10.54 %) "Semi-critical" units, where the stage of ground water extraction is between 70 % and 90 % and 4951 (73.39 %) units are categorized as "safe" blocks. 127 (1.88 %) assessment units are categorized as "saline".

NAQUIM

- The NAQUIM 2.0 studies are designed to focus on different priority areas. In the year 2024-25, a total of 39 studies were conducted under NAQUIM 2.0 an area of 23724 sq. km spanning across the Country.
- In AAP 2024-25, 63 Public Interaction Programme (PIP) were conducted for dissemination of outputs of NAQUIM and other related studies and create awareness. They also involves area specific discussions on ground water issues and possible solutions.

Water Conservation/Artificial Recharge

- CGWB is providing technical guidance in site selection and design of artificial recharge structures and capacity building of MGNREGS officials in selected 09 Blocks of 08 States as a pilot study. So far, the work has been completed in Andhra Pradesh, Telangana and Tamil Nadu (one block each).CGWB has proposed to construct 07 Check Dam and 12 Recharge well in Kala Amb Valley, Sirmaur district, H.P to the Ministry.
- The District Recharge Plan will be prepared on the scale of 1:50,000. During 2023, MoJS has advised to prepare the DRPs of the 150 water stressed district identified during JSA 2023, in which 50 districts were common for both JSA (2019) and JSA (2023). Therefore, out of the total 350 DRPs identified both in JSA 2019 and JSA 2023, 343 DRPs have been finalized and remaining are under finalization.

Central Ground Water Authority

- CGWA evaluated applications from Industries/ Infrastructure Units / Mining Projects for grant of NOC for ground water withdrawal as per provisions of the Notified guidelines dated 24.09.2020 and amendments dated 29.03.2023 thereto. A total **of 1334** nos. of new NOCs were **issued** and **1954** nos. of NOCs **renewed** during the 2024-25. In addition, **exemption** was given in respect of **4658** nos. **new** applications and **90 renewal** cases. **Thus**, total **8,036** NOC's/exemptions were issued during the year 2024-25.
- The new portal of issuing NOC to ground water users by CGWA i.e. BhuNeer APP was launched on 19th September 2024 in 8th India Water Week. The portal has been developed to serve a large user base across the country and supports functionalities such as user registration, authentication, real-time transaction processing, reporting, and integration with third-party systems, including payment gateways and digital records repositories.

Public Investment Board

- As per the AAP 2024–25, a total of 2,826 Piezometers have been constructed across various states, along with the installation of 298 DWLRs.
- A total of 567 Exploratory Wells (EWs) and Observation Wells (OWs) have been constructed across various states under different packages.

RGNGWTRI

- During the AAP 2024-2025, 52 Block level (Tier-III) trainings on "Local Ground Water- Related Issues and Participatory Ground Water Management" were organized to build capacity at the village/block level. All 52 trainings were organized in physical/offline mode, in which 6420 trainees have participated including 2315 women participants.
- In the year 2024RGNGWTRI, Raipur has published NAQUIM Special Volume - showcasing the outcomes of the NAQUIM 1.0 in the Bhujal News, Volume 33 (2023-24, No 1-4).
- RGNGWTRI has completed one batch of training on Water Auditors Certification culminating into 3 Water Audit Certification Courses. A total of 13 water auditor certification has been awarded during 2024-25 totaling it to 46 Certified Water Auditors.
- RGNGWTRI as a part of Rajbhasha initiative – regularly organizes quarterly Sanjyukt Rajbhasha Workshop in association with the other Central Govt. Organizations. In the 2nd Half Yearly Meeting of TOLIC, RGNGWTRI was awarded the First Prize in recognition of the activities carried out by RGNGWTRI in propagating Rajbhasa in daily Official work.
- The first publication (Volume-1) in Rajbhasha named प्रशिक्षण भारती was published in the AAP 2024-25 at RGNGWTRI.

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1. Overview of Central Ground Water Board

1.1 Introduction

Central Ground Water Board (CGWB) is an Organization under Department of Water Resources, River Development & Ganga Rejuvenation, Ministry of Jal Shakti, Government of India and it is the national apex organization for all aspects of ground water surveys, exploration, development and management.

Scientific studies related to groundwater (hydrogeological studies) started in the mid -19th century with the establishment of the Geological Survey of India (GSI) in 1851. In 1954 a dedicated organization named Exploratory Tubewell Organization (ETO) was set up for exploring ground water. In 1970, ETO was renamed as Central Ground Water Board (CGWB). In 1972, the ground water wing of Geological Survey of India (GSI) was merged with CGWB.

Central Ground Water Board is also discharging the functions as Central Ground Water Authority (CGWA) to regulate and control the development and management of ground water in the country since 1997. In the same year (1997), the Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) was established as a part of CGWB.

1.2 Vision

Sustainable Development and Management of Ground Water Resources of the Country.

1.3 Mandate

CGWB's mandate is "To develop and disseminate technologies, monitor and implement National policies for the scientific and sustainable development and management of India's Ground Water resources, including its exploration, assessment, conservation, augmentation, protection from pollution and distribution, based on principles of economic and ecological efficiency and equity"

1.4 Organizational Set up

Central Ground Water Board (CGWB) operates as a subordinate office of Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti Government of India. CGWB is a multi-disciplinary scientific organization consisting of hydrogeologists, geophysicists, chemists, hydrologists, hydrometeorologists, engineers, and other streams.

The Board is headed by the Chairman and has five Members. The extended Board of CGWB consists of five permanent members representing Central Water Commission, Ministry of Jal

Shakti, Ministry of Environment, Forest & Climate Change, Oil and Natural Gas Commission (ONGC). The organizational structure of the CGWB is illustrated in **Figure 1.1**.

Total sanctioned strength of the Board is 4017, which includes 472 Group A officers, 985 Group B officers and 2560 Group C employees.

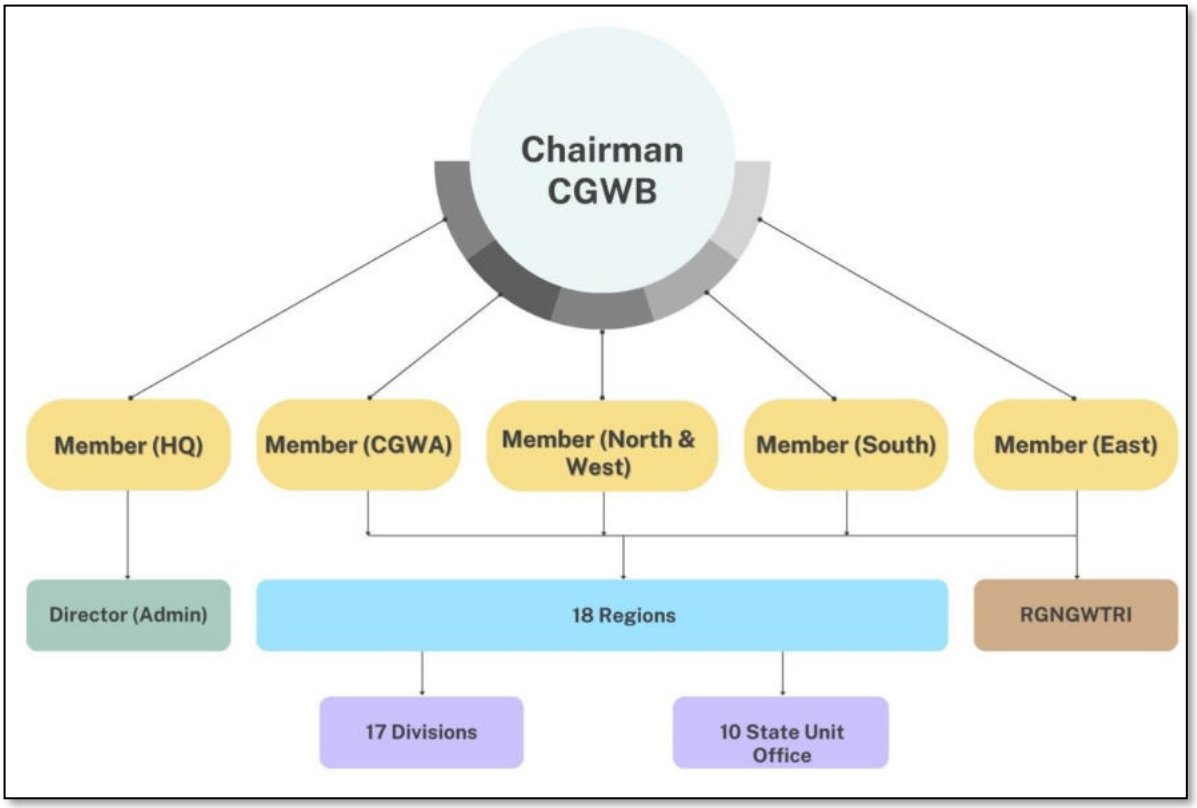

Fig. 1.1: Organizational setup of CGWB

1.5 Offices of Central Ground Water Board

CGWB has its headquarters at Bhujal Bhawan, Faridabad, Haryana; the Central Ground Water Authority (CGWA) operates from its office in New Delhi. Field activities are performed through 18 Regional Offices, 10 State Unit Offices and 17 Divisional Offices located in various part of the Country. The Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) located in Raipur, Chhattisgarh, serves as the training arm of CGWB. The location and jurisdiction of the CGWB offices are mentioned below in **Table 1.1** and **Fig 1.2**.

Table 1.1: Regions & Regional Offices of CGWB

S.No.	Name of Office	Location	Jurisdiction
Central Headquarters (CHQ)			
1	CGWB, CHQ	Faridabad	Entire Country
Central Ground Water Authority (CGWA)			
2	CGWA	New Delhi	Entire Country
Rajiv Gandhi National Ground Water Training and Research Institute			
3	RGNGWTRI	Raipur	Entire Country
North West Himalayan Region (NWHR)			
4	North West Himalayan Region	Jammu	U.T. of Jammu & Kashmir and Ladakh
5	Division-VII	Jammu	U.T. of Jammu & Kashmir and Ladakh
North Himalayan Region (NHR)			
6	North Himalayan Region	Dharamshala	Himachal Pradesh
7	Division-XVII	Dharamshala	Himachal Pradesh
North Western Region (NWR)			
8	North Western Region	Chandigarh	Punjab, Haryana, Delhi & U.T. of Chandigarh
9	State Unit Office (SUO)	Delhi	Delhi
10	Division-II	Ambala	Punjab, Haryana, NCT of Delhi & U.T. of Chandigarh
Uttarakhand Region (UR)			
11	Uttarakhand Region	Dehradun	Uttarakhand
Northern Region (NR)			
12	Northern Region	Lucknow	Uttar Pradesh
13	Division-III	Varanasi	Parts of Uttar Pradesh
14	Division-XVI	Bareilly	Part of Uttar Pradesh & Uttarkhand
15	State Unit Office	Prayagraj	Part of Uttar Pradesh
Western Region (WR)			
16	Western Region	Jaipur	Parts of Rajasthan
17	Division-XI	Jodhpur	Rajasthan
18	State Unit Office	Jodhpur	Part Rajasthan
Mid-Eastern Region (MER)			
19	Mid-Eastern Region	Patna	Bihar, Jharkhand
20	Division-V	Ranchi	Bihar and Jharkhand
21	State Unit Office	Ranchi	Jharkhand
Eastern Region (ER)			
22	Eastern Region	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands

23	Division-XV	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands
North Eastern Region (NER)			
24	North Eastern Region	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura
25	Division-VII	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura
26	State Unit Office	Shillong	Meghalaya
27	State Unit Office	Agartala	Tripura
28	State Unit Office	Itanagar	Arunachal Pradesh
West Central Region (WCR)			
29	West Central Region	Ahmedabad	Gujarat, U.T of Daman & Diu and U.T. of Dadra & Nagar Haveli
30	Division-I	Ahmedabad	Gujarat, Daman & Diu and & U.T. of Dadra & Nagar Haveli
North Central Region (NCR)			
31	North Central Region	Bhopal	Madhya Pradesh
32	Division-XII	Bhopal	Madhya Pradesh
North Central Chhattisgarh Region (NCCR)			
33	North Central Chhattisgarh Region	Raipur	Chhattisgarh
34	Division-XIII	Raipur	Chhattisgarh
South Eastern Region (SER)			
35	South Eastern Region	Bhubaneshwar	Odisha
36	Division-X	Bhubaneshwar	Odisha
Central Region (CR)			
37	Central Region	Nagar	Maharashtra
38	Division-VI	Nagpur	Maharashtra
39	State Unit Office	Pune	Parts of Maharashtra
Southern Region (SR)			
40	Southern Region	Hyderabad	Andhra Pradesh & Telangana
41	Division-IX	Hyderabad	Andhra Pradesh & Telangana
42	State Unit Office	Visakhapatnam	Parts of Andhra Pradesh
South Western Region (SWR)			
43	South Western Region	Bengaluru	Karnataka & Goa
44	Division-XIV	Bengaluru	Karnataka, Goa and Parts of Kerala
45	State Unit Office	Belagavi	Parts of Karnataka and Goa

South East Coastal Region (SECR)			
46	South East Coastal Region	Chennai	Tamil Nadu and U.T. of Puducherry
47	Division-IV	Chennai	Tamil Nadu, U.T. of Puducherry and Parts of Kerala
Kerala Region (KR)			
48	Kerala Region	Thiruvananthapuram	Kerala and U.T. of Lakshadweep

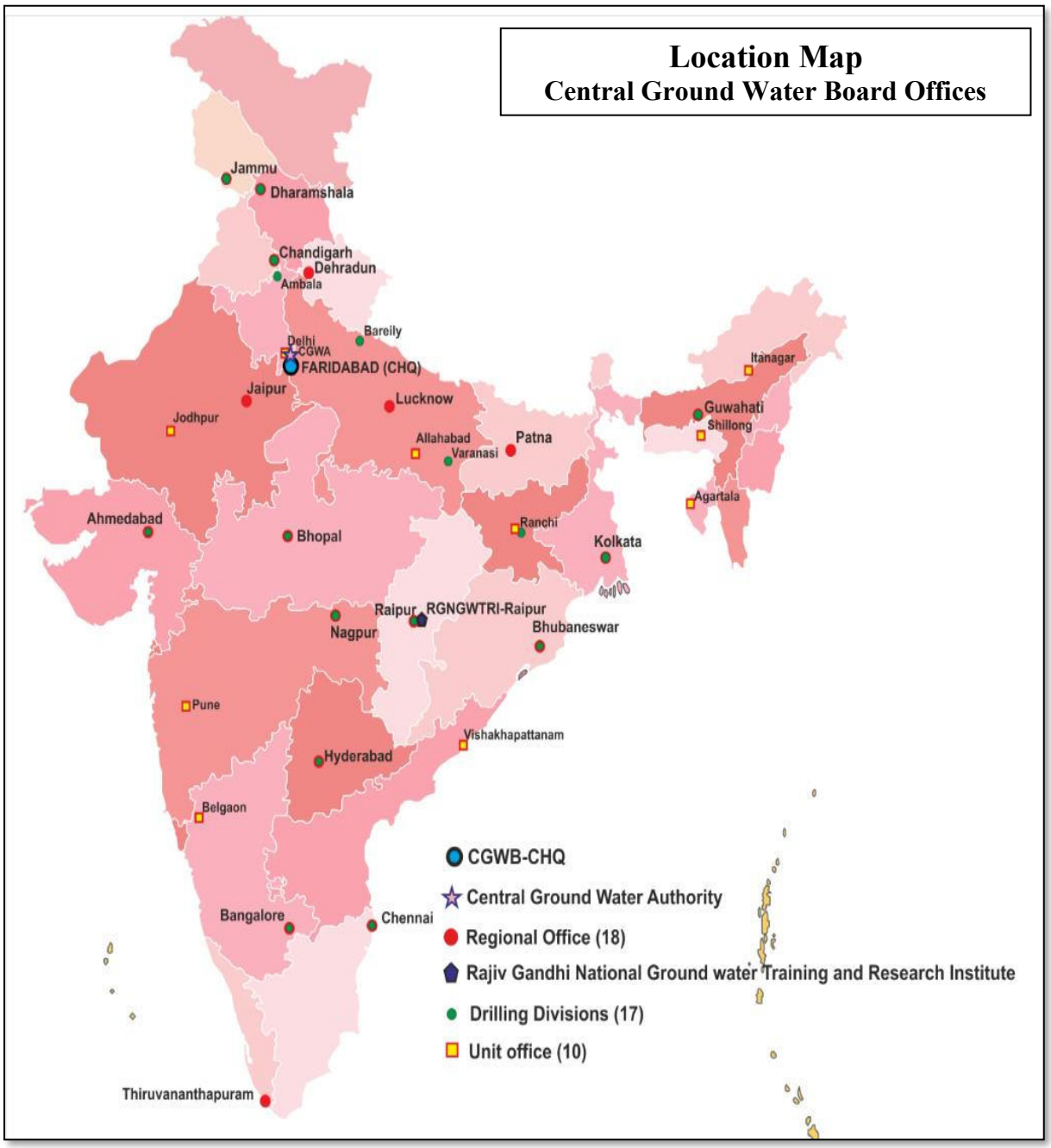

Fig. 1.2: Location and Jurisdiction of the CGWB offices

1.6 Major Activities of Central Ground Water Board

Activities of CGWB involve scientific studies for monitoring assessment and management of ground water resources of the country.

Central Sector Scheme 'Ground Water Management and Regulation (GWMR)' is the major scheme that CGWB is implementing. Major activities under the scheme include Aquifer mapping, Exploratory drilling, Monitoring of water level and quality, Resource assessment, Aquifer Rejuvenation, Ground water regulation through CGWA, Outreach activities etc. Rajiv Gandhi National Ground Water Training and Research Institute implements a three tiered training programme for training and capacity building of groundwater professionals and other stakeholders. In addition to above, CGWB also is an implementing agency of National Hydrology Project (NHP). CGWB also implements specific components of other schemes of DoWR, RD & GR like the Ground Water component of the PMKSY – HKKP scheme. Details of the activities and achievements of CGWB in the year 2024-25 are given in the subsequent chapters.

1.7 Aquifer systems of India

Aquifer System is an outcome of the extensive hydrogeological studies carried out by Central Ground Water Board (CGWB) include Hydrogeological information collected by CGWB through its survey, investigations and exploration were collated, analyzed and integrated to define the extent and characteristics of aquifer systems.

Based on the hydrogeological characteristics, the entire country has been classified into 14 Principal Aquifer Systems and 42 major aquifers as shown in **Figure 1.3**.

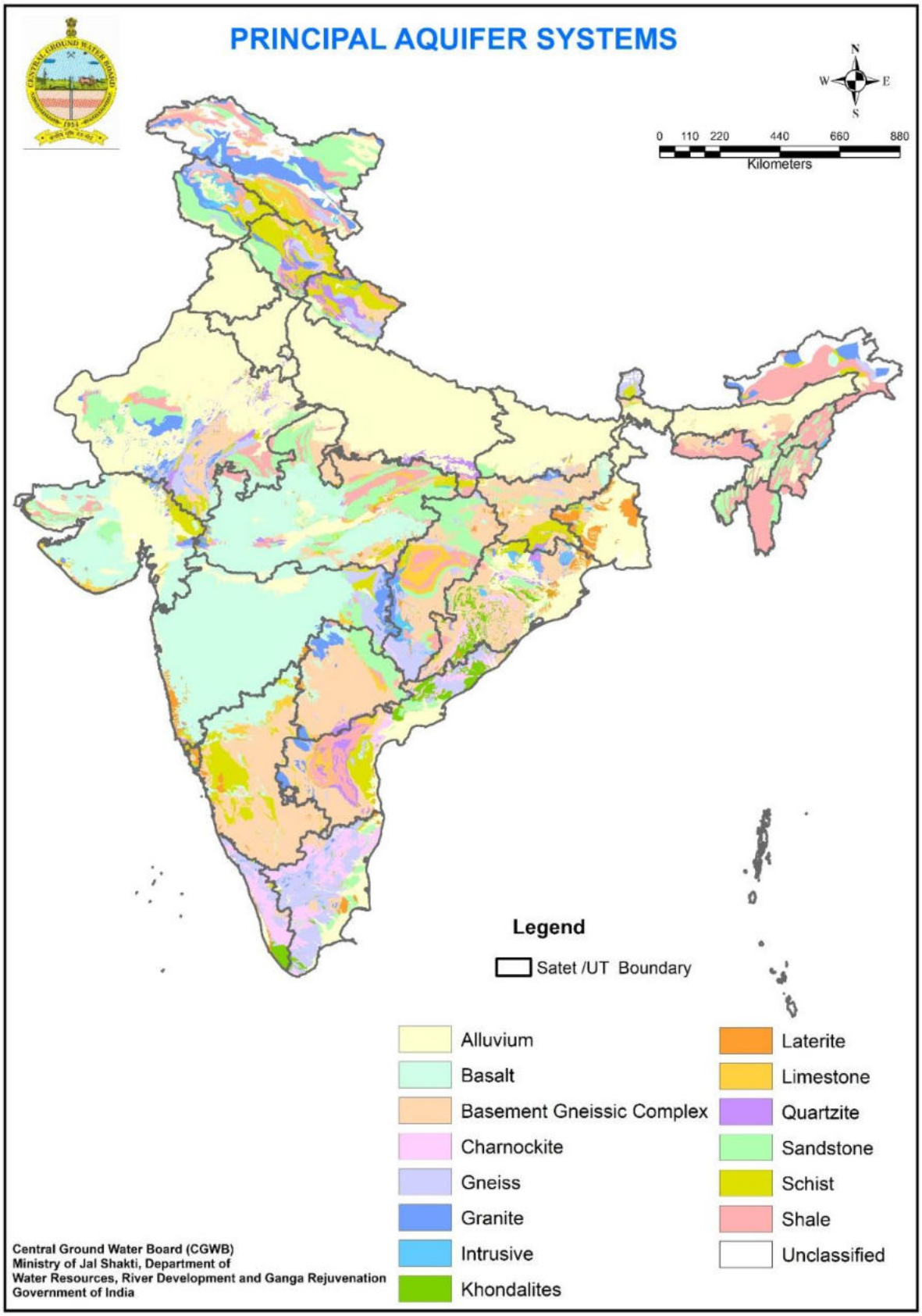

Fig. 1.3: Principal Aquifers of India

2. Schemes, Projects, and Activities

2.1 Ground Water Management and Regulation (GWMR) Scheme

2.1.1 Introduction

Ground Water Management and Regulation scheme is a continuing Central Sector Scheme, which is being implemented since 2007-08 by Central Ground Water Board (CGWB). One of the major activities under the scheme is National Aquifer Mapping & Management (NAQUIM) Programme, under which it was targeted to cover ~25 lakh sq km of mappable area and it has been covered by 31st March 2023. In addition to above other activities like ground water level & quality monitoring, assessment of dynamic ground water resources, regulation and control of ground water withdrawal, demonstrative recharge projects etc. have also been implemented.
To ensure continuity of the activities, the scheme has been approved for implementation till 31st March 2026 and includes activities like monitoring, assessment and regulation of groundwater resources and strengthening of infrastructure for technological up gradation.

2.1.2 Major Activities under the Scheme

- Aquifer Mapping
- Interventions for Aquifer Rejuvenation and Springshed Mapping
- Construction of Piezometers
- Monitoring of ground water levels
- Monitoring of ground water quality
- Assessment of ground water resources
- Regulation and control of ground water extraction
- Outreach Activities

2.2 Public Investment Board

2.2.1 Introduction

In addition to the above ongoing activities under GWMR scheme, a project has been approved by the Public Investment Board (PIB) for data generation in National Aquifer Mapping and Management (NAQUlM). The project will be implemented as a part of the GWM&R Scheme during the period 2022-2026.

2.2.2 Major Activities envisaged under this project

- Construction of 7000 piezometers along with installation of 7000 Digital Water Level Recorders (DWLRs) with telemetry system
- Heli-borne surveys for high resolution mapping in ~ 3 lakh sq km area in arid parts of NW India.
- Data Generation for Aquifer Mapping through construction of Wells (1135 wells in 11 States)

2.3 Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) - non-scheme

2.3.1 Introduction

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) is the training wing of Central Ground Water Board, Ministry of Jal Shakti, Govt. of India and functions as a `Centre of Excellence' with the national role of capacity building of CGWB, groundwater professionals from Central and State Government Organizations, Public Sector Undertakings, Non-Governmental Organizations, Academic institutions, and other stakeholders through three-tier training -Tier I (National Level), Tier II (State Level) and Tier III (Block level) trainings and is located in Sector 23, Naya Raipur, Chhattisgarh.

The institute was established at Raipur in the year 1997, for training and capacity building of groundwater professionals, and sub-professionals in core and applied fields of groundwater sciences. Due to varied hydrogeological settings in the country and excessive demands of groundwater, it has become imperative to train professionals on capacity building for the sustainable development of groundwater. The institute endeavors: (i) to be a center of excellence in Training, Research and Development in the groundwater sector; (ii) to provide training to groundwater professionals and sub-professionals in various fields of groundwater; (iii) to train NGOs, Panchayati Raj Institutions and other stakeholders in groundwater; (iv) to train various stakeholders for taking up groundwater monitoring and data collection work for various national groundwater schemes under the Participatory Ground Water Management Program (v) to undertake Research and Development works in Ground Water Sector.

2.4 National Hydrology Project (NHP)

2.4.1 Introduction

National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of the Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with an aim to improve the extent, quality, accessibility of water resources information, decision support system for floods basin level resource assessment/ planning strengthen the capacity of targeted water resources professionals and management institutions in India, and a total outlay of Rs. 3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs. 39.7674 Crore for the establishment of the National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources Data. The project spans over a period of 08 years starting from 2016-17 to 2024-25, which has been extended for a period of one-and-half years from April 2024 to Sep 2025.

2.5 Pradhan Mantri Krishi Sinchai Yojana (PMKSY)

2.5.1 Introduction

Pradhan Mantri Krishi Sinchai Yojana- Har Khet Ko Paani- Ground Water (PMKSY-HKKP-GW) scheme, launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, envisages to provide irrigation facility for small and marginal farmers in areas having sufficient potential for future development of ground water.

The scheme provides financial assistance to States for assured ground water irrigation to small and marginal farmers with priority to SC/ST and women farmers. The funding pattern is in the ratio of 90:10 (C:S) in case of NE/Hilly areas and 60:40 (C:S) in case of other areas. The scheme is applicable only in areas having stage of ground water development less than 60%, average rainfall more than 750 mm rainfall and having shallow ground water levels (less than 15 m below ground level).

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Arunachal Pradesh, Assam, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, and Uttarakhand. All the 13 projects have been successfully completed. Under these thirteen projects, construction of 29779 wells was targeted with the creation of 88679 Ha of command area benefitting 67930 small & marginal farmers.

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3. Ground Water Management and Regulation Scheme

3.1 Ground Water Level Monitoring

3.1.1 Introduction

Groundwater regime monitoring is one of the key activities of Central Ground Water Board (CGWB). Monitoring of groundwater regime is an effort to obtain information on groundwater level and chemical quality through representative monitoring and sampling. The primary objective of groundwater monitoring is to record the response of various natural and anthropogenic stress on the groundwater regime which impacts the recharge and discharge parameters with reference to geology, climate, physiography, land use pattern and hydrologic characteristics on a regional scale. The key applications of the groundwater data acquired during regime monitoring in the country are:

- Volumetric estimation of groundwater resources.
- To assess the impact of groundwater recharge and draft on long term basis.
- To categorise area into overexploited, critical, semi-critical and safe based on long term water level trend analysis.
- In deciding the depth of water well drilling as well as depth of lowering the pumps.
- Help in design, implement and monitor the effectiveness of groundwater management, protection and conservation program
- To identify groundwater quality affected areas for taking remedial measures.
- To plan groundwater recharge interventions and study the impact in time and space.

CGWB currently has a network of 26351 (Dug Wells: 16431, Piezometers: 8680, Hand Pump: 966, spring: 274) groundwater observation wells throughout the Country. The state wise breakup is given below in **Table 3.1**.

The groundwater regime monitoring started in the year 1969 by Central Ground Water Board. The groundwater levels are measured manually four times a year during the months of January, March/April/ May, August and November coinciding with crop season and onset of monsoon as well as to capture deepest and shallowest water level in the hydrological year.

Table 3.1: State/UT -Wise Status of Groundwater Monitoring Wells

S.No.	State/ UT	Number of GW Monitoring Stations as on March 2025				
		Dug Well	Piezometers	Handpump	Spring	Total Groundwater Monitoring Stations (2025)
1	Andhra Pradesh	656	790	0	0	1446
2	Arunachal Pradesh	26	4	0	8	38
3	Assam	383	91	8	5	487
4	Bihar	977	117			1094

5	Chhattisgarh	1006	269	0	0	1275
6	Delhi	22	115	0	0	137
7	Goa	81	54	0	0	135
8	Gujarat	781	500	0	0	1281
9	Haryana	150	345	0	0	495
10	Himachal Pradesh	129	66	0	27	222
11	Jammu & Kashmir	270	87	0	26	383
12	Jharkhand	466	147	0	0	613
13	Karnataka	1316	946	0	0	2262
14	Kerala	1379	271	0	24	1674
15	Madhya Pradesh	1378	476	0	0	1854
16	Maharashtra	1857	318	0	0	2175
17	Manipur	4	0	0	2	6
18	Meghalaya	70	13	0	21	104
19	Mizoram	7	0	0	20	27
20	Nagaland	80	1	0	20	101
21	Odisha	1489	291			1780
22	Punjab	115	346	0	0	461
23	Rajasthan	647	842	0	0	1489
24	Sikkim	0	4	0	0	4
25	Tamil Nadu	766	713	0	0	1479
26	Telangana	270	1001	0	0	1271
27	Tripura	104	13		10	127
28	Uttar Pradesh	1082	453			1535
29	Uttarakhand	35	12	196	111	354
30	West Bengal	725	352	762	0	1839
UTs						
1	Andaman & Nicobar	111	0	0	113	
2	Chandigarh	1	23	0	0	24
3	DNH and DD	40	4			44
4	Puducherry	8	14	0	0	22
TOTAL		**16431**	**8680**	**966**	**274**	**26351**

The database thus generated forms the basis for planning the groundwater development and management programme. This data is used for assessment of groundwater resources and changes in the regime consequent to various development and management activities.

3.2 Ground Water Quality Monitoring

3.2.1 Introduction

Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of ground water samples. The Chemical laboratories are well equipped with sophisticated instruments like Isotope-Ratio Mass Spectrometer (IRMS), Inductively Coupled Plasma Mass-Spectrometer (ICP-MS), Atomic Absorption Spectrophotometer (AAS) etc. as

detailed below in **Table 3.2**. Ground water sampling for quality monitoring is done twice a y ear during the pre-monsoon and post-monsoon period. In addition to it ground water samples are also collected and analyzed as a part of other scientific studies. CGWB has also taken up sampling and analysis of ground water samples for Uranium content across the country.

Table 3.2: Status of Instruments in Regional Chemical Laboratories of CGWB

S.No.	Instrument/ Equipment	No. of Instruments/Equipment in 16 Regional Chemical Laboratory of CGWB
1	pH Meter	1-2 in each Laboratory
2	EC Meter	1-2 in each Laboratory
3	Flame- Photometer	1-2 in each Laboratory
4	UV/Visible-Spectro-photometer	1 in each Laboratory
5	Atomic Absorption Spectro-photometer	1 in each Laboratory (2 each at Lucknow & Kolkata)
6	Ion Chromatograph	1 in each Laboratory
7	Ion meter	1 in each Laboratory
8	ICP-MS	1 each in Lucknow, Chandigarh, Guwahati & Hyderabad laboratory
9	IRMS	1 in Hyderabad Laboratory
10	TOC	1 each in Lucknow, Chandigarh, Kolkata, Guwahati & Chennai laboratory
11	LSC	1 each in Lucknow, Chandigarh, Bengaluru, Nagpur laboratory
12	Ultra-Water Purification System	1 in each Laboratory
13	Fluorimeter	1 in each Laboratory

In the AAP 2024-25, a total of 64696 groundwater samples have been analyzed by CGWB which included 32853 for Basic parameters, 31843 for Heavy Metals. The details of water samples analyzed by different chemical laboratories during 2024-25 are presented in **Table 3.3** and its graphical presentation is shown in **Figure 3.1**.

Table 3.3: Region wise Ground Water Samples analyzed during 2024-25

S.No.	Regional Labs	Number of Samples analysed during 2024-25 (01.04.2024 to 31.03.2025)								
		NHS Basic	NAQUIM 2.0 Basic	Other Basic	Total Basic	NHS Heavy Metals	NAQUIM 2.0 Heavy Metals	Other Heavy Metals	Total Heavy Metals	Total
1	NWHR, Jammu	180	232	41	453	730	61	13	804	1257
2	NWR, Chandigarh	1098	41	1508	2647	1348	62	1949	3359	6006
3	WR, Jaipur	932	0	750	1682	1864	0	454	2318	4000
4	WCR, Ahmedabad	731	112	108	951	1416	180	42	1638	2589
5	NCR, Bhopal	1191	223	954	2368	668	429	895	1992	4360
6	NCCR, Raipur	506	762	445	1713	506	690	471	1613	3326
7	CR, Nagpur	1417	1557	514	3488	831	962	343	2136	5624
8	NR, Lucknow	1828	204	593	2625	1842	99	1414	3355	5980
9	MER, Patna	962	569	275	1806	0	0	0	0	1806
10	ER, Kolkata	1352	343	232	1927	1352	434	286	2072	3999
11	NER, Guwahati	1129	195	409	1733	2284	390	824	3498	5231
12	SER, Bhubaneshwar	1083	618	81	1782	1083	858	2	1943	3725
13	SR, Hyderabad	2126	1981	793	4900	0	1153	1564	2717	7617
14	SWR, Bangalore	695	822	157	1674	1398	1060	113	2571	4245
15	SECR, Chennai	613	803	363	1779	313	287	95	695	2474
16	KR, Trivandrum	730	151	444	1325	730	155	247	1132	2457
Total		**16573**	**8613**	**7667**	**32853**	**16365**	**6820**	**8712**	**31843**	**64696**

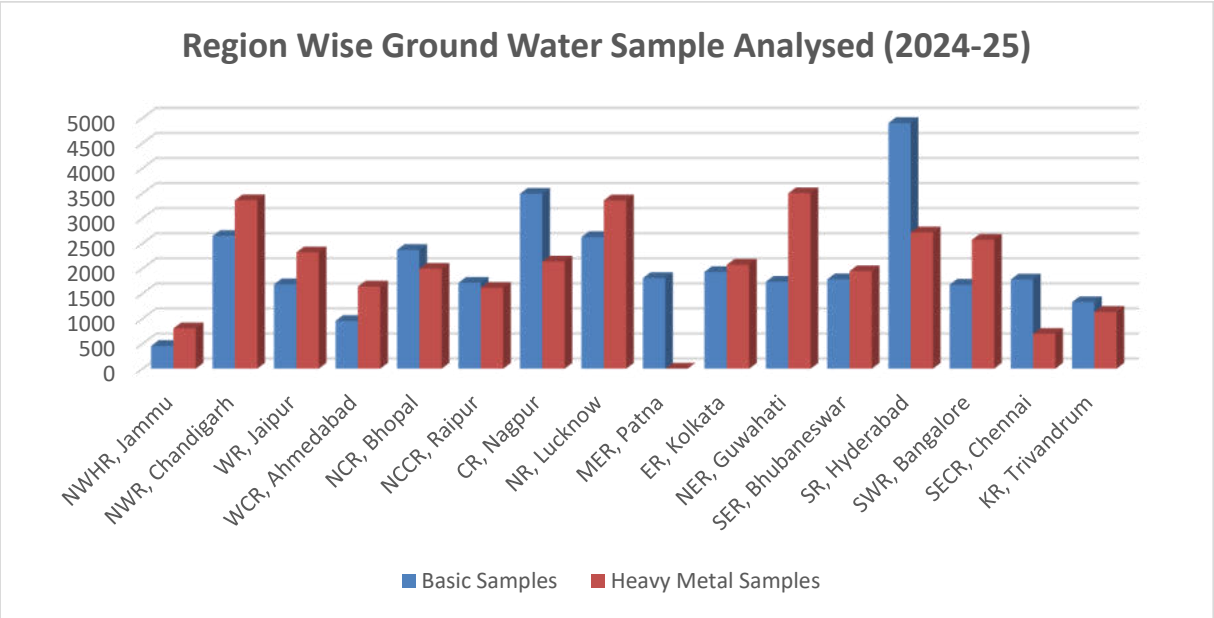


Fig. 3.1: Ground Water Samples analyzed during 2024-25

3.2.2 Regional Chemical Laboratories

16 Regional Chemical Laboratories – NWHR Jammu, NWR Chandigarh, WR Jaipur, WCR Ahmadabad, NCR Bhopal, NCCR Raipur, CR Nagpur, NR Lucknow, MER Patna, ER Kolkata, NER Guwahati, SER Bhubaneshwar, SR Hyderabad, SWR Bangalore, SECR Chennai and KR Trivandrum.

3.2.3 NABL accredited laboratories

14 NABL Accredited Laboratories - NWHR Jammu, NWR Chandigarh, WR Jaipur, NCR Bhopal, NCCR Raipur, CR Nagpur, NR Lucknow, MER Patna, ER Kolkata, NER Guwahati, SER Bhubaneshwar, SR Hyderabad, SECR Chennai and KR Trivandrum.

3.3 Geophysical Studies

3.3.1 Introduction

Geophysical investigations employ non-intrusive methodologies to derive proximal inferences concerning the hydrogeological properties and spatial architecture of subsurface aquifer systems. These techniques facilitate the acquisition of critical subsurface data without perturbing the in-situ geological milieu, thereby maintaining the hydrodynamic and structural integrity of the aquifer units under investigation.

By leveraging a diverse suite of geophysical modalities (e.g., electrical resistivity tomography, seismic refraction/reflection, ground-penetrating radar), researchers can quantitatively assess the petrophysical attributes of aquifers, including porosity, hydraulic conductivity, and fluid saturation. This approach not only augments the comprehension of groundwater resource dynamics but also provides foundational data for optimized groundwater resource management and conservation strategies, ensuring the long-term sustainability of these essential potable water reservoirs.

3.3.2 Surface and Sub-Surface Geophysical Studies

Geophysical studies are essential for understanding aquifer characteristics and their subsurface arrangement, offering indirect insights through non-invasive techniques. The Central Ground Water Board (CGWB) utilizes an in-house facility for various geophysical investigations, employing both surface and subsurface (well logging) methods extensively in groundwater exploration and water well construction. The findings from these geophysical studies are then integrated with hydrogeological investigations to create a comprehensive understanding.

3.3.2.1 Surface Geophysical Investigations

Surface geophysical investigations are crucial for:

- Groundwater exploration
- Aquifer mapping
- Water supply investigations
- Recharge site selection
- Specialized studies

These investigations help in mapping subsurface lithology, which in turn influences how groundwater occurs, distributes, and moves. CGWB began regularly using geophysical techniques for groundwater investigations in the 1970s.

Traditional methods like Electrical Resistivity (ER) surveys are standard practice at CGWB for both soft and hard rock formations. They help:

- Identify groundwater-bearing zones
- Pinpoint locations for borehole construction
- Contribute to proposals for artificial recharge structures
- Map bedrock topography and sandy horizons in non-perennial channels
- Identify the Fresh/Saline interfaces

More recently, Transient Electromagnetic (TEM) techniques have been introduced to identify subsurface layer parameters like resistivity and thickness, similar to ER surveys. While providing comparable information, TEM surveys are faster than conventional ER surveys. CGWB is also currently conducting advanced 2-D Imaging surveys using Multi Electrodes Resistivity Meters for special studies and detailed aquifer mapping.

3.3.2.2 Subsurface Borehole Geophysics

Subsurface borehole geophysics plays a vital role in groundwater exploration by providing valuable insights into:

- Lithology in terms of granular and clay zones
- Fractures
- Water quality (fresh/saline)

Its main purpose is to map the subsurface arrangement of aquifers. In drilled pilot boreholes, borehole-geophysical logging is used to determine the characteristics and thickness of different lithological and hydrogeological units. This information is particularly critical in regions of India where saline or brackish water-bearing aquifers are often mixed with freshwater aquifers. Understanding this helps in the proper placement of casing and screens in water-supply wells, ensuring effective water sourcing and addressing challenges related to groundwater salinity.

A Region wise list of existing Geophysical Instruments is presented in **Table 3.4**.

Table 3.4: Region wise details of Geophysical Instruments in CGWB

S.No.	Region	Geophysical Logger	Resistivity Meter	TEM
1	ER-Kolkata	Uptron Logger- Not Working and Logger Van already surveyed off	CRM-500 Auto-C-2nos (Working)	Terra TEM 24-Working
2	KR-Trivandrum	Not available	SSR-MP-ATS -1 (Working)	Terra TEM 24-Working
3	NCCR-Raipur	Not available	SSR-MP-ATS -1 (Working)	Terra TEM 24-Working
4	NCR - Bhopal	Not available	SSRMP ATS & CRM 500 AUTO C (Working)	-
5	WCR-Ahmadabad	Uptron Logger- Not Working	SSR-MP-ATS (Working)	
6	NWHR-Jammu	Uptron logger -Not Working	ABEM terrameter SAS 300 (Working)	-
7	NHR-Dharmshala	-	-	No Geophysicist Posted
8	UR - Dehradun	Not available	SSR-MP ATS (Working)	-
9	NER-Guwahati	Not available	SSR-MP ATS (Working)	-
10	CR- Nagpur	Not available	SSR-MP-ATS (Working)	-
11	MER - Patna	Uptron Logger- Not working	SSR-MP-ATS & CRM-500 AUTO C (Working)	Terra TEM 24-Working
12	SER - Bhubaneshwar	Uptron Logger & Logger van-Not Working	CRM 20 (Working)	-

S.No.	Region	Geophysical Logger	Resistivity Meter	TEM
13	NWR - Chandigarh	Uptron Logger & Logger van- Not Working	CRM-500 Auto-C (Working) & CRM 500 (Not Working)	-
14	SR - Hyderabad	Not available	ABEM Terrameter LS (ERT), ABEM Terrameter LS (1D, VES), SSR-MP1, & SSR MP ATS (All working)	Terra TEM 24-Working
15	SWR- Bangalore	Not available	SSR-MP-ATS (Working)	Terra TEM 24-Working
16	SECR- Chennai	Not available	SSRMP ATS & CRM AUTO D (Working)	Terra TEM 24-Working
17	WR - Jaipur	Uptron EL-600 Logger working	Aquameter CRM-500 Auto C and DDR4-MP. (Working)	Terra TEM 24-Working
18	NR - Lucknow	Not available	ABEM Terrameter LS working but maximum current is limited to 199 mA. ABEM Terrameter LS (2D) is working but its maximum depth is limited to 70 m because one relay board is not working	Terra TEM 24-Working
19	SUO-Delhi	Not available	Aquameter CRM AUTO C (Working)	-

During the AAP 2024-25, CGWB through its field officers carried out 2230 Vertical Electrical Sounding (VES), 1633 Transient Electro-magnetic (TEM) survey for NAQUIM 2.0, short term water supply investigations, special studies. The achievements of geophysical surveys; VES & TEM are presented in **Table 3.5** & **Table 3.6** respectively. A total 1593 boreholes logging conducted in in-house and outsourcing. A special study was also carried out at 14 locations using advanced technology (2D Imaging) for FMDA study in Faridabad flood plains of the Yamuna River in Haryana, wherein representative field photographs documenting the geophysical surveys are illustrated in **Fig. 3.2**.

Geophysical Survey (Electrical Resistivity Tomography) for FMDA study in Faridabad in flood plain of River Yamuna in Haryana.	
TEM survey at Kerala	VES Survey at Mersi, North Goa
TEM Field Demonstration in Tier-II Program, Mysoure University, Karnataka	VES survey in Pratappara Village, Nandod, Narmada District, Gujarat
2-D Imaging at Cantonment, Jhansi, U.P.	Feasibility Survey for Army Establishment, Laddakh

Fig. 3.2 Representative Field Photographs of Geophysical Surveys

Table 3.5: Region wise achievements Geophysical Activities (VES)

S.No.	State/UT	Region	Annual Target (2024-25)	Progress as on 31.03.2025
1	Maharashtra	CR	140	147
2	Andaman and Nicobar Islands	ER	40	70
3	West Bengal	ER	70	0
4	Kerala	KR	50	50
5	Lakshadweep	KR	0	0
6	Bihar	MER	25	30
7	Jharkhand	MER	25	27
8	Chhattisgarh	NCCR	200	220
9	Madhya Pradesh	NCR	175	185
10	Assam	NER	100	95
11	Arunachal Pradesh	NER	0	0
12	Meghalaya	NER	0	0
13	Manipur	NER	0	0
14	Mizoram	NER	0	0
15	Nagaland	NER	10	14
16	Tripura	NER	0	13
17	Uttar Pradesh	NR	100	109
18	Jammu and Kashmir	NWHR	75	75
19	Ladakh	NWHR	0	
20	Haryana	NWR	100	54
21	Punjab	NWR	50	39
22	Tamilnadu	SECR	300	311
23	Odisha	SER	150	174
24	Andhra Pradesh	SR	125	87
25	Telangana	SR	125	176
26	Delhi	SUO	60	45
27	Goa	SWR	20	20
28	Karnataka	SWR	115	118
29	Uttarakhand	UR	75	75
30	Gujarat	WCR	75	76
31	Rajasthan	WR	20	20
	Total		**2225**	**2230**

Table 3.6: Region wise achievements Geophysical Activities (TEM)

S.No.	State/UT	Region	Annual Target (2024-25)	Achievement as on 31.03.2025
1	Uttar Pradesh	NR	300	285
2	West Bengal	ER	310	200
3	Bihar	MER	100	107
4	Jharkhand	MER	50	51
5	Chhattisgarh	NCCR	200	210
6	Rajasthan	WR	50	6
7	Kerala	KR	150	150
8	Tamil Nadu	SECR	100	119
9	Andhra Pradesh	SR	45	48
10	Telangana	SR	30	36
11	Goa	SWR	150	150
12	Karnataka	SWR	250	271
	Total		**1735**	**1633**

3.3.3 High resolution aquifer mapping and management in arid areas of India:

Central Ground Water Board (CGWB) has initiated high resolution mapping of aquifers using state of art technology heliborne geophysical survey in parts of the arid areas spread over the states of Rajasthan, Gujarat and Haryana. The study is aimed at establishing aquifer geometry, demarcation of de-saturated and saturated aquifers, fresh/saline aquifers, identification of paleochannels network, identification of potential sites for ground water withdrawal and identification of sites for water conservation structures, MAR sites etc. The phase-I heliborne survey project of an area of 97165 sq.km and 40313line km flight line has been completed covering 92 Blocks in Rajasthan, Gujarat and Haryana states. In Rajasthan, 47 Blocks spanning in parts of 8 districts were addressed, with 1 Block each in Bikaner, Sikar, and Jaisalmer districts, 7 Blocks in Churu district, 9 blocks in Ganganagar district, 8 blocks in Jalore district, 10 blocks in Pali district, and 10 blocks in Jodhpur district. In Gujarat, the areas covered included 11 blocks in Rajkot, 6 blocks in Jamnagar, 5 blocks in Morbi, 10 blocks in Surendranagar, and 4 blocks in Devbhumi Dwarka districts, while in Haryana, 5 blocks in Kurukshetra and 4 blocks in Yamunanagar districts were surveyed. The deliverables such as granular zones, saturated, de-saturated, saline/fresh aquifers, suitable sites for drilling, expected water level, suitable zones for artificial recharge and type of recharge structures at Gram Panchayat level are being extracted along the flight line from the heliborne survey results. A coffee table book was also prepared on “High Resolution Aquifer Mapping in Arid Regions of North Western India” and inaugurated by Hon’ble Minister of State, Ministry of Jal Shakti Govt. of India.

In alignment with the strategic objectives set by the Ministry, Gram Panchayat-level reports for 68 blocks were successfully completed at the Central Headquarters (CHQ) in a mission-mode approach. This extensive exercise was carried out over a span of five months, from November 2024 to 31 March 2025, with the dedicated deployment of officers drawn from various regional offices. Their coordinated efforts ensured the timely and efficient delivery of reports in alignment with the stipulated goals. Deliverables at the Gram Panchayat level were extracted for 92 blocks, covering approximately 10,000 villages across around 4,500 Gram Panchayats. Groundwater potential sites were recommended for dugwells/borewells in Rajasthan (3,850 sites), Gujarat (2,320 sites), and Haryana (323 sites). Additionally, Managed Aquifer Recharge (MAR) sites were identified in Rajasthan (1,052), Gujarat (1,153), and Haryana (323), along with recommendations for the appropriate types of recharge structures.

Inauguration of Coffee Table Book by Hon'ble Minister of State, Ministry of Jal Shakti Govt of India on *High Resolution Aquifer Mapping in Arid Regions of North Western India* on the occasion of 8th India Water Week 2024 (19.09.2024).

3.3.3.1 Trainings on Heliborne Transient Electromagnetics

The training on Heliborne Geophysical Studies, proposed by the Central Ground Water Board (CGWB), was conducted in three modules: a 3-week program at Aarhus University, Hydrogeophysics Group (HGG), Denmark (June 10–28, 2024); a 2-week module at Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), CGWB, Raipur (February 3–14, 2025); and a 1-week session at CGWB, Central Headquarters (CHQ), Faridabad (March 3–7, 2025).

A total of 15 CGWB officers (including 3 females and 12 males) participated in the training. The delegation comprised 9 Geophysicists, 5 Hydrogeologists, and 1 Engineer, all professionals engaged in groundwater-related work. The training aimed to enhance their understanding of heliborne geophysical surveys and the Heliborne Electromagnetic Method.

Participants actively engaged in hands-on sessions, learning techniques for data acquisition, processing, and interpretation using Aarhus Workbench software. The knowledge gained is expected to contribute significantly to future groundwater management initiatives in India.

Photographs of Trainings on Heliborne Transient Electromagnetics

Module-1: Training at Aarhus University Denmark (10-28 June, 2024)

Module-2: Training at RGI, CGWB, Raipur (03-14 February, 2025)

Module-3: Training at CGWB, CHQ, Faridabad (03-07 March, 2025)

3.4 Exploratory Drilling

3.4.1 Introduction

Ground Water Exploration aided by drilling is one of the major activities of the Board. It is aimed at delineation of aquifers in different hydrogeological setups and determination of their hydraulic parameters. The exploratory drilling operations have enabled demarcation of aquifers both in lateral and vertical extensions and evaluation of various aquifer parameters, designing of suitable structures and assessment of their yield potential in various hydrogeological

settings. These studies have helped in identifying areas worthy for future ground water development. Ground Water exploration contributes to a large extent in guiding the States to implement ground water development schemes.

Groundwater exploration is being carried out by the Board through its fleet of 73 drilling rigs (22 Direct Rotary, 49 Down the Hole and 02 Dual Rotary). The region/division/state wise deployment of rigs during AAP 2024-25 is mentioned in **Table 3.7** and the division-wise details of pumping test units is given in **Table 3.8**.

During the year 2024-25, Central Ground Water Board under its in-house Ground Water Exploration programme constructed 750 wells (EW- 199, OW- 68, PZ- 477, RW-6) (**Table 3.9**). Out of 199 exploratory wells, 168 wells were constructed in hard rock, 31 wells in alluvium and formations 15 exploratory wells constructed in Tribal areas of the country.

The details of State/ Division / Region wise distribution of boreholes drilled/ completed during the year is presented in **Table 3.10** & **3.11**.

The Board since inception has drilled 48638 bore holes (including 10349 bore holes through outsourcing), as on 31.03.2024, to identify worthy areas for ground water development in the country. The statement showing State-wise distribution of boreholes drilled/completed till March, 2025 in the country is presented in **Table 3.12**.

Table 3.7: Region/Division/State wise Deployment of Rigs during AAP 2024-25

Region	Division	State	Number Of Rigs			
			DR	DTH	DUR	T
NWHR, Jammu	VIII, Jammu	Jammu & Kashmir	0	2	1	3
NWR, Chandigarh	II, Ambala	Punjab	1	0	0	1
		Haryana	2	0	0	2
WR, Jaipur	XI, Jodhpur	Rajasthan	2	3	0	5
WCR, Ahmedabad	I, Ahmedabad	Gujarat	3	2	0	5
NCR, Bhopal	XII, Bhopal	Madhya Pradesh	0	3	0	3
NCCR, Raipur	XIII, Raipur	Chhattisgarh	0	4	0	4
CR, Nagpur	VI, Nagpur	Maharashtra	0	5	0	5
NR, Lucknow	III, Varanasi	Eastern UP	3	2	0	5
	XVI, Bareilly	Western UP	3	0	0	3
MER, Patna	V, Ranchi	Bihar	1	2	0	3
		Jharkhand	0	1	0	1
ER, Kolkata	XV, Kolkata	West Bengal	3	2	0	5
NER, Guwahati	VII, Guwahati	Assam	2	1	0	3
		Nagaland	0	1	0	1
SER, Bhubaneshwar	X, Bhubaneshwar	Orissa	1	4	0	5

SR, Hyderabad	IX, Hyderabad	Andhra Pradesh	0	2	0	2
		Telangana	0	3	0	3
SWR, Bangalore	XIV, Bangalore	Karnataka Goa	0 0	4 1	0 0	4 1
SECR, Chennai	IV, Chennai	Tamil Nadu	1	4	0	5
KR, Trivandrum		Kerala	0	1	0	1
UR, Dehradun	XVI, Bareilly	Uttarakhand	0	0	1	1
NHR, Dharamshala	XVII, Dharamshala	Himachal Pradesh	0	2	0	2
Total			22	49	2	73
Abbreviation:	**DR: Direct Rotary**	**DUR: Dual Rotary**	**DTH: Down The Hole**			

Table 3.8: Status of Division wise Pumping Units of CGWB

Division	No. of Pumping Units
I. Ahmedabad	**1**
II. Ambala	**1**
III. Varanasi	**3**
IV. Chennai	**1**
V. Ranchi	**3**
VI. Nagpur	**1**
VII. Guwahati	**1**
VIII. Jammu	**1**
IX. Hyderabad	**1**
X. Bhubaneshwar	**2**
XI. Jodhpur	**2**
XII. Bhopal	**1**
XIII. Raipur	**1**
XIV. Bengaluru	**1**
XV. Kolkata	**3**
XVI. Bareilly	**2**
XVII. Dharamshala	**0**

Table 3.9: State wise Wells constructed by CGWB during 2024-25 (till 31th March 2025)

State	Tentative Target 2024-25					Cumulative Achievement 2024-25 (Since April 2024)				
	EW	OW	PZ	RW	T	EW	OW	PZ	RW	T
West Bengal	0	0	27	2	29	0	0	29	0	29
Bihar	10	0	10	1	21	4	4	5	1	14
Jharkhand	5	0	5	1	11	2	2	8	0	12
Assam	6	0	10	0	16	7	0	8	0	15
Arunachal Pradesh	0	0	0	0	0	0	0	0	0	0
Nagaland	2	0	4	0	6	5	0	3	0	8
Manipur	0	0	0	0	0	0	0	0	0	0
Meghalaya	0	0	0	0	0	0	0	0	0	0
Tripura	0	0	0	0	0	0	0	0	0	0
Odisha	4	2	56	0	62	5	2	57	0	64
Chhattisgarh	10	0	35	0	45	6	6	33	0	45
Uttar Pradesh	17	10	27	0	54	17	11	18	0	46
Uttarakhand	1	1	0	0	2	1	1	0	0	2
New Delhi	0	0	0	0	0	0	0	0	0	0
Punjab	2	2	10	0	14	2	2	13	0	17
Haryana	3	3	4	0	10	3	3	4	0	10
Chandigarh	0	0	0	0	0	0	0	0	0	0
Jammu & Kashmir	0	0	15	0	15	0	0	15	0	15
Himachal Pradesh	0	0	10	0	10	0	0	10	1	11
Rajasthan	7	3	40	0	50	8	3	33	0	44
Gujarat	9	0	25	0	34	19	1	17	0	37
Daman & Div	0	0	0	0	0	0	0	0	0	0
Madhya Pradesh	0	0	25	0	25	1	1	28	0	30
Maharashtra	29	6	50	0	85	24	8	53	0	85
Dadra & Nagar Haveli	0	0	0	0	0	0	0	0	0	0
Andhra Pradesh	10	6	6	0	22	17	7	9	0	33
Telangana	12	6	20	0	38	23	6	47	0	76
Tamil Nadu	28	10	42	0	80	38	6	38	1	83
Goa	5	3	4	0	12	6	3	2	0	11
Karnataka	8	4	36	0	48	11	2	38	2	53
Kerala	0	0	10	0	10	0	0	9	1	10
Total	168	56	471	4	699	199	68	477	6	750

Table 3.10: Division- wise wells constructed by CGWB during 2024-25

Division	Target 2024-25					Cumulative Achievement 2024-25 (since April 2024)					Achievement (%)
	EW	OW	PZ	RW	T	EW	OW	PZ	RW	T	
I. Ahmedabad	9	0	25	0	34	16	1	17	0	34	100.00%
II. Ambala	5	5	14	0	24	5	5	17	0	27	112.50%
III. Varanasi	12	10	17	0	39	14	10	7	0	31	79.49%
IV. Chennai	28	10	52	0	90	38	6	47	2	93	103.33%
V. Ranchi	15	0	15	2	32	6	6	13	1	26	81.25%
VI. Nagpur	29	6	50	0	85	24	8	53	0	85	100.00%
VII. Guwahati	8	0	14	0	22	12	0	11	0	23	104.55%
VIII. Jammu	0	0	15	0	15	0	0	15	0	15	100.00%
IX. Hyderabad	22	12	26	0	60	40	13	56	0	109	181.67%
X. Bhubaneshwar	4	2	56	0	62	5	2	57	0	64	103.23%
XI. Jodhpur	7	3	40	0	50	11	3	33	0	47	94.00%
XII. Bhopal	0	0	25	0	25	1	1	28	0	30	120.00%
XIII. Raipur	10	0	35	0	45	6	6	33	0	45	100.00%
XIV. Bangalore	13	7	40	0	60	17	5	40	2	64	106.66%
XV. Kolkata	0	0	27	2	29	0	0	29	0	29	100.00%
XVI. Bareilly	6	1	10	0	17	4	2	11	0	17	100.00%
XVII. Dharamshala	0	0	10	0	10	0	0	10	1	11	110.00%
Total	168	56	471	4	699	199	68	477	6	750	107.29%

Table 3.11: Region-wise wells constructed by CGWB during 2024-25

Region	Tentative Target 2024-25					Achievement 01.04.24 to 31.03.2025					% Achievement
	EW	OW	PZ	RW	T	EW	OW	PZ	RW	T	
NWHR Jammu	0	0	15	0	15	0	0	15	0	15	100.00%
NWR Chandigarh	5	5	14	0	24	5	5	17	0	27	112.50%
WR Jaipur	7	3	40	0	50	8	3	33	0	44	88.00%
WCR Ahmedabad	9	0	25	0	34	19	1	17	0	37	108.82%
NCR Bhopal	0	0	25	0	25	1	1	28	0	30	120.00%
NCCR Raipur	10	0	35	0	45	6	6	33	0	45	100.00%
CR Nagpur	29	6	50	0	85	24	8	53	0	85	100.00%
NR Lucknow	17	10	27	0	54	17	11	18	0	46	85.18%
MER Patna	15	0	15	2	32	6	6	13	1	26	102.94%
ER Kolkata	0	0	27	2	29	0	0	29	0	29	100.00%
NER Guwahati	8	0	14	0	22	12	0	11	0	23	104.50%

SER Bhubaneshwar	4	2	56	0	62	5	2	57	0	64	103.23%
SR Hyderabad	22	12	26	0	60	40	13	56	0	109	181.67%
SWR Bangalore	13	7	40	0	60	17	5	40	2	64	106.66%
SECR Chennai	28	10	42	0	80	38	6	38	1	83	103.33%
KR Trivandrum	0	0	10	0	10	0	0	9	1	10	100.00%
UR Dehradun	1	1	0	0	2	1	1	0	0	2	100.00%
NHR Dharamshala	0	0	10	0	10	0	0	10	1	11	110.00%
Total	168	56	471	4	699	199	68	477	6	750	107.29%
Abbreviation:	**EW: Exploration Well**			**OW: Observation Well**			**PZ: Piezometer Well**			**RW: Recharge Well**	

Table 3.12: State/UT wise cumulative Boreholes Drilled by CGWB till 31st March 2025

S.No	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	RW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs							
A.	**STATES**												
1	Andhra Pradesh	456	122	201	779	1012	452	362	9	4	0	1839	2618
2	Arunachal Pradesh	0	0	0	0	46	10	0	1	1	0	58	58
3	Assam	17	20	0	37	480	214	86	16	42	0	838	875
4	Bihar	151	39	16	206	337	203	93	10	514	1	1158	1364
5	Chhattisgarh	414	15	116	545	912	302	257	0	28	0	1499	2044
6	Goa	13	1	0	14	64	21	16	0	31	0	132	146
7	Gujarat	420	116	8	544	1215	558	572	27	255	0	2627	3171
8	Haryana	75	55	80	210	418	281	255	23	170	0	1147	1357
9	Himachal Pradesh	0	0	0	0	272	49	48	1	0	0	370	370
10	Jammu & Kashmir	21	0	0	21	503	124	80	8	114	0	829	850
11	Jharkhand	275	28	16	319	509	260	88	4	71	0	932	1251
12	Karnataka	804	146	0	950	1665	748	473	7	5	2	2900	3850
13	Kerala	10	0	13	23	616	217	273	16	13	1	1136	1159
14	Madhya Pradesh	725	57	117	899	1475	753	259	8	149	0	2644	3543
15	Maharashtra	92	2	13	231	1872	592	314	2	166	0	2946	3177

				7									
16	Manipur	0	0	0	0	29	14	1	0	2	0	46	46
17	Meghalaya	0	0	0	0	122	40	4	2	8	0	176	176
18	Mizoram	0	0	0	0	3	3	0	0	0	0	6	6
19	Nagaland	0	0	0	0	20	6	4	0	3	0	33	33
20	Orissa	595	40	67	702	1819	463	234	21	191	0	2728	3430
21	Punjab	121	105	0	226	237	220	137	20	14	0	628	854
22	Rajasthan	809	195	338	1342	1533	571	675	93	591	0	3463	4805
23	Sikkim	0	0	0	0	31	9	0	0	0	0	40	40
24	Tamil Nadu	425	206	452	1083	1513	460	487	13	93	1	2567	3650
25	Tripura	0	0	0	0	64	31	1	5	22	0	123	123
26	Telangana	373	78	154	605	860	550	665	5	27	0	2107	2712
27	Uttarakhand	28	12	0	40	78	8	4	1	129	0	220	260
28	Uttar Pradesh	694	274	137	1105	1186	809	276	40	501	0	2812	3917
29	West Bengal	263	80	125	468	616	317	239	12	82	6	1266	1740
Total (A)		6781	1591	1977	10349	19507	8285	5903	344	3226	6	37271	47620
B.	UNION TERRITORIES												
1	Andaman & Nicobar	0	0	0	0	46	13	0	1	0	0	60	60
2	Chandigarh	0	0	0	0	9	17	14	2	0	15	57	57
3	Dadra & Nagar Haveli	0	0	0	0	14	1	0	0	0	0	15	15
4	Delhi	0	0	0	0	149	64	196	13	0	380	802	802
5	Daman & Diu	0	0	0	0	0	0	7	0	0	0	7	7
6	Pondicherry	0	0	0	0	30	20	8	5	0	14	77	77
Total (B)		0	0	0	0	248	115	225	21	0	409	1018	1018
GRAND TOTAL (A+B)		6781	1591	1977	10349	19755	8400	6128	365	6	3635	38289	48638

3.4.2 Pumping Test of Exploratory Well

After construction, a well's specific capacity is increased to achieve optimum well life and stop sand from pouring into the well. In order to develop effective well design, yield capability evaluation, and well spacing criteria, pumping experiments are then carried out to evaluate aquifer properties, such as transmissivity and storage coefficient, as well as well parameters, such as specific capacity and well efficiency. Total of 106 tubewells were tested for their aquifer parameters during the year 2024-25 (**Table 3.13**).

Table 3.13: Status of Pumping Test of Wells by CGWB in 2024-25

DIVISION	STATE	No. of wells tested for aquifer parameters	Balance No. of wells to be tested (Backlog)
I. Ahmedabad	Gujarat	0	**44**
II. Ambala	Haryana	2	**5**
	Punjab	2	**7**
	Delhi	0	**0**
	Chandigarh	0	**0**
III. Varanasi	Uttar Pradesh	7	**104**
IV. Chennai	Tamil Nadu	13	**15**
	Kerala	0	**0**
V. Ranchi	Bihar	2	**19**
	Jharkhand	6	**45**
VI. Nagpur	Maharashtra	9	**6**
VII. Guwahati	Assam	10	**11**
	Meghalaya	0	**2**
	Tripura	0	**2**
	Arunachal Pradesh	1	**1**
	Nagaland	0	**1**
VIII. Jammu	Jammu & Kashmir	0	**9**
IX. Hyderabad	Andhra Pradesh	6	**0**
	Telangana	7	**6**
X. Bhubaneshwar	Odisha	3	**14**
XI. Jodhpur	Rajasthan	21	**30**
XII. Bhopal	Madhya Pradesh	3	**12**
XIII. Raipur	Chhattisgarh	5	**24**
XIV. Bengaluru	Karnataka	2	**6**
	Kerala	0	**2**
XV. Kolkata	West Bengal	6	**0**
XVI. Bareilly	Uttar Pradesh	1	**31**
	Uttarakhand	0	**5**

DIVISION	STATE	No. of wells tested for aquifer parameters	Balance No. of wells to be tested (Backlog)
XVII. Dharamshala	Himachal Pradesh	0	**0**
TOTAL		**106**	**401**

3.4.3 High Yielding Wells

Ground Water Exploration is carried to study the sub-surface hydrogeological set up of the aquifer to evaluate aquifer parameters. During 2024-25, CGWB under its scientific exploratory drilling programme has explored high yielding aquifers in various parts of the country based on hydrogeological studies coupled with remote sensing and geophysical techniques. Such studies will help in identifying similar ground water sources in other parts of the state having similar hydrogeological conditions and in guiding the state agencies to adopt the follow up action with regard to ground water development for drinking water supply and meeting other demands. During the year 2024-25, the Board has constructed 133 high yielding wells. High Yielding Wells explored during 2024-25 are indicated in **Table 3.14** below.

Table 3.14: Region wise details of high yielding wells constructed during 2024-25

Region-wise High Yielding Wells (AAP 2024-25)						
S. no	**Region**	**Name of Village / location of High discharge well**	**District**	**Lat**	**Long**	**Discharge in (LPS)**
1	**NCR Bhopal**	PALIYA	INDORE	22.829	75.792	18.97
2		Bharkachha Kalan	Raisen	22.821	78.060	7.1
3		MANAVTA NAGAR	INDORE	22.724	75.928	5.538
4		CITY FOREST (BICHOLI HAPSI)	INDORE	22.707	75.949	7.1
5		Badwar	Umariya	23.632	80.733	7.8
6		Bijauri	Katni	23.688	80.641	5.5
7		Bijauri	Katni	23.688	80.641	3.3
8		Dadraudi	Umariya	23.627	80.867	3.1
9		Raksha	Umariya	23.746	81.171	21.5
10		Raksha	Umariya	23.745	81.171	4.3
11		Dodka	Umariya	23.672	81.143	6.8
12		Naungawan Khas	Umariya	23.862	81.147	27
13		Funga	Annupur	23.181	81.810	6.7
14		Sobhapur	Narmadapuram	22.777	78.261	6.3
15		Banapur	Narmadapuram	22.468	77.481	5.1
16		Dumar	Narmadapuram	22.945	78.545	7.1

17		Barwara	Narsinghpur	22.941	79.274	5.1
18		Paduwa	Jabalpur	23.161	79.844	7.1
19		Archha	Jabalpur	23.192	79.833	5.7
20		NAWASA	DHAR	22.660	75.348	3.7
21		JATWAN	Jabalpur	23.256	79.941	7.1
22		NUNJA	Jabalpur	23.421	80.173	3.4
23		KHEDA	DHAR	23.034	75.210	19.3
24		Nawali	Bhind	26.432	78.393	3.3
25		Chidi	Bhind	26.213	78.970	12.2
26		Parkhi	Narmadapuram	22.799	78.451	4.5
27		Kapuri	Narmadapuram	22.849	78.527	8.3
28		Guran	Indore	22.930	75.886	3.4
29		Daudkhedi	Indore	23.145	75.762	4.3
30		Balgarh	Dewas	22.944	76.049	7.8
31		Kumedi	Indore	22.783	75.862	19.5
32		Parwaliya (Shallow Pz)	Jaora	23.737	75.115	4.8
33		Parwaliya (Deep Pz)	Jaora	23.737	75.115	4
34		Pipliya Jodha	Jaora	23.811	75.120	10
35		Ayana	Jaora	23.621	74.995	7
36	**NCCR Raipur**	Bhoyna	Dhamtari	20.664	81.588	28.16
37		Bhoyna	Dhamtari	20.664	81.588	18.39
38		Sankara	Dhamtari	20.755	81.622	15.17
39		Sankara	Dhamtari	20.755	81.622	12.5
40		Kasawahi	Dhamtari	20.627	81.491	10.066
41		Balodgahan	Gurur	20.648	81.466	14.3
42		Balodgahan	Gurur	20.648	81.466	14.3
43		Indira nagar stadium	Korba	22.359	82.703	15
44		Dhaneli EW	Balod	20.745	81.409	7
45	**NR Lucknow**	Chakelwa	Sonbhadra	23.995	83.158	8
46		Chouna	Sonbhadra	24.007	83.170	18.5
47		Forest Colony	Sonbhadra	24.125	83.048	18.5
48	**Andhra Pradesh**	Kadiri	Sri Sathya Sai	14.127	78.154	11.72
49		Kodihalli	Sri Sathya Sai	13.801	77.547	2.43
50		Pulamathi	Sri Sathya Sai	13.785	77.532	3.16
51		Pulamathi	Sri Sathya Sai	13.785	77.532	1.74
52		Bevinahalli	Sri Sathya Sai	13.772	77.467	13.77
53		Bevinahalli	Sri Sathya Sai	13.772	77.467	9.86
54		Lepakshi	Sri Sathya Sai	13.805	77.605	0.78
55		Ramachandrapuram	Sri Sathya Sai	17.436	78.167	13.77
56		Ramachandrapuram	Sri Sathya Sai	17.436	78.167	10.76
57		Kanchi Samudram	Sri Sathya Sai	17.436	78.167	3.74

58		Gollapuram	Sri Sathya Sai	13.725	77.525	13.77
59		Kanchi Samudram	Sri Sathya Sai	13.776	77.608	2.83
60		Gollapuram	Sri Sathya Sai	13.725	77.525	13.77
61		Sirivaram	Sri Sathya Sai	13.754	77.579	2.43
62		Thimmaganipalli	Sri Sathya Sai	13.857	77.585	1.74
63		Pulakunta	Sri Sathya Sai	17.280	78.418	2.43
64		Maluguru	Sri Sathya Sai	17.433	78.238	2.43
65		Kotipi	Sri Sathya Sai	13.773	77.514	1.18
66		Kondur	Sri Sathya Sai	13.844	77.608	2.83
67	**Telangana**	Shivarampally-2	Rangareddy	17.328	78.433	6.7
68		Nagireddiguda-2	Rangareddy	17.298	78.328	4.25
69		Tukkuguda-1	Rangareddy	17.249	78.475	0.78
70		Tukkuguda-2	Rangareddy	17.248	78.475	1.74
71		Changomul	Vikarabad	17.257	78.025	0.78
72		Ramachandrapuram	Sangareddy	17.512	78.288	8.19
73		Ramachandrapuram	Sangareddy	17.512	78.288	4.25
74		Aloor	Rangareddy	17.323	78.064	2.83
75		Aloor	Rangareddy	17.323	78.064	0.78
76		K.V.Kanchanbagh	Hyderabad	17.340	78.503	1.44
77		Tellapur-1	Sangareddy	17.478	78.286	1.44
78		Tellapur-2	Sangareddy	17.478	78.286	0.78
79		Orella	Rangareddy	17.355	78.133	2.07
80		Orella	Rangareddy	17.355	78.133	1.18
81		Sahebnagar	Rangareddy	17.322	78.576	2.07
82		Sahebnagar	Rangareddy	17.322	78.576	2.43
83		L.B.Nagar-1	Rangareddy	17.340	78.556	5.39
84		Almasguda-2	Rangareddy	17.310	78.540	4.25
85		Bhulkapur	Rangareddy	17.436	78.167	4.25
86		Bhulkapur	Rangareddy	17.436	78.167	3.16
87		Bhulkapur	Rangareddy	17.436	78.167	1.74
88		Thorrur-1	Rangareddy	17.436	78.167	9.86
89		Adibatla-1	Rangareddy	17.230	78.550	2.43
90		Kothagadi	Vikarabad	17.369	77.903	3.16
91		Peerzadiguda -1	Rangareddy	17.369	77.903	1.18
92		Sathamrai-1	Rangareddy	17.280	78.418	1.18
93		Kothagadi	Vikarabad	17.369	77.903	5.39
94		Sathamrai-2	Rangareddy	17.369	77.903	2.43
95		Devula Thanda-1	Sangareddy	17.433	78.238	0.78
96		Devula Thanda-2	Sangareddy	17.433	78.238	6.7
97		Burhampalle	Vikarabad	17.332	77.952	1.44
98		Yerravalle	Vikarabad	17.409	77.880	3.74

99		Yerravalle	Vikarabad	17.409	77.880	3.74
100	**SWR Bangalore**	GSI Office premises, Kumaraswamy Layout, Bengaluru-EW	Bengaluru Urban	12.911	77.568	3.75
101		K.V Panamburu-EW	Dakshin Kannada	12.945	74.821	13.77
102		K.V Panamburu-OW	Dakshin Kannada	12.945	74.821	24
103		PDO office, Udigala	Chamrajanagra	11.876	76.839	6.93
104		Govt. High School, Heggadahalli	Nanjangud	12.106	76.604	4.26
105		Old WRD Office Premises, Tivim VP	North Goa	15.619	73.853	6.1
106		Permude-EW (Billawa Samaj Land)	Dakshin Kannada	13.000	74.877	9.8
107		Permude-OW (Billawa Samaj Land)	Dakshin Kannada	13.000	74.877	18
108		Kuttetoor-EW (Kengali Over head tank land)	Dakshin Kannada	13.004	74.846	9.8
109		Kuttetoor-OW (Kengali Over head tank land)	Dakshin Kannada	13.004	74.846	11.72
110	**ER Kolkata**	Bankata Junior Basic School	Bankura	23.092	86.905	12.00
111		Teghori Junior High School	Bankura	23.230	86.891	6.90
112		Khoyrasole PHED Campus	Birbhum	23.793	87.145	7.10
113		Subarnapur PHED Campus	Nadia	22.961	88.572	11.90
114	**KR Trivanthpuram**	Kizhakkambalam/Govt. LP School	Ernakulam	10.035	76.409	18 (Saline)
115		Kalamassery/CUSAT Campus-Marine Sciences Department	Ernakulam	10.041	76.330	8.00
116	**CR, Nagpur**	Umari (B.)_EW	Nagpur	21.420	78.813	17.92
117		Umari (B.)_OW	Nagpur	21.420	78.813	12.18
118		Manjari_EW	Solapur	17.506	75.248	17.92
119		Manjari_OW	Solapur	17.506	75.248	17.92
120		Mendashingi_EW	Solapur	17.399	75.255	12.18
121		Mendashingi_OW	Solapur	17.399	75.255	3.17
122		Shindivihir_EW	Wardha	21.039	78.555	7.73
123		Belgaon_EW	Wardha	21.232	78.337	7.73

124		Belgaon_OW	Wardha	21.232	78.337	5.94
125		Film City_EW	Mumbai Sub Urban	19.161	72.886	12.19
126		Film City_OW	Mumbai Sub Urban	19.161	72.886	9.86
127		Selgaon_EW	Wardha	21.195	78.457	3.17
128		Waki Gherdi_EW	Solapur	17.341	75.280	9.84
129		Waki Gherdi_OW	Solapur	17.341	75.280	7.84
130		Raiwadi_Pz	Nagpur	21.532	78.942	4.43
131		Pardi_Pz	Wardha	21.182	78.299	4.42
132		Vinchur_Pz	Nashik	20.106	74.230	7.76
133		Dongargaon_Pz	Bhandara	78.813	73.544	4.43
134		Dhanoli_Pz	Wardha	21.059	78.605	5.94
135	**NER, Guwahati**	UDALI BLOCK OFFICE	NAGAON	25.910	92.990	3
136		JUGIJAN BLOCK OFFICE	NAGAON	26.010	92.860	3.3
137		TARUN FIELD	MORIGAON	26.250	92.340	5
138		MAYONG HR. SEC SCHOOL	MORIGAON	26.260	92.040	6.15
139		Silphukuri Cooperative Rice Godown	MORIGAON	26.330	92.540	4.04
140		Bhurbandha Aancholik Navjyoti Sangha playground	MORIGAON	26.310	92.360	5
141		Nongmensong EW	East Khasi Hills	25.580	91.910	6.49
142		Nongmensong OW	East Khasi Hills	25.580	91.910	14.74
143		Panbazar	Kamrup Metro	26.190	91.750	4
144	**WR, Jaipur**	Ramsisar Sika block Fatehpur	Sikar			10
145		Jamalsar, block Bikaner	Bikaner			6
146		Rosava, block Fathepur	Sikar			10
147		Govindpura, block Fathepur	Sikar			6
148		Shekhisar, block Fathepur	Sikar			11

3.5 Ground Water Resource Assessment

3.5.1 Introduction

The Dynamic ground water resources (as in 2024) of the entire country have been assessed jointly by CGWB and State Ground Water Departments under the supervision of the State level Committees. The dynamic ground water resources are also known as Annual Ground Water Recharge, since it gets recharged every year from rainfall and other sources (secondary sources) such as applied irrigation water, surface water bodies, water conservation structures, etc.

As a part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction are assessed for each assessment unit (block/taluka/mandal/tehsils/firka etc.). The 'Stage of Ground Water Extraction' is then computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe (<= 70 %), Semi-Critical (>70 % and<=90 %), Critical (>90 % and <=100%) and Over-Exploited (>100 %).

A Central Level Expert Group (CLEG) was constituted by the Department of Water Resources, RD & GR, Ministry of Jal Shakti for over-all supervision of the re assessment of ground water resources in the entire country with the reference year 2022. The groundwater resources assessment at the State/UT Levels was carried out jointly by Central Ground Water Board and State Nodal/Ground Water Departments under the guidance of respective State Level Committees (SLC) with over-all supervision of CLEG. The Report on "National Compilation of Dynamic Ground Water Resources, 2024" was released by Honourable Minister of Jal Shakti on 31stDecember, 2024. Ground Water Resource Assessment - 2024 (GWRA-2024) has been carried out through a software/web-based application "India-Groundwater Resource Estimation System (IN-GRES)" developed by CGWB through IIT-Hyderabad (http://ingres.iith.ac.in). The pictorial categorization of the assessment units of the entire country is shown in **Figure 3.3** and its corresponding details are shown in **Table 3.15** below.

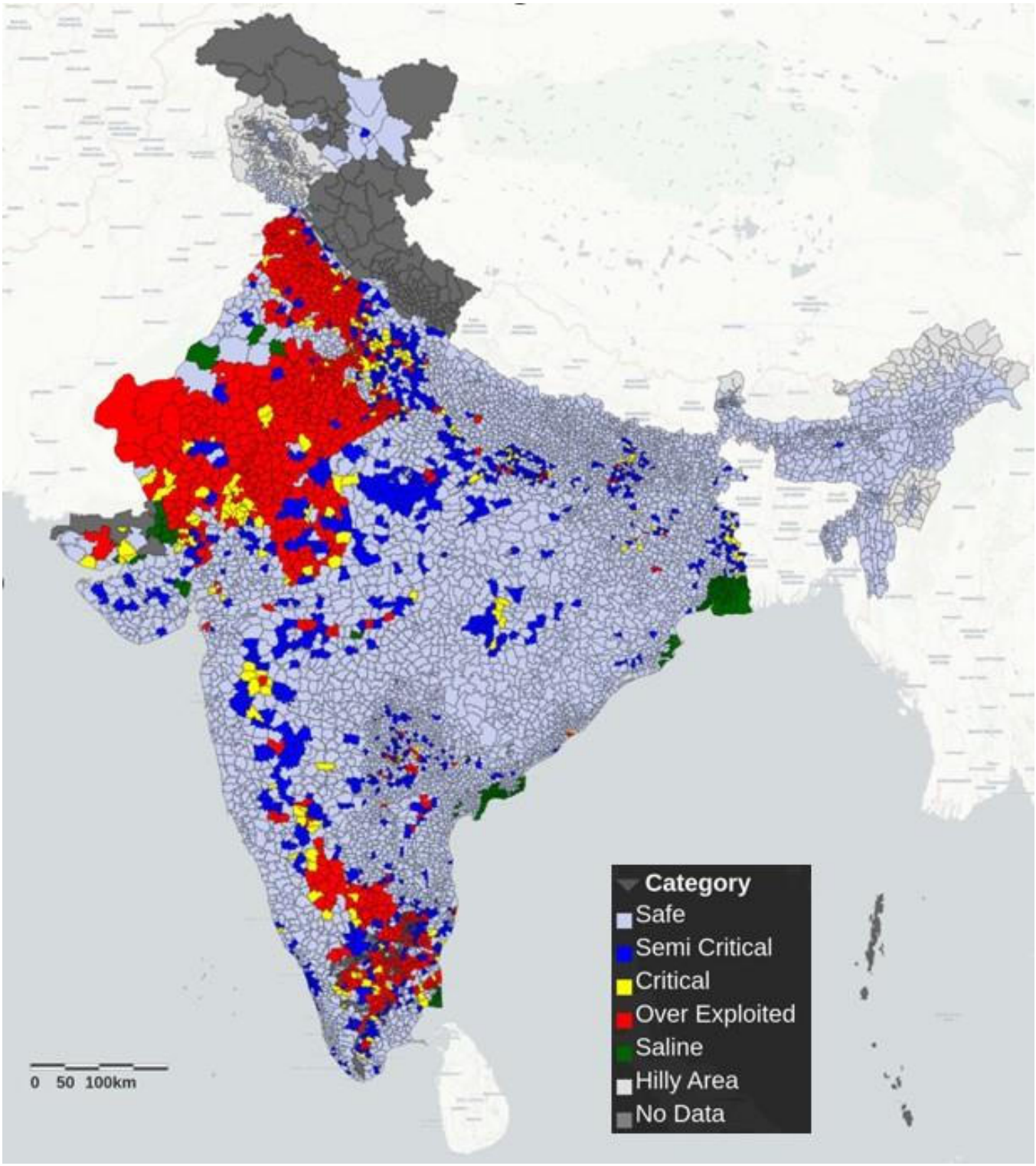


Fig. 3.3: Categorization of Assessment Units

Table 3.15: State/UT-wise Categorization of Assessment Units, as per GWRA- 2024

CATEGORIZATION OF BLOCKS/ MANDALS/ TALUKAS IN INDIA (2024)												
S.No.	State/Union Territories	Total No. of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
	States		Nos.	%	Nos.	%	Nos.	%	Nos.	%	Nos.	%
1	Andhra Pradesh	679	591	87.0	38	5.6	2	0.3	9	1.3	39	5.74
2	Arunachal Pradesh	42	42	100								
3	Assam	245	244	99.59	1	0.41						
4	Bihar	535	473	88.41	49	9.16	9	1.68	4	0.75		
5	Chhattisgarh	146	120	82.19	21	14.38	5	3.42				
6	Goa	12	12	100								
7	Gujarat	252	184	73.02	24	9.52	10	3.97	22	8.73	12	4.76
8	Haryana	143	36	25.17	8	5.59	11	7.69	88	61.54		
9	Himachal Pradesh	10	10	100								
10	Jharkhand	263	240	91.25	12	4.56	6	2.28	5	1.9		
11	Karnataka	237	144	60.76	33	13.92	15	6.33	45	18.99		
12	Kerala	152	120	78.95	29	19.08	3	1.97				
13	Madhya Pradesh	317	225	70.98	61	19.24	5	1.58	26	8.2		
14	Maharashtra	359	302	84.12	41	11.42	7	1.95	8	2.23	1	0.28
15	Manipur	9	9	100								
16	Meghalaya	40	40	100								
17	Mizoram	26	26	100								
18	Nagaland	52	52	100								
19	Odisha	314	299	95.22	9	2.87					6	1.91
20	Punjab	153	22	14.38	12	7.84	4	2.61	115	75.16		
21	Rajasthan	302	37	12.25	21	6.95	27	8.94	214	70.86	3	0.99
22	Sikkim	38	38	100								
23	Tamil Nadu	313	127	40.58	55	17.57	20	6.39	106	33.87	5	1.60
24	Telangana	620	490	79.03	85	13.71	13	2.10	32	5.16		
25	Tripura	59	59	100								
26	Uttar Pradesh	836	566	67.7	165	19.74	46	5.5	59	7.06		
27	Uttarakhand	20	16	80	4	20						
28	West Bengal	345	239	69.28	36	10.43	10	2.9			60	17.39
29	Andaman And Nicobar	9	9	100								
30	Chandigarh	1	1	100								
31	Dadra and Nagar Haveli and Daman and Diu	3							3	100		

CATEGORIZATION OF BLOCKS/ MANDALS/ TALUKAS IN INDIA (2024)												
S.No.	**State/Union Territories**	**Total No. of Assessed Units**	**Safe**		**Semi-Critical**		**Critical**		**Over-Exploited**		**Saline**	
32	**Delhi**	34	5	14.71	2	5.88	13	38.24	14	41.18		
33	**Jammu And Kashmir**	149	149	100								
34	**Ladakh**	18	17	94.44	1	5.56						
35	**Lakshadweep**	5	4	80	1	20						
36	**Puducherry**	8	3	37.50	3	37.50			1	12.50	1	12.50
	Grand Total	**6746**	**4951**	**73.39**	**711**	**10.54**	**206**	**3.05**	**751**	**11.13**	**127**	**1.88**
Note-												
Blocks- Arunachal Pradesh, Assam, Bihar, Chhattisgarh, Haryana, Jharkhand, Kerala, Madhya Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Rajasthan, Sikkim, Tripura, Uttar Pradesh, Uttarakhand, West Bengal, Andaman & Nicobar Island, Jammu & Kashmir, Ladakh, Lakshadweep												
Taluks- Goa, Gujarat, Karnataka, Maharashtra, Tamil Nadu, Puducherry												
Mandals- Andhra Pradesh, Telangana												
District- Dadra & Nagar Haveli, Daman & Diu												
Valley- Himachal Pradesh												
UT- Chandigarh												
Tehsil- Delhi												

3.6 National Aquifer Mapping and Management Programme (NAQUIM)

3.6.1 Introduction

Central Ground Water Board (CGWB) is implementing National Aquifer Mapping and Management program (NAQUIM), which envisages mapping of aquifers (water bearing formations), their characterization and development of Aquifer Management Plans to facilitate sustainable management of ground water resources. NAQUIM was initiated in 2012 as a part of the GWMR plan scheme with the objectives to delineate and characterize the aquifers and develop plans for ground water management. Out of 33 lakh km^2 of the entire country, entire mappable area of 25 lakh km^2 has been covered under this programme. NAQUIM outputs are shared with various stakeholders including the District Authorities. Public Interaction Programmes (PIP) are also being conducted in various parts of the country to promote outreach on groundwater management.

3.6.2 Objectives

The National Aquifer Mapping and Management Programme (NAQUIM) was initiated and implemented by Central Ground Water Board under the scheme of Ground Water Management & Regulation during XII plan (2012-17). The Programme was designed as a follow-up of the recommendations of four Working Groups of the erstwhile Planning Commission namely

"Sustainable Ground Water Development and Management", "Minor Irrigation and Watershed Development", "Urban and Industrial Water Supply and Sanitation "and "Data Base Development and Management".

With an aim to support effective management of ground water resources in the country, the NAQUIM programme was taken up with the following objectives:

- Delineation of aquifers in terms of their horizontal and vertical extent.
- Characterization of aquifers in terms of their hydrogeological properties, resource availability and quality
- Preparation of thematic aquifer maps
- Developing management strategies to ensure sustainability of ground water resources.

3.6.3 NAQUIM 2.0

NAQUIM 2.0 was initiated in 2023 with broad objectives of:

i) providing information in higher granularity with a focus on increasing density of dynamic data like ground water level, ground water quality etc.
ii) providing issue based scientific inputs for ground water management upto Panchayat level,
iii) providing printed maps to the users and
iv) putting in place a strategy to ensure implementation of the recommended strategies.

The NAQUIM 2.0 studies are designed to focus on different priority areas. In the year 2024-25, a total of 39 studies were conducted under NAQUIM 2.0 an area of 23724 sq. km spanning across 30 States/UTs as summarized in **Table 3.16**.

Table 3.16: Priority area wise NAQUIM 2.0 studies taken up in AAP 2023-24

S.No.	State	Regions	No of Studies	Priority Area type	District/Block
1	Maharashtra	**CR**	**4**	OCS	Sangli (Jath block)
2	Maharashtra	CR		OCS	Nagpur (Katol & Saoner Block)
3	Maharashtra	CR		Poor WQ	Chandrapur (Warora block)
4	Maharashtra	CR		Urban Agglomerate	Mumbai
5	SUO DELHI	**Delhi**	**2**	Poor WQ	Okhla landfill
6	SUO DELHI	Delhi		Poor WQ	Bhalsawa
7	Andaman & Nicobar UT	**ER**	**4**	Urban Agglomerate	South Andaman
8	Sikkim	ER		Springshed	Gangtok
9	West Bengal	ER		Industrial Areas	Purba Medinipur

10	West Bengal	ER		Poor Availability	Purulia
11	Kerala	**KR**	**5**	Industrial Areas	Palakkad
12	Kerala	KR		OCS	Kasargod
13	Kerala	KR		Springshed	Idukki
14	Kerala	KR		Urban Agglomerate	Trivandrum
15	Lakshadweep UT	KR		Poor WQ	Minicoy and Amni
16	Bihar	**MER**	**4**	Poor Availability	Study 1. Mohanpur and adjacent block of Gaya District.
17	Bihar	MER		Poor Availability	Study 2.Meskaur and adjacent block of Nawada District
18	Bihar	MER		Poor WQ	Study 3.Madanpur and adjacent blocks of Aurangabad district
19	Jharkhand	MER		OCS	Koderma, Domchanch and Jainagar Blocks of Koderma district
20	Chhattisgarh	**NCCR**	**5**	Industrial Areas	Korba
21	Chhattisgarh	NCCR		OCS	Balod
22	Chhattisgarh	NCCR		OCS	Dhamtari
23	Chhattisgarh	NCCR		OCS	Durg
24	Chhattisgarh	NCCR		Urban Agglomerate	Raipur urban
25	Madhya Pradesh	**NCR**	**4**	Industrial Areas	Ujjain
26	Madhya Pradesh	NCR		OCS	Ratlam (Alot block)
27	Madhya Pradesh	NCR		Springshed	Dindori, Anuppur
28	Madhya Pradesh	NCR		Urban Agglomerate	Bhopal
29	Arunachal Pradesh	**NER**	**6**	Urban Agglomerate	Itanagar Capital Complex, Papum Pare
30	Assam	NER		Industrial Areas	Bamunimaidan and North Guwahati
31	Assam	NER		Poor Availability	Guwahati City, Kamrup Metro,
32	Assam	NER		Springshed	Cachar and Hailakandi Districts
33	Meghalaya	NER		Urban Agglomerate	Greater Shillong, East Khasi
34	Tripura	NER		Springshed	Dhalai
35	HP	**NHR**	**1**	Springshed	Kangra
36	Uttar Pradesh	**NR**	**4**	Industrial Areas	Unnao

37	Uttar Pradesh	NR		OCS	Baghpat
38	Uttar Pradesh	NR		Poor WQ	Siddharth Nagar
39	Uttar Pradesh	NR		Urban Agglomerate	Kanpur Nagar
40	Jammu & Kashmir UT	**NWHR**	**4**	Springshed	Ramban
41	Jammu & Kashmir UT	NWHR		Springshed	Banihal
42	Jammu & Kashmir UT	NWHR		Urban Agglomerate	Srinagar Urban Area
43	Laddakh UT	NWHR		Springshed	Kargil District
44	Haryana	**NWR**	**4**	Industrial Areas	Gurugram
45	Haryana	NWR		OCS	Samalkha and Bapoli Blocks of Panipat District
46	Punjab	NWR		Deeper Aquifer	Sangrur and Bhiwanigarh Blocks of Sangrur District
47	Punjab	NWR		Poor WQ	Ludhiana City
48	Odisha	**SER**	**5**	Coastal	Balasore Sadar block including Chandipur
49	Odisha	SER		Industrial Areas	Part of Jharsuguda and Sambalpur District
50	Odisha	SER		Mining Area	Sukinda
51	Odisha	SER		OCS	Khordha and Nayagarh
52	Odisha	SER		Urban Agglomerate	Sambalpur
53	Andhra Pradesh	**SR**	**9**	Urban Agglomerate	Vishakhapatnam, Anakkapalli
54	Andhra Pradesh	SR		Urban Agglomerate	Guntur, Vijayavada, Amaravati
55	Tamil Nadu	SR		Mining Area	Ariyalur
56	Tamil Nadu	SR		OCS	Thiruppur
57	Tamil Nadu	SR		OCS	Theni
58	Tamil Nadu	SR		Poor WQ	Salem
59	Telangana	SR		Poor Availability	Vikarabad,Rangareddy
60	Telangana	SR		Urban Agglomerate	Hyderabad, Parts of Rangareddy, Medchal
61	Telangana	SR		Urban Agglomerate	Rangareddy (Parts)
62	Goa	**SWR**	**5**	Poor Availability	South Goa
63	Karnataka	SWR		OCS	Bengaluru Rural
64	Karnataka	SWR		Poor WQ	Dharwad
65	Karnataka	SWR		OCS	Ramnagar
66	Karnataka	SWR		Urban Agglomerate	Bengaluru city

67	Uttarakhand	**UR**	**1**	Autoflow	Udhamsingh Nagar (Jaspur, Gadarpur, Bazpur and Kashipur Block)
68	Gujarat	**WCR**	**5**	Coastal	Navsari
69	Gujarat	WCR		OCS	Banskantha
70	Gujarat	WCR		Poor WQ	Vadodara
71	Gujarat	WCR		Urban Agglomerate	Ahmedabad &Daskaroi
72	Gujarat	WCR		Urban Agglomerate	Gandhinagar
73	Rajasthan	**WR**	**4**	Industrial Areas	Pali
74	Rajasthan	WR		Industrial Areas	Neemrana
75	Rajasthan	WR		Urban Agglomerate	Jaipur
76	Rajasthan	WR		Urban Agglomerate	Jodhpur
		Total	**76**		

Out of these 76 studies taken-up, 69 studies were successfully completed during the year 23-24, while remaining 8 studies were continued in 2024-25. A total 39 studies were taken up in AAP 2024-25, including 8 continuing studies of AAP 2023-24. The completed 39 studies include 8 priority areas covering 23724 sq. km. across the country. The details of priority area and State wise NAQUIM 2.0 studies completed are given in **Table 3.17** & **3.18**.

Table 3.17: Priority area wise NAQUIM 2.0 studies completed during 2024-25

Priority Area	No. of Primary Theme	Regions taken up the Studies
Auto-flow areas	1	UR
Coastal areas	1	SWR
Industrial areas	5	NWHR, SER, SWR, UR, WCR
Mining areas	1	SER
Water Stressed/OCS areas	12	CR, ER, KR, MER, NCCR, NR, SECR, SUO Delhi, SWR
Poor WQ areas	12	CR, ER, KR, MER, NCCR, NR, SECR, SUO Delhi, SWR
Springshed areas	3	KR, NER
Urban Agglomerate areas	4	CR, ER, SR, SWR
Total	**39**	

Table 3.18: State wise NAQUIM 2.0 studies completed during AAP 2024-25

S.No.	Region	State	Area (sq km)	District (s)	Title of the study / Priority Type
1	CR	Maharashtra	600	Mumbai and Mumbai Sub Urban **	Urban Agglomerates
2	CR	Maharashtra	756	Parts of Mohadi and tumsar blocks, Bhandara districts	Poor GW quality areas
3	CR	Maharashtra	1772	Sangola block, Solapur	Water stressed areas/OCS
4	CR	Maharashtra	713	Karanja block, Wardha	Water stressed areas/OCS
5	ER	Andaman & Nicobar UT	15	South Andaman	Water stressed areas/OCS
6	ER	Sikkim	10	Namchi	Urban Agglomerates
7	ER	West Bengal	459	Nadia	Poor GW quality areas
8	KR	Kerala	258	Kasargod**	Water stressed areas/OCS
9	KR	Kerala	550	Idukki**	Springshed areas
10	KR	Kerala	892	Alappuzha	Poor GW quality areas
11	MER	Bihar	710	Nalanda	Water stressed areas/OCS
12	MER	Jharkhand	644	(Lesliganj, Satbarwa, Pandwa, Medininagar blocks) – Palamu District	Poor GW quality areas
13	NCCR	Chhattisgarh	456	Devbhog, Gariyaband	Poor GW quality areas
14	NCCR	Chhattisgarh	427	Kunkuri, Jashpur	Poor GW quality areas
15	NCCR	Chhattisgarh	720	Jagdalpur, Bastar	Poor GW Quality areas
16	NCR	Madhya Pradesh	400	Ratlam	Water stressed areas
17	NER	Nagaland	896	Kohima	Springshed
18	NER	Tripura	900	Dhalai**	Springshed
19	NR	Uttar Pradesh	736	Sonbhadra district*	Poor GW quality areas
20	NWHR	Jammu & Kashmir	200	Kathua: Kathua, Nagri &Barnoti Blocks	1.Industrial Cluster,
21	NWR	Haryana	663	Bhiwani	Water stressed areas/OCS
22	NWR	Haryana	235	Charkhi Dadri	Water stressed areas/OCS
23	SECR	Tamil Nadu	985	Salem &Namakkal districts	Water stressed areas/OCS
24	SECR	Tamil Nadu	890	Krishnagiri	Poor GW quality areas
25	SER	Odisha	180	Jajpur**	Mining areas
26	SER	Odisha	890	Angul, Dhenkanal	Industrial clusters
27	SR	Andhra Pradesh	373	Hindupur and Lepakshi	Water stressed areas
28	SR	Telangana	779	Parts of Hyderabad, Medchal Malkajgiri &	Urban Agglomerates

				Sangareddy**	
29	SR	Telangana	745	Parts Vikarabad & Rangareddy**	Water stressed areas
30	SUO	Delhi	100	South west, west, North east and North	Poor GW quality areas
31	SUO	Delhi NCR	165	Faridabad*	Sustainable drinking water supply for Faridabad City. (collaboration with FMDA)
32	SWR	Goa	478	North Goa	Coastal areas
33	SWR	Karnataka	630	Chikballapur	Poor GW quality areas (Tank filling scheme)
34	SWR	Karnataka	21	Dakshin Kannada	Industrial clusters
35	SWR	Karnataka	711	Bengaluru Urban**	Urban Agglomerate
36	UR	Uttarakhand	337	Haridwar	Industrial clusters
37	UR	Uttarakhand	2000	Udham Singh Nagar	Autoflow Zones
38	WCR	Gujarat	800	Ahmedabad	Industrial clusters
39	WCR	Gujarat	628	Narmada	Water stressed area
	Total	**39 Studies**	**23724**		

**8 continuing studies of AAP 2023-24
*Studies taken up in collaboration with other agencies

3.7 Ground Water Conservation and Artificial Recharge

3.7.1 Introduction

Groundwater conservation and Artificial recharge (AR) are critical components in the sustainable management of water resources. The practice of conserving groundwater involves implementing strategies to reduce water wastage, protect aquifers from contamination, and promote the efficient use of this resource. The integration of groundwater conservation and artificial recharge practices is essential for ensuring long-term water security. By adopting a holistic approach that combines policy measures, community engagement, and technological innovations, stakeholders can effectively manage groundwater resources. Furthermore, collaborative efforts among government agencies, local communities, and environmental organizations can lead to the development of sustainable practices that safeguard groundwater resources for future generations, ultimately contributing to the resilience of ecosystems and human societies alike.

3.7.2 Convergence with MGNERGS in Water Stressed areas

CGWB is providing technical guidance in site selection and design of artificial recharge structures and capacity building of MGNREGS officials in selected 09 Blocks of 08 States as a pilot study (**Table 3.19**). The work has been taken up under the MGNREGS fund. So far, the work has been completed in Andhra Pradesh, Telangana and Tamil Nadu (one block each) and is in progress in the rest of the blocks of the states. Glimpses of the AR structures constructed are given in **Figure 3.4** & **3.5**.

Table 3.19: Details of water stressed areas taken up under MGNREGS

S.No.	State	District	Block	Target (No. of Structures)	Status of Work (No. of structures constructed)
1	Andhra Pradesh	Prakasam	Peddaravedu	1253	Project Completed
2	Telangana	Nallagonda	Munugode	2174	Project Completed
3	Tamil Nadu	Krishnagiri	Mathur	20	Project Completed
4	Karnataka	Chickballalpur	Gowribidanur	103	36
5	Rajasthan	Bhilwara	Madalgarh	56	47
6	Rajasthan	Sirohi	Reodhar	126	81
7	Haryana	Kurukshetra	Babain	15	13
8	Madhya Pradesh	Ujjain	Badnagar	436	353
9	Maharashtra	Amravati	Warud	1798	422

Fig 3.4: The Pond constructed in Babain Block, Kurukshetra District, Haryana

Fig 3.5: Artificial Recharge Structures at Chickballalpur District, Karnataka

3.7.3 Implementation of Master Plan for Artificial Recharge 2020

- CGWB have prepared a conceptual document on "Master Plan for Artificial Recharge to Ground Water in India" in 2020 to work out the feasibility of various structures for the different terrain conditions of the country and estimated costs, the document provides a broad outline of the project and expected investments which include artificial recharge in both rural and urban areas. A total 14, 175, 144 structures at an estimated cost of Rs 1.33 lakh Cr have been envisaged for rainwater harvesting and artificial recharge in the country.
- One district in each state has been identified for implementation of Master Plan for Artificial Recharge-2020 on pilot basis. The implementation of Master Plan for Artificial recharge has been initiated in 33 States/UTs which are Andhra Pradesh, Assam, Arunachal Pradesh, Bihar, Chhattisgarh, Chandigarh, Delhi, Gujarat, Haryana, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Puducherry, Rajasthan, Sikkim, Tamil Nadu, Telangana, Tripura, Uttar Pradesh, Uttarakhand, West Bengal, UT of Andaman and Nicobar, Jammu & Kashmir, Ladakh. The photographs of the AR structures constructed are provided in **Figure 3.6** to **3.8**.
- As NAQUIM 1.0 has already been completed in March 2023, the recommendations made under NAQUIM 1.0 have to be incorporated in updation Master Plan for Artificial Recharge-2020. The said updation of Artificial Recharge-2020 Master Plan has been completed.

Fig. 3.6: Artificial Recharge Structures at South Andaman District, Andaman & Nicobar

Fig. 3.7: Artificial Recharge Structures at Kasargod District, Kerela

Fig. 3.8: Artificial Recharge Structures at Nuapada District, Odisha

3.7.4 Groundwater Augmentation through Artificial Recharge in water-stressed areas of Rajasthan (2021- continuing activity)

During 2021-22, CGWB has taken up artificial recharge for groundwater augmentation in certain water stressed areas of Jaisalmer, Jodhpur & Sikar districts of Rajasthan.

- **Phase-I:** Construction of Concrete Gravity Dam at Bastawa Mata (Block Balesar) and Zoned Earth Fill Dam with clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan with the objective of artificial recharge to groundwater aquifer systems has been taken up and completed the targets. The physical and financial details of the dams are tabulated in **Table 3.20**.
  - The construction of Zoned Earth Fill Dam with Clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan has been completed and inaugurated by the Hon'ble Minister of Jal Shakti on 16.03.2024. The photographs of the Zoned Earth Fill Dam with Clay Core at Indroka shown in **Figure 3.9**.
  - The concreting work of Concrete Gravity Dam at Bastawa Mata (Block Balesar) in Jodhpur district, Rajasthan has been completed and the ancillary works are in progress. The photographs of the Concrete Gravity Dam at Bastawa Mata are shown in **Figure 3.10**.

Table 3.20: Details of AR Structure to be constructed in Phase I

S.No.	Location Details	Type of Artificial Recharge Structure	Fund Allocated
1	Indroka, Mandore, Jodhpur	Zoned Earth Fill Dam with Clay Core	Rs. 15.21 Cr
2	Bastawa Mata, Balesar, Jodhpur	Concrete Gravity Dam	Rs. 19.42 Cr
		Total	Rs. 34.63 Cr

Fig. 3.9: Zoned Earth Fill Dam with Clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan

Fig. 3.10: Concrete Gravity Dam at Bastawa Mata (Block Balesar) in Jodhpur district, Rajasthan

- **Phase-II:** Under this phase, it is proposed to construct Stone Masonary Check Dams (MCD), Anicuts, Concrete Check Dams (CCD & Recharge shafts) at 101 sites in parts of Balesar, Shekhla, Lohawat, Tinwari and Phalodi blocks of Jodhpur district and Sankra block of Jaisalmer and Shrimadhopur block of Sikar district of Rajasthan. The cost of the project is Rs 74.32 Cr.

- Total 83 Water Harvesting Structures (WHS) have been constructed, out of which 74 WHS have been handed over to WRD, Govt. of Rajasthan and the construction of remaining 18 WHS has been foreclosed due to local problems. The details of the total WHS structures constructed so far are provided in **Table 3.21** and glimpse of the WHS constructed under Phase II are shown in **Figure 3.11**.

Table 3.21: Details of the structures constructed under Phase II

Description	Packages								Total
	I		II				III	IV	
	Jodhpur		Jodhpur				Jaisalmer	Sikar	
	Balesar	Sekhala	Tinwari	Sekhala	Phalodi	Lohawat	Sankra	Srimadhopur	
Proposed Structures	56	1	6	9	1	17	4	7	**101**
Work Completed	42	1	6	9	0	16	2	7	**83**

Check Dam, Bada Kotecha, block Tinwari, district Jodhpur

Check Dam, Village Bari, block Tinwari, district Jodhpur

Check Dam, Bhalu Laxmangarh, Block Shekhala, district Jodhpur

Recharge Shafts, Mundru, Block Shrimadhopur, district Sikar

Fig. 3.11: Artificial Recharge Structures constructed under Phase II

- **Phase III (Extended PhaseII):** In order to improve the groundwater scenario in Rajasthan as implemented in Phase II and also to observe the concentrated improvements in ground water condition, the Phase III (Extended Phase-II) has been extended in Jodhpur, Jaisalmer, Sikar, Jhunjhunu and Alwar districts of Rajasthan. Further, the cumulative

recharge impact may be seen at these places. It is proposed to construct Masonry Check Dams (MCD), Anicuts, Concrete Check Dams (CCD), Model Talab etc. at 53 sites. The estimated cost of the project is 61.82 Cr.

- A total of 49 WHS has been constructed and remaining 04 WHS is under progress. The details of the total WHS structures constructed so far are provided in **Table 3.22** and glimpse of the WHS constructed under Phase III are provided in **Figure 3.12**.

Table 3.22: Details of AR structures constructed under Phase II (extended)

Description	Jodhpur District	Jaisalmer District	Alwar District	Jhunjhunu District	Barmer District	TOTAL
Proposed Structures	22	24	1	2	4	53
Work Completed	33	7	1	1	7	49

Check Dam, Dhelana, district Jodhpur

Check Dam Arna Jharna, Jodhpur

Fig. 3.12: Water Harvesting Structures constructed under Phase III

3.7.5 AR structures in Kala Amb valley, Sirmaur, Himachal Pradesh

Kala Amb Valley in Sirmaur district of Himachal Pradesh falls under ground water dependent area and the ground water is mainly utilized for drinking and industrial development. It has been observed during the past decade that the large withdrawal of ground water in the industrial sector has resulted in depletion of groundwater level in the area. It has also been observed that during the monsoons, runoff quickly flows out of the area with less natural recharge to ground water due to sloping areas. As surplus runoff volume in streams and rivulets are available, appropriate structures across these streams at selective locations may facilitate the harvesting of water which in turn enhances the ground water recharge in downstream areas.

- CGWB has proposed to construct 07 Check Dam and 12 Recharge well in Kala Amb Valley, Sirmaur district, H.P to the Ministry. The proposal was approved by the Ministry and the same was conveyed vide letter no: T-15014/4/2021-GW section,

dated 22.11.2023 with the estimated cost of Rs.19,92,73,968. The Ministry was also conveyed that the work shall be executed directly under the supervision of RD/EE office of CGWB in the lines of NAQUIM project.

- A meeting was convened on 28th, June, 2024 with Dr. Ashwani Kumar, IAS, Special Secretary, Jal Shakti Vibhag (JSV), Government of Himachal Pradesh for smooth execution of the project, proposing a bipartite MOU between CGWB and JSV, HP and also obtaining NOC for the proposed AR structures. Meanwhile, tender document was prepared and submitted to the ministry. To execute the work, various committees were constituted and issued the Office Memorandum (OM).
- The draft Memorandum of Understanding (MoU) was prepared in accordance with Jal Shakti Vibag (JSV), Government of Himachal Pradesh in various levels and finally, submitted to Ministry of Jal Shakti for the concurrence.
- A Team of officers from CHQ, CGWB Faridabad has visited Kala Amb Valley along with WAPCOS and official from CGWB, NHR, Dharamshala on 23.10.2024 for explaining and pin pointing the surveyed AR sites in Kala Amb Valley. The photographs of the field visit is provided in the **Figure 3.13**.

Fig. 3.13: Field visit by the Officers of CGWB. regarding pinpointing of the AR sites Kala Amb Valley

3.7.6 District Recharge Plan Maps

District Recharge Plans (DRP) were to be prepared for the 255-water stressed district of the country which was identified during Jal Shakti Abhiyan (JSA) 2019 which will be shared to the State Government department/implementing agencies and District authorities. The District Recharge Plan will be prepared on the scale of 1:50,000.

During 2023, MoJS has advised to prepare the DRPs of the 150 water stressed district identified during JSA 2023, in which 50 districts were common for both JSA (2019) and JSA (2023). Therefore, out of the total 350 DRPs identified both in JSA 2019 and JSA 2023, 343 DRPs have been finalized and remaining are under finalization.

3.7.7 Construction/Renovation of Artificial Recharge Structures/Rain water Harvesting Structures in the premises of MoJS offices

The Ministry of Jal Shakti (MoJS) has mandated the installation of rainwater harvesting (RWH) structures across all its offices nationwide, including those under the Central Ground Water Board (CGWB). To facilitate this, MoJS has coordinated with CGWB offices for the construction and renovation of artificial recharge and rainwater harvesting structures within their premises. Meetings have been organized with other MoJS offices to finalize the list of such structures that have been constructed or renovated. A total of 76 no. Rainwater Harvesting/Artificial Recharge Structure constructed/renovated in the premises of all the CGWB Offices. The photographs of some of the structures are shown in **Figure 3.14**.

Recharge Pit with Recharge Tube Wells at CGWB, ER, Kolkata

Recharge Pit with Recharge Borewell at CGWB, CHQ, Faridabad

Recharge Pit at CGWB, SR, Hyderabad

Recharge Pit with Recharge Borewell at RGNGWTRI, Raipur

Fig. 3.14: Rain water Harvesting/Artificial Recharge Structure constructed/renovated in the premises of all the CGWB Offices

3.8 Regulation and Control of Ground Water Extraction by Central Ground Water Authority (CGWA)

3.8.1 Regulation and Control of Ground Water Extraction

Central Ground Water Authority, constituted under Section 3 (3) of the Environment (Protection) Act, 1986 has been regulating groundwater development in the country The Authority regulates in 19 States/ UTs (15 States/ UTs. Have their own Acts and 2 States are regulating under some GO). The Authority issues No Objection Certificate for ground water abstraction.

- Exercise of powers under section 5 of the Environment (Protection) Act, 1986 for issuing directions and taking such measures in respect of all the matters referred to in sub-section

(2) of section 3 of the said Act.

- To resort to penal provisions contained in sections 15 to 21 of the said Act.
- To regulate and control, management and development of groundwater in the country and to issue necessary regulatory directions for the purpose.
- Exercise of powers under section 4 of the Environment (Protection) Act, 1986 for the appointment of officers.

Important activities of CGWA during the period mentioned are given below:

3.8.2 Guidelines for Ground Water Regulation

Ministry of Jal Shakti notified the revised guidelines on 24.09.2020 and for regulation and control of ground water development in the country. These guidelines have Pan India applicability. Further the guidelines have been amended on 29.03.2023 The salient features of guidelines are as follows:

- Seeking of NOC is exempted for individual domestic consumers (in both rural and urban areas) drawing groundwater for drinking/ domestic use, Rural drinking water supply schemes, Armed Forces and Central Armed Police Forces Establishments, Agricultural users and Micro and Small Enterprises drawing < 10 KLD of groundwater, All industries/ mining projects/ infrastructure projects drawing ground water only for drinking/ domestic purposes up to 5 Cum /day in all assessment units, Residential Apartments and Group Housing Societies: (a) For drinking water and domestic uses, drawing ground water up to 20 m3/day subject to the conditions mentioned in Para 2.0 of the guidelines. (b) Dwelling units for Economically Weaker Sections (EWS) under Government schemes.
- No new industries except those MSME based on ground water are permitted in Over-exploited assessment units.
- Expansion of existing industries except MSME located in Over-exploited assessment units involving increase in quantum of ground water abstraction is not permitted.
- Industries drawing > 100 KLD of ground water in OE, Critical and Semi-critical assessment units are required to submit impact assessment reports along with the application.

- Industries drawing > 100 KLD of ground water in OE, Critical and Semi-critical assessment units are required biennial (once in two years) water audit and reduce fresh groundwater consumption by 20% over the next three years.
- No NOC is granted for groundwater extraction for water parks, theme parks, amusement parks in OE assessment units. Commercial infrastructure projects requiring ground water for drinking /domestic use shall also be covered under this category. Further, the Indicative list of location specific Infrastructure projects as per Annexure VI of the guidelines.
- All ground water users seeking NOC are required to pay ground water abstraction charges in Critical, Semi-critical and Safe assessment units and ground water restoration charges in OE assessment units. The charges so collected are to be utilized by States/ UTs on specific water conservation measures, specific supply/ demand side interventions.
- No charges are to be paid by users drawing saline ground water.
- Users drawing groundwater illegally are liable to pay Environmental Compensation.
- Penal provision has been kept for ensuring compliance of various conditions of NOC.
- Bulk water suppliers have also been brought in the ambit of NOC.
- Installation of Sewage Treatment Plants is mandatory for all new residential apartments/ Group Housing Societies where ground water requirement is more than 20 m3 /day. STP treated water is to be used by such users for flushing, green belt etc.
- All users seeking NOC are required to install tamper proof flow meters along with telemetry on all the ground water abstraction structures.
- In over-exploited assessment units, use of ground water for construction activity is permitted only if no treated sewage water is available within 10 km radius of the site.

3.9 Processing of Applications for Grant/ Renewal of No Objection Certificate (NOC) for Ground Water Withdrawal

3.9.1 Processing of Applications and Issuance of NOCs/ Exemptions

- CGWA evaluated applications from Industries/ Infrastructure Units / Mining Projects for grant of NOC for ground water withdrawal as per provisions of the Notified guidelines dated 24.09.2020 and amendments dated 29.03.2023 thereto. A total **of 1334** nos. of new NOCs were **issued** and **1954** nos. of NOCs **renewed** during the 2024-25. In addition, **exemption** was given in respect of **4658** nos. **new** applications and **90 nos renewal** cases. **Thus**, total **8,036** NOC's/exemptions were issued during the year 2024-25.

3.9.2 Evaluation of Impact Assessment Reports/ Comprehensive Hydrogeological Reports

- **As part of the application processing**, Impact Assessment Reports (**IAR**)/ Comprehensive Hydrogeological Reports (**CHR**)/ Modelling reports submitted by the Project Proponents were also evaluated by internal and external Expert Appraisal Committees (**EAC**). CGWA held **48 meetings** of **external** EAC and **70 meetings** of **internal** EAC, wherein total **119** and **271 reports** were reviewed/ evaluated respectively.

3.9.3 Representations for Review of Charges Imposed

- The committee to consider representations of Project Proponents for revision of charges (GW Charges/ EC/ Penalty) held **3 meetings** during the year, wherein **17 representations** were dealt with.

3.9.4 Monitoring of Compliance of Conditions Stipulated in the NOC

- CGWA has developed an online self-compliance format for the project proponent to upload the data as per compliance conditions defined in the NOC and the same is being monitored by the Regional Offices for ensuring the compliance conditions by the project proponent. Random site inspection is also carried out during the year to check the compliance of NOC issued by CGWA.
- Show-cause Notices are also being issued to units on a regular basis, which have been found not to have fully complied with the NOC conditions. Orders for sealing of bore/tube wells and/ or disconnection of electricity supply through the concerned DCs/ DMs were also issued in respect of units, which did not give satisfactory replies to the show-cause notice.
- While processing applications for renewal of NOC, compliance of NOC conditions by the proponents was ensured. Penalty provisions are kept for non-compliance conditions of NOC issued. Provision of Rs. 1 lakh under Section 15 of Environment (Protection) Act of 1986 was imposed on the violators whose NOC was issued prior to 24.09.2020, i.e prior to issuance of guidelines i.
- Separate penalty provisions are kept as per different compliance conditions in the guidelines issued after 24.09.2024 as per table 16.1. Renewal NOCs were issued after the compliance was completed and penalty was deposited by such offenders to the concerned Regional Office of CGWB/CGWA, New Delhi.

3.10 Important Meetings for Regulation of Groundwater/ Review of CGWA

3.10.1 Meetings By Hon'ble Minister, Jal Shakti

- The Hon'ble Minister, MoJS took meeting of CGWA Officers on **22.07.2024** to review the status of VIP References - Aava Mineral Water case and Eco Green case.
- The Hon'ble Minister, MoJS took meeting of CGWA on **24.10.2024** to review the status of VIP References - Aava Mineral Water case and Eco Green case.

3.10.2 Meetings by Secretary/ Additional Secretary

- The Secretary, DoWR took review meeting of CGWA activities on **24.12.2024** and **26.03.2025**, particularly to review the status of disposal of NOCAP applications.
- The Hon'ble Minister, MoJS took meeting of CGWA on **24.10.2024** to review the status of VIP References - Aava Mineral Water case and Eco Green case.
- The AS (Admin, IC & GW), DoWR took review meeting of CGWA Activities on **10.04.2024, 25.07.2024, 17.12.2024** and **19.02.2025**.

3.10.3 Meetings with State Ground Water Authorities (SGWAs)

Meetings were held under the chairmanship of the Chairman, CGWA in hybrid mode with **9 SGWAs** of **northern** States/ UTs on **25.06.2024** and **27.02.2025**, and with **8 SGWAs** of **southern** States/ UTs on **08.08.2024** and **28.02.2025** to discuss the status of implementation of MoJS guidelines and other relevant issues pertaining to ground water regulation. The status of following aspects of ground water regulation in the States/ UTs was reviewed by the Chairman and suitable directions/ advice/ suggestions were given: **i)** syncing of State/ UT Guidelines with MoJS guidelines**; ii)** status of clearance of NOC Applications; **iii)** measures taken for preventing illegal extraction of groundwater by Bulk Water Suppliers; **iv)** mandatory RWH Rules in State including prevention of groundwater from contamination from faulty implementation of RWH schemes and; **v)** constitution of Nodal Department/ Agency for implementation of Hon'ble Supreme Court order in the matter of Children falling in Borewell**.**

3.10.4 Other Important Meetings

- A meeting was held between Member Secretaries of CGWA and CPCB to discuss the action on recommendations of Parliamentary Standing Committee on water Resources in its 22nd and 25th reports. Scientists from CPCB and CGWA also attended the meeting. As per recommendations, the deliberations were held on following points.
  - Steps for Prevention of GW Pollution by CPCB as Centralized Body
  - Submission of Periodic/ Quarterly report of ETP/ CETP status to CGWA
  - Issuance of CTO/ CTE/ CCA *vis-à-vis* GW NOC.

3.10.5 50th meeting of the Board of Central Ground Water Authority

- 50th meeting of the Central Ground Water Authority (CGWA) was held on **27.06.2024** under the chairmanship of Dr S K Ambast, Chairman, CGWA.
- Amendments to extant guidelines were proposed to the board in order to remove conflicts/ discrepancies in guidelines, as also to introduce provisions for certain scenarios experienced during the processing of applications. Concurrence of Board was sought on these proposed amendments for further submission to Ministry. **Out of total 10 amendments proposed**, following **9 amendments were concurred**: **1)** provision for six month time-window for applying for groundwater NOC after obtaining project approval; **2**) removal of discrepancy of piezometer construction for infrastructure project; **3**) increasing ground water charges by double, triple and so on in case of continued non-compliance by PP as regard to flow-meter installation; **4**) merging inter-dependent penalties towards non-installation of piezometer/ flow-meter and non-submission of related data; **5)** addition of application fee for Modification Application; **6)** EC applicable on annual excess ground water extraction; **7)** defining non-commercial infrastructure projects for exemption from EC; **8)** consider definitive recommendations of Water Audit report to decide reduction in ground water quantum in renewal NOC; **9)** criterion/ SOP for EC imposition in cases of existing projects having alternative supply earlier and now requiring ground water. **The proposed amendment** regarding **charges for commercial complex** as integrated part of the Residential Townships was referred to the next Board meeting for **further deliberation**. The Board also vetted revised criteria for EAC of IARs/ CHRs.

3.10.6 51st meeting of the Board of Central Ground Water Authority

- 51st meeting of the Central Ground Water Authority (CGWA) was held on **27.06.2024** under the chairmanship of Dr S K Ambast, Chairman, CGWA.
- Amendments to extant guidelines were proposed to the board in order to remove conflicts/ discrepancies in guidelines, as also to introduce provisions for certain scenarios experienced during the processing of applications. Concurrence of Board was sought on these proposed amendments for further submission to Ministry. Following proposed **amendments were concurred in Main Guidelines, Saline GW Extraction Guidelines** and **Bulk Water Suppliers Guidelines**: **1)** inclusion of rates for fresh water extraction from Saline category Assessment Unit, subject to usage only for drinking domestic purpose; **2)** inclusion of rates of saline ground water extraction by Packaged Drinking Water Units and Bulk Water Suppliers; **3)** revised ceiling and purpose of groundwater extraction by Bulk Water Suppliers**; 4)** modification in requirement of IAR/ Modelling studies for saline ground water extraction/ fresh water extraction from saline assessment unit to remove ambiguity; **5)** addition of Residential Complex/ Apartments/ Group Housing Societies/ Petrol Pump/ CNG Station/ Piped Gas distribution facility or any other similar facility, Solar Power Plant/ Station, Warehouses for e-marketing in the list of location-specific infrastructure (for new projects in OE areas); **6)** charges for commercial units as integral part of Residential Townships/ Swimming Pools. In addition, SOP for EC imposition criterion in revival cases was approved by the Board. Slight modification in EAC criteria approved in 50th meeting was also approved by the Board.
- The Board approved the revised **Policy for Accredited Ground Water Professionals**, with some minor modifications/ suggestions.

3.10.7 52nd meeting of the Board of Central Ground Water Authority

- 52nd meeting of the Central Ground Water Authority (CGWA) was held on **06.03.2025** under the chairmanship of Dr S K Ambast, Chairman, CGWA.
- The representatives of Assam and Delhi Government presented their draft guidelines for ground water extraction for approval of the Board. The draft guidelines were principally agreed to and vetted by the Board with suggested modifications.
- Board also agreed to the proposal that EC may not be imposed in those cases where revival is done within the period corresponding to advance deposit, i.e even when the revival is outside the given time-window of sixty days after rejection.

3.10.8 Court Cases - Submission of Reply/ Reports

- CGWA submitted timely replies/ reports/ affidavits etc. in compliance to directions in various court matters (**Hon'ble Supreme Court/ Hon'ble High Courts/ Hon'ble NGT**). Total **79 replies/ reports/ affidavits** have been submitted to various courts during 2024-25. A few very important ongoing matters, such as 'Groundwater Usage by Cricket Stadiums'; 'UN Report Regarding Depleting Ground Water'; 'Arsenic and Fluoride in Groundwater'; 'Extraction of Ground Water by Kutchh Lignite Thermal Power Station, Gujarat'; 'Illegal extraction of ground water in NCT of Delhi'; 'Extraction of Ground Water without NOC by RIICO' etc. were dealt with during the year. As part of the joint committee in matter involving Hitech City, Ghaziabad before Hon'ble Supreme Court, CGWA officers, carried out inspection of the project along

with CPCB, CPCB and GWD, UP officials and submitted the report about observation pertaining to ground water extraction by the project. The report has been submitted by CPCB to Hon'ble Supreme Court.

3.10.9 On Site Inspection by CGWB/ CGWA

- On-site inspections were carried out by the Regional Offices of CGWB to check the compliance of NOCs accorded by CGWA before recommending the renewal applications to CGWA, New Delhi. Necessary show- cause notices were issued to the project proponents who have not complied with the conditions of the NOC issued by CGWA.
- A team **headed by Member Secretary (CGWA)**, comprising members of **External Expert Committee (EAC)** and CGWA Officers conducted study tours and site inspections of mining and industrial projects in Chhattisgarh and Rajasthan. Meeting of External EAC was held on-site for evaluation of Impact assessment Report for heavy extraction of groundwater by India's largest onshore oil field of Cairn Energy in Barmer district, Rajasthan.
- CGWA officers from New Delhi also carried out site inspections of **02 industrial Units** and **03 Mining Units** in **Rajasthan, Uttarakhand, Chhattisgarh, Odisha** and **Gujarat state** in year 2024-25.

3.10.10 Development of Advanced NOCAP Software

- The new portal of issuing NOC to ground water users by CGWA i.e. BhuNeer APP was launched on 19th September 2024 in 8th India Water Week. The portal has been developed to serve a large user base across the country and supports functionalities such as user registration, authentication, real-time transaction processing, reporting, and integration with third-party systems, including payment gateways and digital records repositories. The 08-level processing of application in NOCAP portal has been reduced to 05 stages i.e. Evaluation, EAC, Recommendation & Approval Stage (1-3 parts) for smooth processing and early disposal of applications. Additionally, a Query Module also developed for one-to-one communication between applicant and CGWA user for any clarification and update in application or documents. The portal is live since 30.10.2024 for filing of new and renewal application. Additional modules are also being developed in order to complete migration from NOCAP to BhuNeer APP and improved requirements in respect of project proponents and CGWA. The modules in pipeline for development are Grievance module, Payment Modification (head transfer /refund), Expansion in project, Modification request by project proponent, Surrendering /Cancelling NOC, Amnesty Scheme, Telemetry data Integration, Advance Report Generation, Revision of charges (EC/Penalty etc.) etc.
- In order to **phase out** the old **NOCAP** Portal, all applications on NOCAP Portal were disposed of by the end of March, 2025. The rejected applications, however, will be revived on NOCAP Portal only.

3.10.11 Implementation of OM regarding relaxation in Environmental Compensation for MSME, relaxation in EC on Saline Water and relaxation in charges for User Id etc.

- The OM dated 12.03.2024 providing relaxation in Environmental Compensation (**EC**)

for one year period was under implementation till **11.03.2025**. The beneficiaries were MSME extracting groundwater up to 100 KLD. The OM also provides relaxation in EC on saline ground water extraction by keeping earlier Public Notice in suspension for one year. In addition, charges for User ID change and other minor corrections have also been reduced for one year vide above OM. The OM was issued to ease the financial burden on entities abstracting groundwater by offering concessions on penalties and administrative fees, likely to promote compliance and facilitate proper regulation of groundwater resources. Relief to the tune of **≈₹ 22.20 Crore** was provided in **418 cases** with following break-up different heads: **a) ≈₹ 19.03 Crore** relief in EC in **389** cases; **b) ≈₹ 3.14 Crore** relief in EC towards saline ground water extraction in **34** cases and; **c) ≈₹ 0.03 Crore** relief in charges for Log-in ID change in **118** cases.

3.11 Trainings/ Workshop-cum-Awareness Program

3.11.1 Trainings/ Workshops Conducted by CGWA/ Lectures Delivered by CGWA Officers

- Central Ground Water Board, West Central Region organized 1-day workshop on 24.01.2025 jointly with Gujarat Chamber of Commerce & Industry (GCCI) on "***Central Ground Water Authority Guidelines and BhuNeer APP***"**.** The Member Secretary, CGWA along with officers of CGWA, New Delhi delivered talks to create awareness and promote better understanding for CGWA NOC procedure and NOC Compliances at GCCI, Ashram Road, Ahmedabad.
- Total Eight (**8**) Nos in-house two-day Training-cum-Workshop were organized at CGWA, New Delhi for officers from 10 Regions (CR, ER, MER, NCCR, NCR, NER, SER, UR, WCR, WR,) of 19 CGWA-regulated States/ UTs. The two-day sessions in two batches were organized in each quarter of the year. These sessions aimed at making officers conversant about the newly developed features of NOCAP and BhuNeer APP and also to resolve technical issues faced by the regional offices. Officers of CGWA, New Delhi also provided clarifications/ explanations on queries about provisions of guidelines during these sessions.
- The Member Secretary, CGWA along with CGWA officers sensitised industry representative about the OM dated 12.03.2024 providing relaxation in Environmental Compensation (EC) through webinars organized in association with industrial associations – CII (06.09.24), ASSOCHAM (20.09.24) and FICCI (30.09.24).
- Ms Shilpi Gupta, Scientist D delivered following lectures as guest faculty at RGNGWTRI to **ILTC** batch/ participants of **Certification Course for Water Auditors**: **(i) 4 sessions** on 'Need of Water Auditing' and 'SOP on Water Audit Report and its Various Components' during 17.12.2024-20.12.2024.
- Ms Iti Gupta, Scientist D delivered following lectures as guest faculty at RGNGWTRI to **ILTC** batch: **(i)** 'Formation and Function of CGWA: An overview' on 13.11.2024, **6 sessions** on CGWA Guidelines, SOP for issuance of NOC, NOCAP modules, Preparation of CHR and modelling reports, Hands on Application processing on NOCAP during 02.12.2024-04.12.2024**; (ii)** 'Ground Water Regulation & Control' on 19.03.2024**; (iii) 5 sessions** on 'Overview of NOCAP & Bhuneer APP, Processing of Applications for Industry, Infrastructure & Mining Projects on 20.03.2025.

3.11.2 Trainings/ Workshops Attended by CGWA Officers

- Officers of CGWA actively participated in Workshop on 'Ground Water Clearance for Mine Proposals organized by Ministry of Mines'.
- Shri Avinash Chandra, Assistant Hydrogeologist attended two-week training program on 'Management Aspects and Leadership Traits-MALT' organized by RGNGWTRI, Raipur from 13.01.2025 to 24.01.2025.
- Officers of CGWA attended the 8th India Water Week, 2024 on "Partnerships and Co-operation for Inclusive Water Development and Management" organized by Ministry of Jal Shakti during **17.09.2024 -20.09.2024**.
- Ms. Veikhone Sophia Kay, Scientist C attended "Water Sustainability Conference 2025" organized by Bureau of water Use Efficiency (**BWUE**), National Water Mission, Ministry of Jal Shakti at NMDC Convention Centre, New Delhi on **12.03.2025**.

3.11.3 Policy Documents/ SOPs/ Other Documents Prepared by CGWA

CGWA prepared following Policy Documents/ SOPs/ Other Documents during the year 2024-25 for various purposes.

- Accreditation Policy for Ground Water Professionals, October, 2024 - issued
- SOP for Ground Water Modelling – uploaded on NOCAP
- SOP for Water Audit Report – uploaded on NOCAP
- Compendium of Guidelines/ Public Notices/ Orders issued by CGWA – uploaded on NOCAP
- SOP for Sampling/ Analysis of Arsenic and Fluoride and Ground Water – the SOP was prepared in connection with Hon'ble NGT matter and submitted the Tribunal.
- Proposed Amendments to guidelines – sent to the Ministry. The Ministry constituted a committee for revision of guidelines vide Order dated 12.03.2025.
- A comprehensive proposal for 'Strengthening of CGWA' for more robust compliance and monitoring system was submitted to the Ministry, seeking 110 additional posts for CGWA.

3.11.4 Other Activities by CGWA Officers

- Officers of CGWA participated in **Swachhata hi Sewa** Campaign at Jahapanah City Forest, Greater Kailash on **28.10.2024**.
- **Jal Shatki Abhiyan (JSA-CTR, 2024):** Shri Junaid Ahmad, Scientist C, was designated as Technical Officer (**TO)**, for District North East Delhi. He accompanied the designated CNO to the assigned district during pre-monsoon and post-monsoon to review the status of interventions defined under JSA Activities and provided technical inputs/ guidance for the activities. The report was submitted to National Water Mission.
- **National Water Award (NWA):** As member of the Screening Committee for **evaluation of applications** for NWA, Shri Ashok Kumar Patre, Scientist D evaluated **43 Applications** under industrial category. Ms. Aditi Bhatt, Scientist C, carried out **ground-truthing** of the shortlisted applications under Civil Societies (1 No.) and Schools (8 Nos.) in Delhi and NCR area.

3.12 Technical Assistance to State Governments

3.12.1 Short Term Water Supply Investigations

Table 3.23: Region wise list of short-term investigations carried out in AAP 2024-25

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
1	NWR Chandigarh	Feasibility Study for Construction of Tubewell, At Air Force Station, Baknaur, Ambala Cantt., Ambala	Air Force Station, Baknaur, Ambala Cantt
2		Feasibility Study for Construction of Tubewell, At Air Force Station, Kalpi,	Air Force Station, Kalpi, Ambala Cantt
3		Feasibility Study for Construction of Tubewell AT Chandimandir Cantonment Area, Panchkula, Haryana	Chandimandir Cantonment Area, Panchkula
4		Short Term Investigation Report for Construction of Rain Water Harvesting & Artificial Recharge Structure at Military Station, Nabha, Patiala	Military Station, Nabha, Patiala
5		Feasibility Study for Construction of the Tube well at Air Force Station, Patiala Cantonment, Patiala	Air Force Station, Patiala Cantonment, Patiala
6		Feasibility Study report for Borewell at Pump House No. 08 (Hudson Line) Jalandhar Cantt., Jalandhar, Punjab	Jalandhar Cantt.
7		Feasibility Report at CH-35, MES, Chandimandir Cantonment, Panchkula.	MES, Chandimandir Cantonment, Panchkula.
8		Feasibility Report at KLP (Phase II & Phase III), MES, Chandimandir Cantonment, Panchkula.	MES, Chandimandir Cantonment, Panchkula.
9		Short Term Investigation Report for Construction of Rain Water Harvesting & Artificial Recharge Structure near CWE Chandimandir & Tapan Dwar, Military Engineering Services, Chandimandir Cantonment, Panchkula, Haryana	MES, Chandimandir Cantonment, Panchkula.
10		Feasibility for Construction of one new Tubewell at 33 FAD Dapper	33 FAD Dapper
11		Feasibility Report for Provn of KLP Accn at L-70 Phase II, Ambala Cantonment.	Ambala Cantonment.
12		Feasibility Report at Construction of Junoir Commissioned Officer (JCO) and other Married Accn out of EVI at Ambala Cantonment.	Ambala Cantonment.
13		Feasibility Report for Provn of DEFI	Ambala Cantonment.

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
		MD Accn for Officers/JCO/ OR at various pockets i.e. Harding line, ICC line, Abdhul Hamid Line, Ambala Cantonment.	
14		Feasibility of Bore Hole (1No) & Rain water harvesting & Artificial Recharge structure at Airforce Station, Gurugram city, Haryana.	Airforce Station, Gurugram city, Haryana.
15		Feasibility study for Construction of Rain water harvesting & Artificial Recharge structure at the site of Rigid Pavement at Jalandhar Cantt	Jalandhar Cantt
16		Feasibility report of Construction of 03 Nos New Borewells in replace of Old Borewells at 33 FAD Dapper	33 FAD Dapper
17		Feasibility Study report of construction of Tube wells at Suranussi Cantt, Punjab	Suranussi Cantt, Punjab
18		Feasibility Study report of construction of Tube wells at Jalandhar Cantt, Punjab	Jalandhar Cantt, Punjab
19	**NWR Chandigarh**	'Report on Hydrogeological Investigations to study feasibility for construction of a tubewell in the K-Area Zirakpur near Chandigarh	cantonment Zirakpur
20		Feasibility for construction of tubewell at Chandimandir Cantonment Area, Panchkula, Haryana.	Chandimandir Cantonment Area, Panchkula
21		Feasibility study for construction of Rain water harvesting and artificial recharge structure at OTM accn including ancillaries at Dapper.	33 FAD Dapper
22		Feasibility of construction of tubewell in the premises of Military Station, Ferozpur Cantt, Ferozpur, Punjab	Military Station, Ferozpur Cantt, Ferozpur
23		Feasibility of construction of tubewell in Air Force Station, Ambala Cantt., Ambala	Air Force Station, Ambala Cantt., Ambala
24		Feasibility Study for Construction of Rain Water Harvesting Structure for provn of Auditorium cum Cinema Hall at Patiala Mil Station, Patiala Cantt.	Patiala Mil Station, Patiala Cantt.
25		Feasibility Study for Construction of Rain Water Harvesting Structure in Sector 47 against work "Demolition / Reconstruction of unsafe / Dilapidated	AF STN, Chandigarh

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
		SMQ at AF STN, Chandigarh	
26	NCR Bhopal	Hydrogeological Investigation for Water Supply/ Feasibility report of CM Rise School for construction of bore well in Bori Bujurg and Dawatiya villages of Burhanpur district	Burhanpur District Administration
27		Hydrogeological and geophysical surveys carried out in the premises of army center of education training college, panchmarhi, madhya pradeshfor	Army Center of Education Training College, Panchmarhi, Narmadapuram, Madhya Pradesh
28	NCCR Raipur	Geophysical Investigation of GC-CRPF, Campus at Village Bhilai, Arang, Raipur, Chhattisgarh	GC-CRPF, Arang, Raipur, Chhattisgarh
29	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of trehgam and pathahiri, district kupwara (j&k-ut)	Garrison Engineer 874 Engr Wks sec.
30		Hydrogeological investigation for water supply/feasibility report of z-gali and pharkian gali, district kupwara (j&k-ut)	Garrison Engineer 874 Engr Wks sec.
31		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at STC BSF Udhampur (UT of J&K)	Inspector General STC, BSF Udhampur
32		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at 15 Grenadiers, District Rajouri, (UT of J&K)	Maj, AGE (I) CIF ® C/o 56 APO Rajouri
33		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at AFVH and ALG Ground, District Rajouri, (UT of J&K)	Garrison Engineer 862 Engr Wks Section c/o 56 APO
34		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of Borewell in CBS Sarol Romeo Post, ASC Supply Point and Nagrota Camp, District Rajouri, (UT of J&K)	Garrison Engineer 862 Engr Wks Section c/o 56 APO
35		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of Borewell in Kalal, District- Rajouri (UT of J&K)	The Administrative Commandant Station HQ Noushera

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
36	NWHR Jammu	Hydrogeological Investigation for Water Supply/ Feasibility report for construction of Borewell in Palma, Saaj and Galuthi, District- Rajouri (UT of J&K)	Asst. Garrison Engineer Palma, Rajouri, J&K
37		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of Borewell in BSF Sunderbani, District- Rajouri (UT of J&K)	Assistant Commandant (Works) AE, SHQ, BSF, Sunderbani UT of J&k
38		Feasibility report for construction of Borewell at 22 Ammunition company and 164 artillery regiment, Samba Cantonment (UT of J&K)	Major Quartermaster 22 Gola Barud Sansthan 22 Ammunition Company
39		Feasibility report for construction of tube-well at Kendriya Vidhalaya, Sunjwan, district Jammu (UT of J&K)	Garrison Engineer Military Engineer Service Jammu c/o 56 APO
40		Feasibility report for construction of tube-well at Z-Gali and Pharkian Gali, district Kupwara (UT of J&K)	Garrison Engineer 874 Engr Wks sec C/O 56 APO
41		Feasibility report for construction of borewell at BoP Nawapind and BN HQ, Indreshwar Nagar, district Jammu (UT of J&K)	The Construction Officer HQr 165 Bn, BSF Indershwar Nagar Jammu.
42		Hydrogeological investigation for water supply/feasibility reportof bg water point and hamirpur water point, district poonch (j&k-ut)	Garrison Engineer 881 Engr Wks sec C/O 56 APO
43		Hydrogeological investigation for watersupply/feasibility reportof dharna, mendhar, upperlower garrison and balnoi, district poonch (j&k-ut)	Garrison Engineer 881 Engr Wks sec C/O 56 APO
44		Hydrogeological investigation for water supply/feasibility reportof in parts of contanoment in poonch, district poonch (j&k-ut)	Garrison Engineer 881 Engr Wks sec C/O 56 APO
45		Hydrogeological investigation for water supply/feasibility report of likir, ut of ladakh	HQ 614(r) Aviation Brigade c/o 56 APO
46		Feasibility report for construction of borewell at Rampur, Billawar, district Kathua UT of J&K	The Executive Engineer (C) BSNL, Civil Division Jammu, J&K-UT.
47		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Sikh LI (12 Garwal), UT of Ladakh	Military Engineer Services Garrison Engineer Partapur

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
48	NWHR Jammu	Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Post 09, UT of Ladakh	Military Engineer Services Garrison Engineer Partapur
49		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at 1905 LT REGT, UT of Ladakh	Military Engineer Services Garrison Engineer Partapur
50		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Siachen Base Camp, UT of Ladakh	Military Engineer Services Garrison Engineer (AF) Thoise 860 Engr Wks Section
51		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Thoise Airport Station, UT of Ladakh	Military Engineer Services Garrison Engineer (AF) Thoise 860 Engr Wks Section
52		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Kiari, UT of Ladakh	Military Engineer Services Garrison Engineer 860 Engr Wks Section
53		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Chumathang, UT of Ladakh	Military Engineer Services Garrison Engineer 860 Engr Wks Section
54		Hydrogeological Investigation for Water Supply/ Feasibility report for construction of bore well at Nypma, UT of Ladakh	Military Engineer Services Garrison Engineer 860 Engr Wks Section
55		Hydrogeological investigation for water supply/feasibility report of chusul (sikhli), ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
56		Hydrogeological investigation for water supply/feasibility report of tara (jatli), ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
57		Hydrogeological investigation for water supply/feasibility report of heena (op), ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
58		Hydrogeological investigation for water supply/feasibility report of nyoma, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
59		Hydrogeological investigation for water supply/feasibility report of loma, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
60		Hydrogeological investigation for water supply/feasibility report of demchok, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
61	NWHR Jammu	Hydrogeological investigation for water supply/feasibility report of fukcha, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
62		Hydrogeological investigation for water supply/feasibility report of chisumle, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
63		Hydrogeological investigation for water supply/feasibility report of zursar, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
64		Hydrogeological investigation for water supply/feasibility report of hanle, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
65		Hydrogeological investigation for water supply/feasibility report of gref, dungti, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
66		Hydrogeological investigation for water supply/feasibility report of chumar, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
67		Hydrogeological investigation for water supply/feasibility report of karzok, ut of ladakh	105 Engineer Regiment, Leh, UT of Ladakh
68		Hydrogeological and geophysical investigations for water supply/feasibility report of drass, ut of ladakh	Garrison Engineer Kargil C/O 56 APO
69		Hydrogeological and geophysical investigations for water supply/feasibility report of kargil, ut of ladakh	Garrison Engineer Kargil C/O 56 APO
70		Hydrogeological and geophysical investigations for water supply/feasibility report of budhkharbu, ut of ladakh	Garrison Engineer Kargil C/O 56 APO
71		Feasibility report for construction of dugwell at IIM, Jammu (J&K-UT)	Chief Administrative Office IIM, Jammu.
72		Feasibility report for construction of tubewell at Kendriya Vidyalaya, Jourian, Akhnoor, District Jammu (J&K-UT)	CPWD, Jammu Division, Landoi Bridge, Satwari, Jammu
73		Feasibility report for construction of tubewell at BSNL Compound near Telephone Exchange at Sector 2, Chhani Himmat, Jammu, District Jammu	Executive Engineer (C) BSNL, Civil Division Jammu, J&K-UT.
74		Report on the feasibility for constructing a tubewell in the premises	Air Force

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
	UR Dehradun	of pcda (air force) rajpur road, dehradun district, uttarakhand	
75		Report on the feasibility for constructing a tubewell in the premises of kendriya vidyalay, raiwala cantt., dehradundistrict, uttarakhand	KENDRIYA VIDYALAYA
76		Report on the feasibility for constructing a tubewell in the premises of mt colony, ima, dehradun district, uttarakhand	IMA
77		The scope of rooftop rainwater harvesting/ artificial recharge at clement town cantonment, dehradun district, uttarakhand	Defence
78		Report on the scope of rooftop rainwater harvesting/ artificial recharge at uttarakhand state council for science and technology, dehradun district, uttarakhand	UCOST
79		The Scope of Rooftop Rainwater Harvesting/Artificial Recharge at Nationa; Hydrographic Office, Dehradun District, Uttarakhand	NHO
80	NR Lucknow	Provision of Hanger for first line maintainance at AF STN BKT (Bakshi Ka Talab)	Air Force Station, BKT Lucknow
81		Provision of Rain Water Harvesting Structure at AF STN BKT (Bakshi Ka Talab)	Air Force Station, BKT Lucknow
82		Feasibility Report of Borewell at Varanasi Cantt.	Garrison Engineers MES(East), Prayagraj
83		Rain water Harvesting Pit for provn. Of essential OTM ACCN Lucknow	Garrison Engineers (Project), MES, Lucknow
84		Replacement of BER Borewll with Pumpset including ailled WKS at Pump House No. 28 at Shivaji Lines GE(U) Meerut Cantt.	Garrison Engineers (U), MES, Meerut
85		Selection of TW site for water supply at New Cantonment Allahabad, UP	Garrison Engineers (Project), MES, Lucknow
86		Rainwater Harvesting Pit for CA No: CELZ/LKO/31 OF 2021-22: Provision of certain KLP ACCN at Lucknow	Garrison Engineers MES(West), New Cantt. Allahabad
87		Checking the feasibility of construction of a new tubewell at Memaura Air force Station, Lucknow	Memaura Air force Station, Lucknow
88		Borewell for Provision of 21 Type II	Garrison Engineers E/M,

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
	NR Lucknow	Govt. MD Accn for Def CIV against existing DEFI at 11 GRCC, Lucknow	Lucknow
89		Rainwater Harvesting Pit for Provn of certain DEFI OTM Accn for INF BN No III 71 MTN BDE at Lucknow CA NO CELZ/LKO/15 of 2023-24	Garrison Engineers E/M, Lucknow
90		Borewell for Provision of KLP Accn for AFMSD at Lucknow	Garrison Engineers E/M, Lucknow
91		Borewell for Provision of 300m Baffle Range at AMC Centre and College Lucknow	Garrison Engineers E/M, Lucknow
92	NER Guwahati	Report on short term water supply investigation the military station, sigar (near pasighat), east siang district, arunachal pradesh	Indian Army, Sigar Military Station, Arunachal Pradesh
93		A brief report on hydrogeological investigation at the premises of frontier headquarters, ssb, tezpur, shonitpur district, assam	SSB, Frontier HQ, SSB, Tezpur, Sonitpur, Assam
94		A brief report on hydrogeological investigation at the premises of crpf, manderdisa, dima hasao, assam	CRPF, Radaodisa, Maibong, Dima Hasao, Assam
95		Hydrogeological investigation at krishi vigyan kendra, megagre, william nagar, east garo hills district, meghalaya	Krishi Vigyan Kendra, Megagre, William Nagar, East Garo Hills, Meghalaya
96		A brief report on feasibility for construction of a deep bore well in the military area changsari, kamrup district, assam	Indian Army, Military Area, Changsari, Kamrup, Assam
97		Report on short term water supply investigation in the 9 madras (travancore), hayuliang, anjaw district, arunachal pradesh	Indian Army, Hyuliang, Lohit, Arunachal Pradesh
98		A brief report on feasibility for construction of groundwater abstraction structure in the premises of lalmati model hospital,2 no. Nalapara, boko, kamrup, assam	National Health Mission, Boko, Kamrup, Assam
99		A brief report on feasibility for construction of a deep bore well in the premises of cisf group hq, beharbari, guwahati, kamrup district, assam	CISF, Premises of CISF HQ, Guwahati
100		Report on hydrogeological investigation at military station, jorhat, assam	Indian Army, Jorhat Military Station, Assam

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
101	NER Guwahati	A brief report on hydrogeological investigation in the area of air force, borjhar, kamrup (m), assam	AIR FORCE, Borjhar, Kamrup (M), Assam
102		A brief report on feasibility for construction of a deep bore well at the premises of jawahar navodaya vidyalaya, niangbari, ribhoi, meghalaya	Jawahar Navodaya Vidyalaya, Ribhoy, Meghalaya
103		A brief report on feasibility for construction of a deep tubewell at the proposed site in the premises of hatiali army camp, dibrugarh district, assam	Indian Army, Hatiali Camp, Dibrugarh, Assam
104		A brief report on construction of a ground water abstraction structure in the premises of air force station, tezpur, sonitpur district, assam	AF Station Tezpur, Sonitpur, Assam
105		Hydrogeological investigation for assessment of feasibility for groundwater abstraction in the premises iaas officers' residential quarters, jatia, guwahati, assam	IAAS (Accountant General) Jatia, Guwahati, Kamrup(M), Assam
106		Feasibility report for construction of rain water harvesting & ground water recharge structure in the premises of indian air force station, digaru, kamrup (m) district, assam	Air Force Station, Digaru, Kamrup(M), Assam
107		Report on hydrogeological & geophysical investigation at 3rd assam police commando battalion, daldoli, karbi anglong district, assam	Assam Police Housing Corporation
108		Hydrogeological investigation at the premises of national science, centre khanapara, guwahati, kamrup (m) district, assam	National Sicence Centre, Guwahati, Kamrup(m), Assam
109		Hydrogeological investigation in the premises of 33 battalion, itbp, sonapur, kamrup (m) district, assam	33rd Bn, ITBP, Karchia, Sonapur, Guwahati, Kamrup(m), Assam
110		Report on short-term investigation for evaluating feasible site for construction of tube well in misa army cantonment area	Indian Army, Misa Camp, Nagaon, Assam
111		A brief report on hydrogeological investigation at the premises of arunachal bhawan, rupnagar, guwahati, assam	Addl Resident Commissioner, Arunachal Bhawan, Guwahati, Kamrup(M), Assam
112		Report on short-term investigation for evaluating feasible site for construction	Indian Army, Chabua Army Cantt. Dibrugarh,

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
		of tube well at chabua army cantonment area	Assam
113	**NER Guwahati**	Feasibility report for construction of bore well at pm shri school, jnv, nongpyndeng, nongstoin, west khasi hills district, meghalaya	JNV, West Khasi Hills, Meghalaya
114		Hydrogeological investigation at the premises of assam administrative staff college khanapara, guwahati, kamrup (m) district, assam	Assam Administrative Staff College, Khanapara, Guwahati, Kamrup(M), Assam
115		Water suply investigation for hq eastern air command, upper shillong and air force station, laitkor peak, meghalaya	Indian Air Force
116		1. Report on Water Supply Investigation at CRIDA Research Farm, Central Research Institute for Dryland Agriculture (CRIDA)-Indian Council of Agricultural Research (ICAR), Department of Agricultural Research and Education, Ministry of Agriculture & Farmers' Welfare, Hayath Nagar, Hayath Nagar Mandal, Ranga Reddy District, Telangana	ICAR
117	**SR Hyderabad**	2. Report on Water Supply Investigation at CRIDA Head Office, Central Research Institute for Dryland Agriculture (CRIDA)-Indian Council of Agricultural Research (ICAR), Department of Agricultural Research and Education, Ministry of Agriculture & Farmers' Welfare, Santosh Nagar, Saidabad Mandal, Hyderabad District, Telangana.	ICAR
118		3. Report on Water Supply Investigation at CRIDA Research Farm, Central Research Institute for Dryland Agriculture (CRIDA)-Indian Council of Agricultural Research (ICAR), Department of Agricultural Research and Education, Ministry of Agriculture & Farmers' Welfare, Gungal, Yacharam Mandal, Ranga Reddy District, Telangana.	ICAR
119		4. Report on Water Supply Investigation at Jawahar Navodaya Vidyalaya, Gopanapally Village,	JNV Gopanapally village, Ranga Reddy District,

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
	SR Hyderabad	Nallagandla, Serilingampally Mandal, Ranga Reddy District, Telangana	
120		5. Report On Water Supply Investigation at Begumpet Airport, Begumpet, Hyderabad, Airport Authority of India, Secunderabad Mandal, Hyderabad District, Telangana	Begumpet Airport, Telangana
121		6. Report On Water Supply Investigation at University of Hyderabad, Central University Po, Gachibowli, Seri Lingampally Mandal, Ranga Reddy District, Telangana	University Of Hyderabad, Central University Po, Gachibowli, Seri Lingampally Mandal, Ranga Reddy District, Telangana
122		7. Report On Water Supply Investigation at Adikavi Nannaya University, Rajamahendravaram (Rajahmundry), Velugubanda, Rajanagaram Mandal, East Godavari District, Andhra Pradesh	Adikavi Nannaya University, Rajamahendravaram (Rajahmundry), Velugubanda, Rajanagaram Mandal, East Godavari District, Andhra Pradesh
123	SUO Visakhapatnam	1. Hydrogeological Investigation for providing of Water Supply to Indian Coast Guard Radar Station, Baruva Village, Sompeta Mandal, Srikakulam District, Andhra Pradesh	Indian Coast Guard Radar Station, Baruva Village, Sompeta Mandal, Srikakulam District, Andhra Pradesh
124		2. Hydrogeological Investigations at Indian Coastal Guard, Malkapuram, Visakhapatnam, Andhra Pradesh	Indian Coast Guard, Malkapuram, Visakhapatnam, Andhra Pradesh
125		3. Hydrogeological Investigations at Indian Coastal Guard Station, Kakinada, Andhra Pradesh	Indian Coast Guard Station, Kakinada, Andhra Pradesh
126	SER Bhubaneshawar	Short Term Water Supply Investigation for selection of site for borewells in the campus of KV School (AADC), Gopalpur, Ganjam District, Odisha	Defence Army
127		Short Term Water Supply Investigation for selection of site for borewell inside premises of PM Shri JNV, Angul, Odisha	JNV
128		Short Term Water Supply Investigation for selection of site for borewell inside premises of INS Chilika, Khurda District	Indian Coast Guard
129	ER Kolkata	Short Term Water Supply Investigation for selection of replacement borewell	Defence-Army Military Station Coochbehar, West

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
		site near Pump House No.1 at Coochbehar Military Station, Coochbehar District.	Bengal
130		Short Term Water Supply Investigation for selection of 2 borewell sites at 16-hectare land of Indian Coast Guard Station, Mayabander, North & Middle Andaman district, A&N Islands.	Indian Coast Guard
131		Short Term Water Supply Investigation Survey report at the premises of Indian Coast Guard, Namunagarh, South Andaman district, A&N Islands.	Indian Coast Guard
132		Short Term Water Supply Investigation for replacement of Borewell site at Army Cantt Turf View Complex and OLD IMH, Alipore, Kolkata	Defence-Army Military Station Alipore, Kolkata
133		Short Term Water Supply Investigation at Panagarh Air Force Station, Panagarh, Paschim Barddhaman district, West Bengal	Defence- Air Force Station Panagarh, Paschim Barddhaman, West Bengal
134	**KR Trivananthapuram**	Geophysical study on the feasibility of construction of borewell for rain water harvesting in the premises of Aromatic and Medicinal Plants Research Station (AMPRS), Odakkali, Ernakulam district, Kerala using Electrical Resistivity Survey	Kerala State Government
135		Feasibility of Borewells in Chitralekha Complex, And Vayu Vihar at HQ(SAC), Air Force Station, Akkulam, Thiruvananthapuram	Air Force, Thiruvananthapuram
136	**SUO Delhi**	Hydrogeological investigation report at gpra colony, sec-2, rk puram, new delhi	CGERWA, Mohammadpur
137		Hydrogeological investigation report at delhi cantt (cunpk)	MES, DELHI CANTT.
138		Hydrogeological investigation report at shankar vihar, delhi cantt (cunpk)	MES, DELHI CANTT.
139		A Brief report on Hydrogeological conditions in and around West Ansari Nagar and Ayur Vigyan Nagar	NBCC, Delhi
140		A Brief report on Hydrogeological conditions in and around Hotel Suryaa, New Friends Colony, New Delhi	Hotel Suryaa, New Friends Colony
141		A Brief Technical report on Hydrogeological conditions around Wazirpur and Ashok Vihar area	DJB, Govt. of Delhi

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
142	**SUO Delhi**	A Technical recommendation report in connection with permission of borewells in North West Delhi	BDO, North West Delhi
143		A Technical recommendation report for new borewell construction in North West Delhi	BDO, North West Delhi
144		A Technical recommendation report for new borewell construction in North West Delhi	BDO, North West Delhi
145	**CR Nagpur**	Hydrogeological investigation for the possibilities of rain water harvesting in bro school & centre (gref centre), dighi camp, pune, maharashtra	Bro school & centre (gref centre), dighi camp, pune, maharashtra
146		Short term investigation report for the water supply bore well at village shedeshwar, umred tahsil, nagpur distirct	Village shedeshwar, umred tahsil, nagpur distirct, Maharashtra
147	**SWR Bengaluru**	Short Term Water Supply Investigation for selection of site for borewells	Bangalore University campus, Chikkballapura
148		Feasibility Study for Rain Water Harvesting and Artificial Recharge in Sri Sathya Sai Institute of Higher Medical Sciences, Whitefield, Bengaluru, Karnataka	Sri Sathya Sai Higher Meduical Hospital, White field,
149		Short Term Water Supply Investigation for selection of site for borewells	MEG Centre, Banswadi, Bangalore
150		Short Term Water Supply Investigation for selection of site for borewells	26ED, AirForce, Vimanapura, Bangalore
151		Short Term Water Supply Investigation for selection of site for borewells	
152		Short Term Water Supply Investigation for selection of site for borewells	Survey Of India Residential Campus, Koramangala, Bangalore
153		Short Term Water Supply Investigation for selection of site for borewells	Mysore University, Mysore
154		Short Term Water Supply Investigation for selection of site for borewells	
155		Short Term Water Supply Investigation for selection of site for borewells	Sri Sathya Sai Institute of Higher Medical Sciences, Whitefield, Bengaluru
156		Short Term Water Supply Investigation for selection of site for borewells	ASC centre (North), Military Garrison, Banaswadi, Bangalore
157	**SWR Bengaluru**	Feasibility report on RWH	Dhempe College of Arts and Science, Miramer, Panaji, Goa

S.No.	Region	Title of Short-Term Water Supply Investigation	Agency for which study was taken up
158		Feasibility report on RWH and Artificial Recharge Structures	Air Force, Bengaluru-Varthur
159	**WR Jaipur**	Feasibility Report on RWH & AR at Mil Stn. Jaipur (Piru Vihar)	
160	**NHR Dharamshala**	Hydrogeological investigation for water supply to prithi military station, averipatti, district kullu, himachal pradesh	Garrison Engineer, 863 Engr Wks Section

3.13 Reports in AAP 2024-25

Table 3.24: Region-wise list of Reports in AAP 2024-25

S.No.	Region	Title of the report published	Authors of the Report	Website link
1	**NER Guwahati**	Aquifer Management Plan Bamunimaidan Industrial Estates and Guwahati Oil Refinery, Kamrup Metro, Assam	Dr. Keisham Radhapyari, Scientist- D (Chemical),Wonjano Mozhui, Scientist -D (Hydrogeologist) ,Gopal Sahoo, Scientist-B (Chemical) ,Rinkumoni Barman, Assistant Chemist ,Dr. Snigdha Dutta, STA (Chemical)	https://cgwb.gov.in/cgwbpnm/search?type=2&cat_id=7&state_id=&district_id=&year_of_issue=&name_of_author=&keywords=&search=search
2		Aquifer Management Plan PARTS OF KAMRUP (METROPOLITAN) DISTRICT, ASSAM	Dr. D. J. Khound, Scientist- D (Hg) , Miss. Mophi Mili, Scientist-C (Hg), Shri Tiamenba Longshir, AHG Dr. K. Radhapyari, Scientist-D (Ch), Shri M. Vidyasagar, Scientist-C (GP)	https://cgwb.gov.in/cgwbpnm/search?type=2&cat_id=7&state_id=&district_id=&year_of_issue=&name_of_author=&keywords=&search=search
3		Aquifer Management Plan Itanagar Capital Region, Arunachal Pradesh	Dr D.J. Khound, Scientist-D (HG), Shri Rajat Gupta, Asst. Hydrogeologist, Shri M. Vidyasagar, Scientist-B (GP), Shri Rinku Mani Barman, Asst. Chemist	https://cgwb.gov.in/cgwbpnm/search?type=2&cat_id=7&state_id=&district_id=&year_of_issue=&name_of_author=&keywords=&search=search
4		Aquifer Management Plan Greater Shillong Area, Meghalaya	Ms. D Rabha, Scientist-D (HG) ,Mr. Shasinlo Kent, Scientist-C (HG) , Ms. Anenuo Pienyu, Scientist-C (HG),Dr. Snigdha Dutta, STA (Ch) CGWB NER Guwhati Sh. T. Raja Babu, Scientist-D (GP)	https://cgwb.gov.in/cgwbpnm/search?type=2&cat_id=7&state_id=&district_id=&year_of_issue=&name_of_author=&keywords=&search=search

S.No.	Region	Title of the report published	Authors of the Report	Website link
5	NER Guwahati	Dynamic Ground Water Resources of Arunachal Pradesh, 2024	Rajat Gupta, Asst. Hydrogeologist,Tiamenba Longshir, Asst. Hydrogeologist	https://cgwb.gov.in/cgwbpnm/
6		Dynamic Ground Water Resources of Assam, 2024	Ms Mophi Mili, Scientist C	https://cgwb.gov.in/cgwbpnm/
7		Dynamic Ground Water Resources of Meghalaya, 2024	Ms. D. Rabha, Scientist-D, Mr. Shasinlo Kent, Scientist-C, Ms. Rinku Rani Das, Scientist-C	https://cgwb.gov.in/cgwbpnm/
8		Dynamic Ground Water Resources of Manipur, 2024	Dr. Sorokhaibam Somarendro Singh, Scientist-D, Himangshu Kachari, Scientist-B	https://cgwb.gov.in/cgwbpnm/
9		Dynamic Ground Water Resources of Mizoram, 2024	Arijit Mitra, Scientist-C	https://cgwb.gov.in/cgwbpnm/
10		Dynamic Ground Water Resources of Nagaland, 2024	Vekhosa Kezo, Scientist-D, Anenuo Pienyu, Scientist-C	https://cgwb.gov.in/cgwbpnm/
11		Dynamic Ground Water Resources of Tripura, 2024	Dr. Raja Ram Purohit,Scientist- D, Smt. Ritu K. Oraon, Scientist-C	https://cgwb.gov.in/cgwbpnm/
12	UR Dehradun	Dynamic Ground Water Resources of Uttarakhand, 2024	Ms. Chandreyee De, Scientist C, Ms. Anjali Kushwaha, Scientist C	https://cgwb.gov.in/cgwbpnm/
13		Aquifer Management plan in part of Udham Singh Nagar and Nainital district. Priority type: Autoflow	Sh. Debojyoti Mondal, Scientist C , Sh. Mohd. Arshad, AHG	
14	NCR Bhopal	Dynamic Ground Water Resources Estimation of Madhya Pradesh-2024	Sh. Chittaranjan Biswal, Scientist-C, Sh. Pradip roy, Scientist-B, Ms. Saumya Siddharth, Assistant Geophysicist , Sh Josabanta Jani, Scientist-B	
15		Annual General Report-2023-24, CGWB, NCR, Bhopal	Sh. Chittaranjan Biswal, Scientist-C, Ms Saumya Chaudhary, Scientist-B, Ms. Saumya Siddharth, Assistant Geophysicist , Sh J. Sreemannarayana, STA(HG)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
16	NCR Bhopal	Springshed Management Study (NAQUIM 2.0), Dindori-Anuppur District, Madhya Pradesh	Ms. Lata Udsaiya, Scientist-C, Sh Chittaranjan Biswal, Scientist-C, Ms Vishnavi Parihar, Scientist-B, Ms Saumya Siddharth, Assistant Geophysicist, Ms Adib Khan, STA (Chemist)	
17		National aquifer Mapping and management Plan Report of Water stressed Alot Block, Ratlam district, Madhya Pradesh	Dr. Rakesh Singh, Scientist-D, Sh. Kodali L. Pradeep, Assistant Hydrogeologist, Sh J. Sreemannarayana, STA(HG), Sh Pradip Roy, Scientist-B	
18		National Aquifer Mapping and Management Plan of Bhopal Urban Agglomerate, Madhya Pradesh	Mrs. Paisnee Patel, Sc. C, Sh. Kamlesh Ojha, Sc. C, Ms. Saumya Chaudhary, Sc. B, Sh. Tej Singh, Assistant Chemist and Ms. Saumya Siddhartha, AGP	
19		NAQUIM 2.0 Study in Ujjain Industrial Cluster, Madhya Pradesh	Mrs. Anakha Ajai, Sc. C, Sh. Sumanta Kumar Mohanta, Sc. C, Sh. Pradip Roy, Sc. B and Sh. Jitendra Kumar, STA (Chemical)	
20		Annual GW Quality Report (2024-25)	Dr. V. K. Kulshrestha, Sc. C, Sh. Jitendra Kumar, STA, Ms. Adiba Khan, STA	
21		Ground Water Year Book (2024-25)	Ms. Lata Udsaiya, Sc. C, Ms. Vaishnavi Parihar, Sc. B, Sh. Pradip Roy, Sc. B	
22		Annual Geophysical Report (2024-25)	Sh. Pradip Roy, Sc. B, Ms. Saumya Siddhartha, AGP	
23		Annual Exploration Report (2024-25)	Sh.Chittaranjan Biswal, Sc-C	
24		Aquifer Management Plan Durg Block, Durg District, Chhattisgarh 2023-24	Smt. Prachi Gupta, Sc-C Ms. Gurpreet Kour, Sc – B Sh. Rakesh Dewangan, Sc-C Sh. Nageswara Rao Elisela, AGP	https://cgwb.gov.in/cgwbpnm/publication-detail/1566

S.No.	Region	Title of the report published	Authors of the Report	Website link
25	NCCR Raipur	Aquifer Management Plan Raipur City, Raipur, Chhattisgarh 2023-24	Ms. Priyanka B. Sonbarse, Sc - D Sh. Sarboday Barik, AHG Dr. Ajay Ku. Sinha, Sc-D Dr. Rajnikant Sharma, Sc -	https://cgwb.gov.in/cgwbpnm/publication-detail/1584
26		Aquifer Mapping and Management Plan of Korba Industrial Area, Korba District, Chattisgarh 2023-24	Sh. Uddeshya Kumar, Sc-C Sh. Subham Dash, AHG Sh. Rakesh Dewangan, Sc-C Sh. Suman Bharti, Sc-B	https://cgwb.gov.in/cgwbpnm/publication-detail/1574
27		Aquifer Mapping and Management Plan of Gurur Block, Balod District, Chattisgarh 2023-24	Sh. Suvam Prakash Dash, Sh. Sarboday Barik, Sh. Suman Bharti, Dr. Rajnikant Sharma	https://cgwb.gov.in/cgwbpnm/publication-detail/1573
28		Aquifer Management Plan Dhamtari Block, Dhamtari District, Chhattisgarh 2023-24	Sh. Mukesh Anand, Sc-B Ms. Sweta Mohanty, AHG Dr. Ajay Ku. Sinha, Sc-D Dr. Anita Bind, STA	https://cgwb.gov.in/cgwbpnm/publication-detail/1564
29		Ground Water Year Book 2023-2024	Smt. Priyanka Sonbarse, Gurpreet Kour	
30		Annual General Report 2023-2024	Smt Priyanka Sonbarse, Sh. Jasobanta Jani	
31		Annual Exploration Report 2023-2024	Sh. Uddeshya Kumar, Sh Suman Bharti	-
32		Dynamic Ground Water Resources Estimation of Chhattisgarh - 2024	Sh. Uddeshya Kumar, Sc-C Sh. B. Abhishek, Sc-C Ms. Anusandhya Pradhan, Sci-B Ms. Gurpreet Kour, Sc – B Sh. Suman Bharti, Sc-B	-
33	NCCR Raipur	Ground Water Quality Yearbook of Chhattisgarh state	Sh. Rakesh Dewangan, Sc-C Dr. Rajnikant Sharma, Sc –C Dr. Anita Bind, STA	-

S.No.	Region	Title of the report published	Authors of the Report	Website link
34	**NHR Dharamshala**	Dynamic Ground Water Resources Estimation of Himachal Pradesh -2024	Sh. Manohar Kumar, Sc-B & H.O.O. Sh. Sanjay Pandey, Sc-B Sh. Prasant Kumar Singh, STA (Hg)	-
35		Ground Water Yearbook of Himachal Pradesh 2023-2024	Sh. Prasant Kumar Singh, STA (Hg)	-
36	**NWHR Jammu**	Aquifer Management Plan Srinagar Urban Agglomerate, Jammu & Kashmir	Anurag Khanna – Regional Director M. L. Angurala, Sc-'D, Rayees Ahmad Pir , Sc- B	
37		Aquifer Management Plan of Ramban, Batote, Gool, Sangaldan Block, Ramban District Union Territory of Jammu and Kashmir	Anurag Khanna – Regional Director M. L. Angurala, Sc-'D, K.S. Rawat-Sc-c,Nikhil Kumar - Assistant Hydrogeologist	
38		Aquifer Management Plan in Parts of Kargil District, Union Territory of Ladakh	Anurag Khanna – Regional Director M. L. Angurala, Sc-'D, Naresh Singh Barti - Assistant Hydrogeologist, Nimay Kashyap-Assistant Chemist, Kush Kumar-STA (Hg)	
39		Aquifer Management Plan in Banihal Block, Ramban District Union Territory Of Jammu And Kashmir	Anurag Khanna – Regional Director M. L. Angurala, Sc-'D, K.S. Rawat-Sc-c, Mohd Abid Khan - Assistant Hydrogeologist, Kush Kumar- STA (Hg)	
40	**NR Lucknow**	Dynamic Ground Water Resources of Uttar Pradesh 2024	Sh. S.K Swaroop, Sc-D Dr. R.K. Prasad, Sc-D Sh. Ebadur Rahman, Sc-C Dr. Fakhre Alam, Sc-B Miss Aditi Singh, Sc-B	https://cgwb.gov.in/cgwbpnm/publication-detail/1534

S.No.	Region	Title of the report published	Authors of the Report	Website link
41	NR Lucknow	Aquifer Mapping (NAQUIM 2.0) Report of Kanpur Urban Agglomerate, Uttar Pradesh	Shri Jagdamba Prasad, Sc-D Shri Vipin Mishra, AHG Shri Anirudh Singh, AGP Smt. Kiran Singh, Sc-D (Chem) Dr. R.K.Prasad,Sc-D(Hm)	https://cgwb.gov.in/cgwbpnm/publication-detail/1571
42		Aquifer Management Plan-Bansi, Jogiya, Barhani and Birdpur Blocks, District, Siddharth Nagar, Uttar Pradesh, Priority Type: Poor Water Quality	Shri Edadur Rahman, Sc-C Shri Umareddy Dereddy, Sc-B Dr. Supriya Singh,Sc-B Shri Amitosh Chandra, Sc-B Smt. Renu Mehra, Sc-B Shri Shivam Prajapat,AHG	https://cgwb.gov.in/cgwbpnm/publication-detail/1570
43		Aquifer Mapping (NAQUIM 2.0) Report of Baghpat District, Uttar Pradesh Priority Type: Water Stressed Area	Shri S.M.Hossain, Sc-D Shri Rahul Kuamr, AHG Shri Vidya Bhooshan,AHG Shri Karam Singh, Sc-C(Chem) Shri Anirudh Singh, AGP	https://cgwb.gov.in/cgwbpnm/publication-detail/1579
44		Aquifer Mapping (NAQUIM 2.0) Report of Unnao Industrial Area, Uttar Pradesh	Shri P K Singh,Sc-D Smt. Richa Gautam,Sc-B Shri Amitosh Chandra,Sc-B Dr. Madhavi Rajak, Assistant Chemist	https://cgwb.gov.in/cgwbpnm/publication-detail/1571
45		Groundwater Quality in Shallow Aquifers of Uttar Pradesh (Based on NHS Data of 2023-2024	Smt. Kiran Singh, Sc-D Shri Karam Singh, Sc-C Dr. Supriya Singh, Sc-B Dr. Madhavi Rajak, Assistant Chemist	
46		Groundwater Year Book of Uttar Pradesh (2023-2024)	Dr. R.K.Prasad, Sc-D(Hm) Shri Ankur, STA(Hg)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
47	CR Nagpur	NAQUIM 2.0:Aquifer Management Plan of Jath Block, Sangali District, Maharashtra	Pandith Madhunure ,Sc-D Sh. Singaren Sandeep Purty,Sc-D Ms. Pournima Barahate, Sc-D Ms. Sayelli U. Tembhurne,Sc-C Sh. Amarnath , AHG Sh. Srinivas Pawar, AHG Sh. Anurag TaK , STA Sh. Ujjawal Kumar, STA Sh. G. M. Londhe , Sc-D Sh. S.M. Subasridharan, STA Dr. Devsharan Verma , Sc-C Dr. Sailee Bhange,, STA	
48		NAQUIM 2.0:Aquifer Management Plan of Katol Block, Nagpur District, Maharashtra,	Nelofar,Sc- D Nirmal Kumar Nanda,Sc-C Ashwin Kumar Atey,Sc-C Dr. P Kiran Kumar, Sc-C Shubhangi Lohani, Sc-B	
49		NAQUIM 2.0:Aquifer Management Plan of Saoner Block, Nagpur District, Maharashtra	Smt. Nelofar, Sc. D, Smt. Bhagyashree Sahoo, Sc. B, Sh. Prakash Chandra Maharana, AHg., Sh. V. Venktesham, STA, Dr. P. Kiran Kumar, Sc. C and Sh. Modi Durgaprasad, Sc. C, Hydrologist	
50		NAQUIM 2.0:Geogenic Fluoride Contamination of Groundwater in Warora Block of Chandrapur District Maharashtra	Dr. Devsharan Verma, Sc-D (Chemist) Nirmal Kumar Nanda, Sc-C, Dr. Sailee Bhange, STA, Anurag Tak, STA, Vishal Meena,STA	
51		Ground Water Year book	Sh. Modi Durgaprasad, Sc-C (H),Dr. P.Kiran Kumar, Sc-C (HM) Sh. Amarnath, AHG Ms. Shubhangi Lohani, Sc-B (Hg) Sh. Ashwin Atey, Sc-C (Hg) Sh. Amarnath, AHG Sh. Venkatesham V., STA (Hg)	
52		Annual Report 2023-24	Dr. Pandith Madhnure,Sc-D Nelofar, Sc-D Nirmal Kumar Nanda, Sc-C Ashwin Kumar Atey, Sc-C S. M. Subasridharan, STA (GP)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
53	CR Nagpur	Ground Water Quality Yearbook Maharashtra	Dr. Devsharan Verma, Sc-D, Dr. Sailee Bhange Shri Vishal Meena, STA (Chem) and Rakesh Dewangan, Sc-D	
54		Annual Report 2022-23	Dr. Pandith Madhnure,Sc-D,Ms. Nelofar, Scientist-C Sh. S. M. Subasridharan, STA (GP) Sh. Sandesh Bhange, YP	
55		Report on Dynamic Ground Water Resources of Maharashtra 2023	Sh. N. Varadaraj, Regional Director, CGWB; CR, Nagpur, Dr. Pandith Madhnure, Sc-D, Dr. J. Davithuraj, Sc-D, ,Sh. Modi Durgaprasad, Sc--C,Dr. Pullabhatla Kiran Kumar, Sc-C, Ms. Bhagyashree Sahoo, Sc-B	
56	SR Hyderabad	NAQUIM 2.0 : Report on Aquifer Mapping Plan of Vijayawada, Guntur & Amaravathi Urban Agglomerates of NTR, Krishna and Guntur Districts, Andhra Pradesh	Principal Authors 1. Dr. M. Sudheer Kumar Scientist – D (Hydrogeology) : Team Leader 2. Dr. S. Srinivasa Vittala Scientist – C (Hydrogeology) : Expert 1 (Hydrogeology) 3. Dr. D. Ananth Rao, Asst. Hydrogeologist : Expert 2 (Hydrogeology) 4. Sh. P. Raghavendra Scientist – B (Geophysics) : Expert (Geophysics) 5. Sh. Y. Satya Kumar Scientist – C (Chemist) : Expert (Hydrochemistry)	
57		NAQUIM 2.0 : Report on Aquifer Management Plan of Visakhapatnam and Anakapalli Urban Agglomerations, Visakhapatnam and Anakapalli Districts, Andhra Pradesh	Shri. Lakshmi Narayana Damodara, Scientist-C Shri. Md Sarif Khan, Scientist-C, Smt. Nilima Patra, Assistant Hydrogeologist, Shri. T. Venkata Giri, Scientist-B (GP), Shri. S.K Ratha, Scientist-B (Chemist)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
58	SR Hyderabad	NAQUIM 2.0 : Report on AQUIFER MANAGEMENT PLAN OF MAHESHWARAM, SHAMSHABAD AND KOTHUR BLOCKS, RANGAREDDY DISTRICT, TELANGANA STATE	Ms. Rani V. R., Scientist D (Hg) Sh. Atharva Pawar, Scientist B (Hg) Sh. Upendra Dhonde, Scientist D Ms. Akoju Ramadevi, AGP Sh. T. Venkatagiri, Sc- B Ms. Swati Dhenkula, Asst. Chemist Dr. P. Kiran Kumar, Sc- C	
59		Ground Water Year Book 2023-2024 of Telangana State	Ms. Rani V.R, Scientist-D (Hg) Sh. Bijay Ketan Mohanta, Scientist-C (Hg) Sh. M S Goutham, Scientist-C (Hg) Sh. P. Kiran Kumar, Scientist-C (HM) Dr. S.S. Vittala, Scientist-C (Hg)	
60		Ground Water Year Book 2023-2024 of Andhra Pradesh State	Ms. Rani V.R, Scientist-D (Hg) Ms. Monika, Scientist-C (Hg) Ms. Resma S. Pillai, Scientist-C (Hg) Sh. P. Kiran Kumar, Scientist-C (HM)	
61		Dynamic Ground Water Resources of Telangana 2024	1. CGWB,SR,Hyderabad 2. Telangana State Ground water Department	
62		Dynamic Ground Water Resources of Andhra Pradesh 2024	1.CGWB,SR,Hyderabad & SUO,Visakhapatnam 2. State Ground water and Water Audit Department,Andhra Pradesh.	
63		Ground Water Quality Yearbook 2023-24 of Telangana State	Sh.Y.Satya Kumar (Scientist-C) Chemist, Sh. Sushant kumar Ratha (Scientist-B) Chemist, Ms. Swathi Dhenkula, Assistant Chemist	
64		Ground Water Quality Yearbook 2023-24 of Andhra Pradesh State	Sh.Y.Satya Kumar (Scientist-C) Chemist, Sh. Sushant kumar Ratha (Scientist-B) Chemist, Ms. Swathi Dhenkula, Assistant Chemist	

S.No.	Region	Title of the report published	Authors of the Report	Website link
65	SWR Bengaluru	Naquim 2.0:Aquifer Mapping and Management Plan of Water Stressed Areas of Doddaballapura Taluk, Bangalore Rural District, Karnataka	Anisha K, Scientist-D (HG); Hemalatha V, Assistant Hydrogeologist; Abdul Rajik Khan, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Yogesh M R, STA (Chem); Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP)	
66		Naquim 2.0:Impact Assessment of Tank Filling Scheme by Surface Water in Channapatna Taluk, Ramanagara District, Karnataka	Sreehari Sarangan M S, Scientist-D (HG); Dr Baby Swetha S, Sc-B (HG); Damera Anjaneyulu, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Jagannath Nahak, Sc-B (Chem); Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP); N.Narsimha Naidu, Scientist B (GP)	
67		Naquim 2.0:Aquifer Mapping and Management Plan for Areas with Poor Groundwater Availability in Gaondongrem And Cotigao Village Panchayat, Canacona Taluk, South Goa District, Goa State	Sangita P. Bhattacharjee, Scientist-D (HG); Sushant S. Navarat, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Dr. Punith Raj T S, Assistant Chemist; Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP)	
68		Naquim 2.0:Groundwater Quality Issues and Management Plan in Navalgund Taluk, Dharwad District, Karnataka State	Sangita P. Bhattacharjee, Scientist-D (HG); Tejas Y Mankikar, Sc-C (HG); Sushant S. Navarat, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Sivaswamy R, Assistant Chemist; Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP); Narasimha Naidu N, Scientist B (GP)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
69	**SWR Bengaluru**	Groundwater Year book of Karnataka: 2023-24	H.P Jayaprakash, Sc-D; Anisha.K, Sc-D; Dr.Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
70		Groundwater Year book of Goa: 2023-24	H.P Jayaprakash, Sc-D; Anisha.K, Sc-D; Dr.Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
71		Groundwater Quality in Shallow Aquifer of Karnataka State	Sh. Jagannath Nahak, Sc-B; Dr. Punith Raj. T.S, Assistant Chemist; Sivaswamy R, Assistant Chemist; Yogesh. M.R, STA (Chemical); Lalitha. B.H, STA (Chemical)	
72		Groundwater Quality in Shallow Aquifer of Goa State	Sh. Jagannath Nahak, Sc-B; Dr. Punith Raj. T.S, Assistant Chemist; Sivaswamy R, Assistant Chemist; Yogesh. M.R, STA (Chemical); Lalitha. B.H, STA (Chemical)	
73		Dynamic Ground water Resources of Karnataka-2024	1. CGWB, SWR, Bangalore 2. Ground Water Directorate Minor Irrigation & Ground Water Development Department, Govt. of Karnataka	
74		Dynamic Ground water Resources of Goa-2024	1.CGWB, SWR, Bangalore & SUO, Belgaum 2. Water Resource Department, Govt of Goa	
75		Impact Assessment Study of Multi Arch Check Dams, Gauribidanur Taluk, Chikballapur District (MACD Recommended under NAQUIM Study)	Rahul R Shende, Sc-D; Carolin Louis, Sc-C; Dr.Baby Swetha S, Sc-B	

S.No.	Region	Title of the report published	Authors of the Report	Website link
76	**SWR Bengaluru**	Demarcation of Groundwater Potential in the Coastal Tract with Special Emphasis on Sea Water Intrusion in Candolim, Calangute and Anjuna Villages in Bardez Taluk, Goa State	Sangita P Bhattacharjee, Sc-D Veena R Achutha, Sc-D; Mayank Dixit, Sc-B; Sushanth S Navarat, AHG	
77		Water Supply Investigation for feasibility of drilling Borewell at Belgaum Military station for Military Engineering Services (MES) camp, Belagavi, Karnataka	Sushanth S Navarat, AHG Sangita P Bhattacharjee, Sc-D; Veena R. Achutha, Scientist-D & N.Narsimha Naidu, Sc-B	
78		Report on Periodical Measurements of Groundwater Monitoring Stations in Karnataka State - November 2023	H.P Jayaprakash, Sc-D Lubna Kouser, Sc-C; Deepa Gupta, Sc-B; Hemalatha V, AHG	
79		Report on Periodical Measurements of Groundwater Monitoring Stations in Karnataka State - January 2024	H.P Jayaprakash, Sc-D Lubna Kouser, Sc-C; Deepa Gupta, Sc-B; Hemalatha V, AHG	

S.No.	Region	Title of the report published	Authors of the Report	Website link
80	SWR Bengaluru	Report on Periodical Measurements of Groundwater Monitoring Stations in Goa State - November 2023	H.P Jayaprakash, Sc-D Lubna Kouser, Sc-C; Deepa Gupta, Sc-B; Hemalatha V, AHG	
81		Report on Periodical Measurements of Groundwater Monitoring Stations in Goa State - January 2024	H.P Jayaprakash, Sc-D Lubna Kouser, Sc-C; Deepa Gupta, Sc-B; Hemalatha V, AHG	
82		Baikampady Industrial Cluster Study, Dakshina Kannada District, Karnataka, Under Naquim 2.0	Sakthivel A, Sc-D Deepa Gupta, Sc-B, D Anjaneyulu, AHG	
83		Tank Filling Scheme by Treated Water, Chikkaballapura Taluk, Chikkaballapura District, Karnataka	Anisha, Sc-D Anamika Aich, Sc-B; Hemalatha V, AHG; Abdul Rajik Khan, AHG	
84		Coastal Salinity Study in Bardez and Tiswadi Taluks, North Goa District, Goa State	Sangita P Bhattacharjee, Sc-D Sushanth S Navarat, AHG	
85		Aquifer Mapping and Management Plan of Water Stressed Areas of Doddaballapura Taluk, Bangalore Rural District, Karnataka	Anisha K, Scientist-D (HG); Hemalatha V, Assistant Hydrogeologist; Abdul Rajik Khan, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Yogesh M R, STA (Chem); Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
86	SWR Bengaluru	Impact Assessment of Tank Filling Scheme by Surface Water in Channapatna Taluk, Ramanagara District, Karnataka	Sreehari Sarangan M S, Scientist-D (HG); Dr Baby Swetha S, Sc-B (HG); Damera Anjaneyulu, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Jagannath Nahak, Sc-B (Chem); Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP); N.Narsimha Naidu, Scientist B (GP)	
87		Aquifer Mapping and Management Plan for Areas with Poor Groundwater Availability in Gaondongrem And Cotigao Village Panchayat, Canacona Taluk, South Goa District, Goa State	Sangita P. Bhattacharjee, Scientist-D (HG); Sushant S. Navarat, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Dr. Punith Raj T S, Assistant Chemist; Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP)	
88		Groundwater Quality Issues and Management Plan in Navalgund Taluk, Dharwad District, Karnataka State	Sangita P. Bhattacharjee, Scientist-D (HG); Tejas Y Mankikar, Sc-C (HG); Sushant S. Navarat, Assistant Hydrogeologist; Dr Lubna Kouser, Scientist-C (Hydrologist); Sivaswamy R, Assistant Chemist; Veena R Achuta, Scientist-D (GP); Mayank Dixit, Scientist B (GP); Narasimha Naidu N, Scientist B (GP)	
89		Ground Water Level Bulletin May 2024- Karnataka State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
90		Ground Water Level Bulletin May 2024- Goa State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
91		Ground Water Level Bulletin August 2024- Karnataka State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	

S.No.	Region	Title of the report published	Authors of the Report	Website link
92	**SWR Bengaluru**	Ground Water Level Bulletin August 2024- Goa State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
93		Groundwater Year book of Karnataka: 2023-24	H.P Jayaprakash, Sc-D; Anisha.K, Sc-D; Dr.Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
94		Groundwater Year book of Goa: 2023-24	H.P Jayaprakash, Sc-D; Anisha.K, Sc-D; Dr.Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
95		Annual General Report - 2022_23		
96		Groundwater Quality in Shallow Aquifer of Karnataka State	Sh. Jagannath Nahak, Sc-B; Dr. Punith Raj. T.S, Assistant Chemist; Sivaswamy R, Assistant Chemist; Yogesh. M.R, STA (Chemical); Lalitha. B.H, STA (Chemical)	
97		Groundwater Quality in Shallow Aquifer of Goa State	Sh. Jagannath Nahak, Sc-B; Dr. Punith Raj. T.S, Assistant Chemist; Sivaswamy R, Assistant Chemist; Yogesh. M.R, STA (Chemical); Lalitha. B.H, STA (Chemical)	
98		Annual General Report - 2021_22		
99		Ground Water Level Bulletin November 2024- Goa State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
100		Ground Water Level Bulletin November 2024- Karnataka State	H P Jayaprakash , Sc-D Anisha, Sc-D; Lubna Kouser, Sc-C; Anamika Aich, Sc-C; Hemalatha V, AHG	
101		Feasibility Study for Rain Water Harvesting and Artificial Recharge in Sri Sathya Sai Institute of Higher Medical Sciences, Whitefield, Bengaluru, Karnataka	H P Jayaprakash, Sc-D A.Sakthivel, Sc-D; Veena Achutha, Sc-D; N.Narsimha Naidu, Sc-B; N.C.Sahoo, AHG	

S.No.	Region	Title of the report published	Authors of the Report	Website link
102	**SWR Bengaluru**	Feasibility Investigation for thepPossibilities of Rain-Water Harvesting Structures in Dhempe College of Arts and Science, Panaji, North Goa District, Goa	Sangita P Bhattacharjee, Sc-D Sushanth S Navarat, AHG	
103		Ground Water Quality in and around Hebbal-Nagawara (HN) & Kormangala-Chellaghatta (KC) Valley – Tank Filling Scheme of Karnataka State Government	Sh. Jagannath Nahak, Sc-B Dr. Punith Raj. T.S, Assistant Chemist; Sh. Sivaswamy, Assistant Chemist; Sh. Yogesh. M.R, STA (Chemical); Smt. Lalitha. B.H, STA (Chemical)	
104		Impact Assessment Study of Multi Arch Check Dams, Gauribidanur Taluk, Chikballapur District (MACD Recommended under NAQUIM Study)	Rahul R Shende, Sc-D; Carolin Louis, Sc-C; Dr.Baby Swetha S, Sc-B	
105		Source Sustainability Interventions in Chikkaballapura District, Karnataka State	All Officers	

S.No.	Region	Title of the report published	Authors of the Report	Website link
106	**SWR Bengaluru**	Water Supply Investigation for feasibility of Drilling Borewell at ASC Centre (North) - 1 ATC at Banaswadi Military Garrison, Bangalore Urban District, Karnataka	Abdul Rajik Khan, AHG N.Narsimha Naidu, Sc-B	
107		Dynamic Ground water Resources of Karnataka-2024	1. CGWB, SWR, Bangalore 2. Ground Water Directorate Minor Irrigation & Ground Water Development Department, Govt. of Karnataka	
108		Dynamic Ground water Resources of Goa-2024	1.CGWB, SWR, Bangalore & SUO, Belgaum 2. Water Resource Department, Govt of Goa	
109	**SER Bhubaneswar**	Aquifer Management Plan Bologarh Block (Khordha District) and Nayagarh Block (Nayagarh District), Odisha Priority Type: Water Stressed Areas	Sumita Sarkar, Scientist-D (Team Lead) Purba Bera, Scientist-C, Sanjay Naik, Scientist-C, Rajeshbabau Annavarapu, Asst GP, Kolipaka Karnakar, Asst Chemist	https://cgwb.gov.in/cgwbpnm/publication-detail/1575
110		Aquifer Management Plan Parts of Jharsuguda and Sambalpur Districts, Odisha Priority Type: Industrial Clusters	Shri Tarun Mishra, Shri Ashutosh Choudhary, Shri Satyam Shukla, Shri B N Dehury, Shri Rajesh Babu, Shri Tatikonda Suresh, Shri K K Pahi	https://cgwb.gov.in/cgwbpnm/publication-detail/1580
111		Aquifer Management Plan Sambalpur Urban Agglomerate, Odisha Priority Type: Urban Agglomerate	Sh. Rajeev Kumar Tripathy, Sh. Raj Kishor Mohanty, Sh. Rajesh Babu Annavarapu, Sh. Biranchi Narayan Dehury, Sh. Sudhanshu Kumar Mohanty	https://cgwb.gov.in/cgwbpnm/publication-detail/1581

S.No.	Region	Title of the report published	Authors of the Report	Website link
112	**SER Bhubaneswar**	Aquifer Management Plan Balasore Sadar and Remuna Blocks, Balasore District, Odisha Priority Type: Coastal Study Area	Sh. Prahallad Das, Sh. Binay Kumar Behera, Sh. Karnakar Kollipaka, Sh. Rajesh Babu Annavarapu, Sh. Sambit Kumar Samantaray	https://cgwb.gov.in/cgwbpnm/publication-detail/1582
113		Groundwater Quality Year Book of the Odisha State 2023-2024	Sinosh P.K, Scientist-B B.N. Dehury, Asst Chemist Kolipaka Karnakar, Asst Chemist	
114		Ground Water Year Book - 2024-25	Mausumi Sahoo, Scientist-D Purba Bera, Scientist-C Bibhuti Bhusan Sahoo, Scientist-C	
115		Dynamic Ground Water Resource Estimation-Odisha, 2024	Mausumi Sahoo, Scientist-D Purba Bera, Scientist-C Bibhuti Bhusan Sahoo, Scientist-C	
116		Ground Water Exploration Report of the State of Odisha	Prahallad Das, Scientist-D Sumita Sarkar, Scientist-D Sambit Samantaray, Scientist-D	
117		Geophysical Report of the State of Odisha	Rajesh Babu Annavarapu, Asst GP Bindu Singh, Scientist -B	
118	**ER Kolkata**	Groundwater Year book of West Bengal, Sikkim & Andaman & Nicobar Islands 2023-24	Chirashree Mohanty, Scientist-D (HG) Dr Shaista Khan, Scientist-C (HG) Dr. Nilamoni Barman, Scientist-C (HM)	
119		Groundwater Quality Year book of West Bengal 2023-24	Dr Suparna Datta, ACH Prasanth Yentapalli, ACH Partha Pratim Mondal, STA (CH) Dr Neeru Tiwari, Scientist-B(CH) Atalanta Narayan Chowdhury, Scientist-B (CH)	
120		Groundwater Quality Year book of Andaman & Nicobar Islands 2023-24	Atalanta Narayan Chowdhury, Scientist-B (CH) Dr Suparna Datta, ACH Prasanth Yentapalli, ACH Partha Pratim Mondal, STA (CH) Dr Neeru Tiwari, Scientist-B	

S.No.	Region	Title of the report published	Authors of the Report	Website link
121	ER Kolkata	Dynamic Ground water Resources of West Bengal-2024	Dr Indranil Roy, Scientist-D (HG) Dr Shaista Khan, Scientist-C (HG) Dr. Nilamoni Barman, Scientist-C (HM)	https://cgwb.gov.in/cgwbpnm/publication-detail/1529
122		Dynamic Ground water Resources of Sikkim-2024	Dr Indranil Roy, Scientist-D(HG) Sandip Bhowal, Scientist-B (HG) Dr. Nilamoni Barman, Scientist-C (HM)	
123		Dynamic Ground water Resources of Andaman & Nicobar-2024	Dr Indranil Roy, Scientist-D (Hg) Dr. Nilamoni Barman, Scientist-C (HM)	
124		Ground Water Level Bulletin of West Bengal, Sikkim & Andaman & Nicobar Island-April 2024	Dr Shaista Khan, Scientist-C (HG)	
125		Ground Water Level Bulletin of West Bengal, Sikkim -August 2024	Dr Shaista Khan, Scientist-C (HG)	
126		Ground Water Level Bulletin of West Bengal, Sikkim & Andaman & Nicobar Island-Novembe 2024	Dr Shaista Khan, Scientist-C (HG)	
127		Ground Water Level Bulletin of West Bengal, Sikkim - January2025	Dr Shaista Khan, Scientist-C (HG)	
128		Ground Water Quality Bulletin - Pre-Monsoon 2024	Dr Suparna Datta, ACH Prasanth Yentapalli, ACH Partha Pratim Mondal, STA (CH) Dr Neeru Tiwari, Scientist-B (CH) Atalanta Narayan Chowdhury, Scientist-B	

S.No.	Region	Title of the report published	Authors of the Report	Website link
129	ER Kolkata	Ground Water Quality Bulletin - Post-Monsoon 2024	Dr Suparna Datta, ACH Prasanth Yentapalli, ACH Partha Pratim Mondal, STA (CH) Dr Neeru Tiwari, Scientist-B (CH) Atalanta Narayan Chowdhury, Scientist-B (CH)	
130		Aquifer Management Plan of 07 Northern Blocks of Purulia District, West Bengal.Priority Type: Water stressed areas Areas with poor ground water quality (NAQUIM 2.0)	Dr. Indranil Roy, Scientist- D (HG) Naseema Jamal, Scientist- C (HG) Bibhuti Bhusan Sahu, Scientist- C (HG) Rajesh Kumar Sahoo (AHG) Tapas Kumar Sahoo (AHG) Dr. Nilamoni Barman, Scientist- C (HM) Dr. K. C. Mondal, Scientist- D (GP) Sujit Sarkar, Scientist- D (GP) Dr. Subrata Das, Scientist- C (GP)	https://cgwb.gov.in/cgwbpnm/publication-detail/1568
131		Aquifer Management Plan of Haldia Industrial Area, Purba Medinipur, West Bengal (NAQUIM 2.0)	Chirashree Mohanty, Scientist-D (HG) Dr Shaista Khan, Scientist-C (HG) Dr. Nilamoni Barman, Scientist-C (HM) Sujit Sarkar, Scientist- D (GP) Dr. Subrata Das, Scientist- C (GP) Dr Suparna Datta, ACH	https://cgwb.gov.in/cgwbpnm/publication-detail/1630
132		Aquifer Management Plan of Gangtok Municipal Corporation, Sikkim (NAQUIM 2.0)	Dr. Indranil Roy, Scientist- D (HG) Arapan Kumar Banerjee,Scientist- C (HG) Sujit Sarkar, Scientist- D (GP) Dr. Nilamoni Barman, Scientist-C (HM) Atalanta Narayan Chowdhury, ACH	
133		Port Blair Urban Area, South Andaman District,(NAQUIM 2.0)	Debasish Bagchi, Scientist-D (HG) Sandip Bhowal, ACH Awadhesh Kumar, ACH Dr. Nilamoni Barman, Scientist- C (HM) Sujit Sarkar, Scientist- D (GP) Dr. K. C. Mondal, Scientist- D (GP) Dr Suparna Datta, ACH Prasanth Yentapalli, ACH	

S.No.	Region	Title of the report published	Authors of the Report	Website link
134	**KR Thiruvananthapuram**	Aquifer Management Plan of Amini and Minicoy Islands, UT of Lakshadweep	Roopesh G Krishnan,Scientist D, Dr VST Gopinath, Scientist D & Dr Aneesh Kumar N, Scientist B	
135		Aquifer Management Plan of Thiruvananthapuram Municipal Corporation, Kerala	Mini Chandran, Scientist D, Saritha S, Scientist D, Bindhu J Viju, Scientist D & Dr VST Gopinath, Scientist D	
136		Aquifer Management Plan of Kanjikode Industrial Area, Palakkad District, Kerala	Sreenath G, Scientist D, Arun Kumar AV, Scientist B, Dr Pankaj Bakshe, Scientist B &Dr VST Gopinath, Scientist D	
137		Dynamic Ground water resources of Kerala,2024	Roopesh G Krishnan,Scientist D, Rakhi U.R., Scientist D,Anu V., Scientist C, Dr.Aneesh Kumar V, Scientist C, Arun Kumar AV, Scientist B	
138		Dynamic Ground water resources of UT of Lakshadweep,2024	Sreenath G, Scientist D,Arun Kumar AV, Scientist B	
139		Ground water Quality of Shallow Aquifer in Kerala, 2024	Bindhu J Viju, Scientist D,Dr.Pankaj Bakshe, Scientist B,Dr N Aneesh Kumar, Scientist B	
140		Ground Water Yearbook of Kerala	Rakhi U R, Scientist D, Aneesh Kumar V., Scientist C, Anu V., Scientist C, Sabna T, Scientist C	
141	**SUO Delhi**	Dynamic Ground Water Resources & Ground Water Quality of Delhi, 2024	Sh. V. Praveen Kumar, AHG	https://cgwb.gov.in/cgwbpnm/
142		Aquifer Management Plan,Bhalaswa Landfill Site, N.C.T Delhi	Sh. S.K. Mohiddin Regional Director,Sh. Vandlabu Praveen Kumar Assistant Hydrogeologist (AHG) and Smt. Mamta , Senior Technical Assistant (GP)	https://drive.google.com/file/d/11GApDH0chVHR2jZfUlvgJ9lCtW3wr2RV/view?usp=drive_link

S.No.	Region	Title of the report published	Authors of the Report	Website link
143	SUO Delhi	Aquifer Management Plan Okhla, Landfill Site, N.C.T Delhi	Sh. S.K. Mohiddin Regional Director,Sh. Gyanendra Rai ,Senior Technical Assistant (Hydrogeology), Smt. Mamta , Senior Technical Assistant (GP) and Sh. Chandan Gupta	https://drive.google.com/file/d/1CW_rhzgUP2MWSaRsGfdKLOeDpxq0ojAN/view?usp=drive_link
144		Ground water Quality of Shallow Aquifers of NCT of Delhi (2023-2024)	Smt. Kriti Mishra, Sc-C, Sh Madhukar Singh, Sc-C, Sh. V. Praveen Kumar, AHG, Sh. S. Ashok Kumar, STA-HG	https://cgwb.gov.in/cgwbpnm/publication-detail/1265
145		Ground Water Year Book, NCT Delhi (2023-24)	Smt Prachi Gupta, Sc-D, Smt. Kriti Mishra, Sc-C, Sh. V. Praveen Kumar, AHG, Sh. S. Ashok Kumar, STA-HG, Miss. Chandini Chandran, YP	
146		Report on Hydrogeological and Geophysical study on causes and remedial measures for water logging conditions in Pragati Maidan and Jungpura NCT, Delhi	Sh. SK Mohiddin, Regional Director, Sh. Shashi Kant Singh, SC-D, Sh. G. Vijayakumar, SC-B, Sh. Gyanendra Rai, STA-HG	
147		Remedial Measures for Water Logging Issues and Revival of Abandoned Tube wells in NDMC Area	Sh. SK Mohiddin, Regional Director, Mrs. Prachi Gupta, Sc D, Ms. Chandini Chandran, Hydrogeologist (YP) , Sh. Sandanala Ashok Kumar, STA (HG)	
148		BULLETIN ON BEHAVIOUR OF WATER LEVEL IN NCT, DELHI DURING MAY 2024	Sh. S.K. Mohiddin, Regional Director,Ms. Prachi Gupta, Sc-D,Sh. Vandlabu Praveen Kumar, AHG ,Sh. Gyanender Rai, STA (HG),Sh. Sandanala Ashok Kumar, STA (HG),Ms. Mamta, STA (GP)Miss. Chandini Chandran, Hydrogeologist (YP)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
149	SUO Delhi	BULLETIN ON BEHAVIOUR OF WATER LEVEL IN NCT, DELHI DURING AUGUST 2024	Sh. S.K. Mohiddin, Regional Director,Ms. Prachi Gupta, Sc-D,Sh. Vandlabu Praveen Kumar, AHG ,Sh. Gyanender Rai, STA (HG),Sh. Sandanala Ashok Kumar, STA (HG),Ms. Mamta, STA (GP)Miss. Chandini Chandran, Hydrogeologist (YP)	
150		GROUND WATER LEVEL BULLETIN NOVEMBER 2024	Sh. S.K. Mohiddin, Regional Director,Ms. Prachi Gupta, Sc-D,Sh. Vandlabu Praveen Kumar, AHG ,Sh. Sandanala Ashok Kumar, STA (HG),Ms. Mamta, STA (GP)Miss. Chandini Chandran, Hydrogeologist (YP)	
151		GROUND WATER LEVEL BULLETIN JNAUARY 2025	Sh. S.K. Mohiddin, Regional Director,Ms. Prachi Gupta, Sc-D,Sh. Vandlabu Praveen Kumar, AHG ,Sh. Sandanala Ashok Kumar, STA (HG),Ms. Mamta, STA (GP),Miss. Chandini Chandran, Hydrogeologist (YP)	
152		BULLETIN ON TREND AND HOT SPOT ANALYSIS OF GROUND WATER QUALITY IN SHALLOW AQUIFERS OF NCT OF DELHI DURING MAY 2024	Sh. S.K. Mohiddin, Regional Director,Mrs. Prachi Gupta, Sc-D Dr. Kriti Mishra, Sc-C Sh. Vandlabu Praveen Kumar, AHG Sh. Sandanala Ashok Kumar, STA (HG) Sh. Gyanender Rai, STA (HG) Miss. Chandini Chandran, Hydrogeologist (YP)	
153		BULLETIN ON TREND AND HOT SPOT ANALYSIS OF GROUND WATER IN SHALLOW AQUIFERS OF NCT, DELHI DURING NOVEMBER 2024	Sh. S.K. Mohiddin, Regional Director,Mrs. Prachi Gupta, Sc-D Dr. Kriti Mishra, Sc-C Sh. Vandlabu Praveen Kumar, AHG Sh. Sandanala Ashok Kumar, STA (HG),Miss. Chandini Chandran, Hydrogeologist (YP)	

S.No.	Region	Title of the report published	Authors of the Report	Website link
154	**SUO Delhi**	Inception Report On Influence of Heavily Polluted Najafgarh Drain on the adjacent Aquifer Systems in the National Capital Territory (NCT) of Delhi	Sh. S.K. Mohiddin, Regional Director, Mrs. Kriti Mishra, SC-C, Sh. V. Praveen Kumar, AHG	
155	**WR Jaipur**	Dynamic Ground water Resources of Rajasthan -2024		
156		Ground Water level Bulletin - May 2024		
157		Ground Water level Bulletin - August 2024		
158		Ground Water level Bulletin - November 2024		
159		Ground Water level Bulletin - January, 2025		
160		Ground Water Year Book -2023-2024		
161		Ground Water Quality Year Book 2024		
162		Aquifer Mapping and Management: Study in Jodhpur Urban Agglomerates NAQUIM 2.0	Sh. Hitesh Kamlakar Ramteke, Sc-C (Hg), Miss Reena Borana , Sc-B (Hg), Sh.Ghanshyam Tiwari, AHG, Dr. P.P. Dookia, Sc-B (G.P), Dr. Prerna Mathuria, ACH	
163		Aquifer Mapping and Management: Study in Pali Industrial Cluster NAQUIM 2.0	S.S. Saraswat, Scientist-Scientist-D (Hg), Shashank Pratap Singh, Scientist-C (Hg), Gyan Prakash Tiwari, AHG, Prem Prakash Dookia, Scientist-B, Dr. Jaipal Garg, Scientist-C	

S.No.	Region	Title of the report published	Authors of the Report	Website link
164	**WR Jaipur**	Aquifer Mapping and Management: (2.0) - Jaipur City and Agglomerates Jaipur District, Rajasthan	Smt Preeti Pandey, Sc-C, Ms. Nupur Pant, Sc-B, Sh. Irfan Ali, AHG, Smt. Sunita Devi, Sc-B, Ms. Shivani Shukla, STA	
165		Aquifer Mapping and Management: (2.0)- Neemrana Industrial Area Alwar District, Rajasthan	Vipin Kumar, Scientist-C , Shri P. K. Verma, Sc-C , Shri Prashant Srivas, AHG , Smt. Aruna Saini, ACH , Smt. Sunita Devi, Sc-B	

3.14 (A) Technical Papers - National

Table 3.25: List of National Papers during 2024-25

S.No.	Author/Authors	Topic of Paper	National Journal
1	Muthyala Prasad, Chandra Prakash Dubey, Mulavada Vidyasagar, Sangem Satheesh	Delineation of Subsurface Geological Structures in Meghalaya, Northeast India using Gravity Gradients Analysis	AEG Journal of Geophysics
2	M.Vidyasagar, Muthyala Prasad, T. Rajababu and Raja Ram Purohit	ROLE OF GEOPHYSICAL SURVEYS IN DELINEATING SPRINGSHED IN DHALAI DISTRICT, TRIPURA	8th India Water Week (IWW) 17/09/2024
3	Gopal Sahoo, Keisham Radhapyari, Suparna Datta, Rinkumoni Barman, Snigdha Dutta, Tapan Chakraborty and Biplab Ray	ASSESSMENT OF GROUNDWATER OF MEGHALAYA, INDIA: A COMPARATIVE STUDY USING WQI TECHNIQUE	8th India Water Week (IWW) 17/09/2024
4	Gopal Sahoo, Rinkumoni Barman, Snigdha Dutta, Keisham Radhapyari, Biplab Ray and Tapan Chakraborty	Assessment of Groundwater Quality for Irrigation in the Shallow Aquifers of Dimapur District, Nagaland	National Seminar on Advances in Irrigation Technologies and Management (NSAITM) 30/11/2024
5	Gopal Sahoo, Rinkumoni Barman, Snigdha Dutta,	Assessment of hydrochemical facies and drinking water quality of	Journal of Geological Society of India

S.No.	Author/Authors	Topic of Paper	National Journal
	Suparna Datta, Biplab Ray and Keisham Radhapyari	North-Eastern state of India-Tripura	
6	Snigdha Dutta, Keisham Radhapyari, Rinkumoni Barman, Gopal Sahoo, , Tapan Chakraborty and Biplab Ray	Assessment of Potentially Harmful Elements in Groundwater of Meghalaya, India, the Scotland of the East: Long-term Effects on Drinking Water Quality, Irrigation, and Human Health Risk in India	8th India Water Week (IWW) 17/09/2024
7	Rinkumoni Barman, Keisham Radhapyari, Gopal Sahoo, Snigdha Dutta, Tapan Chakraborty, Biplab Ray	Evaluation of Time Series-Based Artificial Neural Network-ANFIS Model for Groundwater Quality Prediction	8th India Water Week (IWW) 17/09/2024
8	K.P. Singh, Shashi Kanth Singh, N. veera Babu, Sunita Devi, M.Vidyasagar	High resolution AquiferMapping using advance Heliborne Transient Electrmagnetic survey for groundwater assessment and management at Grampanchayat level of Srimadhopur Block,Sikar District, Rajasthan, India	8th India Water Week (IWW) 17/09/2024
9	S Somarendro Singh ans Suresh Kumar	Occurrence of Arsenic in Groundwater Resources of Majuli River Island in Brahmaputra River Basin.	Book chapter in the Ground water Contamination of India (30 April 2024) Springer: In: Sinha Ray, S., Acharyya, A. (eds) Ground Water Contamination in India. Springer, Cham. https://doi.org/10.1007/978-3-031-49092-7_13
10	S Somarendro Singh*1 Tapan Chakraborty*2, Biplab Ray*3, Dhrubajyoti Khanikar*4, Dipjyoti Khound*5, Chao Konseng Gogoi*6, Himangshu Kachari*7	Development of Springs in Hilly Terrains – a case study in West Siang District, Arunachal Pradesh	Proceeding Volume of 10th International GWC 2025-Groundwater Vision- 2047 of NIH (page No.171)
11	S Somarendro	Ground Water Irrigation	Proceeding Volume of

S.No.	Author/Authors	Topic of Paper	National Journal
	Singh*1 Tapan Chakraborty*2, Biplab Ray*3, Dhrubajyoti Khanikar*4, Dipjyoti Khound*5, Chao Konseng Gogoi*6, Himangshu Kachari*7	Schemes under Pradhan Mantri Krishi Sinchayi Yojana (PMKSY) - Har Khet Ko Paani (HKKP) for expanding and improving irrigation system in the North Eastern States of India	National Seminar on AITM- by NERIWALM, 2024 (page No.28)
12	S. Somarendro Singh*1 and Kh Mohon Singh	Groundwater Potentials of Manipur with Special emphasis to Imphal valley, Manipur"	Proceeding Volume of National Multidisciplinary Seminar, 2025 by IRC, Thambalmarik College and MASTEC, Manipur
13	V.K.Kulshreshtha* R. vashishtha*, J. Kumar* A Khan* and S. Kumar	Comprehensive evaluation of Heavy Metal Pollution Index and Seasonal Concentration Dynamics in Groundwater for Drinking Purposes: A Case study of Gwalior District,MP	Bhujal News
14	V.K.Kulshreshtha* , Adiba Khan* and A.K. Biswal	Hydrochemical Characterization of Groundwater and Assessment of Water Quality for Drinking and Agricultural Use in and Around Uranium-Contaminated Areas of Gwalior, Central India	Bhujal News
15	Vaishnavi Parihar*, Lata Udsaiya , Pradip Roy and Ashok Kumar Biswal	Data-Driven Groundwater Management in water stress area of central India: A Machine Learning algorithm based analysis of aquifer dynamics and over-extraction risks using high frequency data	Conference proceedings, International Groundwater Conference 2025, National Institute of Hydrology (NIH), Roorkee
16	L.N. Damodara*, Md. Sarif Khan, Ravi Kumar Gumma, Dr D Anantha Rao, Nilima Patra, G. Krishnamurthy	A COMPREHENSIVE HYDROGEOLOGICAL STUDY IN WATER STRESSED AREAS-WESTERN PRAKASAM DSITRICT, ANDHRA PRADESH, INDIA	National Seminar on "Advances in Geophysics for a Sustainable Future: Energy, Environment, and Natural Resources (AGSEEN) held from 17th to 19th February 2025 at the Department of Geophysics, Andhra University, Visakhapatnam, Andhra Pradesh.

S.No.	Author/Authors	Topic of Paper	National Journal
17	Fakhre Alam,Ebadur Rahman,Karam Singh & S.G.Bhartariya	Assessment of hydrogeological parameters of groundwatre for domestic and irrigational purposes:A case study from Ramganga Basin	Bhujal News
18	Sujatro Ray Chowdhuri, Madhukar Singh, Ebadur Rahman, Sandeep Kumar Swaroop, S.N. Dwivedi1 and Shive Prakash Rai (2025)	Assessment Of Dynamic And Static Ground Water Resources Through Aquifer Mapping In Siddhartnagar District, Uttar Pradesh, India	INTERNATIONAL GROUNDWATER CONFERENCE, Groundwater Vision 2047: Towards Water Security Under Changing Climate March 05-07, 2025, NIH Roorkee
19	Madhukar Singh, Sujatro Ray Chowdhuri, Fakhre Alam, Mayuri Pandey and Kriti Mishra (2025)	Principal Component Analysis And Modified Ground Water Quality Index As Powerful Actionable Tools For Ground Water Quality Assessment And Policy Intervention: A Case Study In Part Of The Gangetic River Basin Of Gorakhpur District, India	INTERNATIONAL GROUNDWATER CONFERENCE, Groundwater Vision 2047: Towards Water Security Under Changing Climate March 05-07, 2025, NIH Roorkee
20		Quantization of radioactive elements Uranium and lead in Ground Water of Uttar Pradesh, India	8th Indian Water Week 2024
21	Supriya Singh, Madhavi Razak, Karam Singh, Kiran Singh and S.G.Bhartariya	Identifying productive well sites in hard rock terrains using Geophysical Techniques: A case study from Sonbhadra District, Uttar Pradesh	8th Indian Water Week 2025
22	Swati P. Dhenkula,Akshay D., Shende Leena Deshpande, Girish R. Pophali	An overview of heavy metals treatment & management for laboratory waste liquid (LWL)	Journal of Environmental Chemical Engineering, Volume 12, Issue 4, August 2024, 113165
23	T. Raja Babu, Dibakar Mohanta, T. Venkatagiri, P. Raghavender, G. Krishnamurthy	Identification of suitable sites for Rain Water Harvesting Structures based on Hydrogeological & Geophysical Investigations at Adikavi Nannaya University Rajahmundry, East Godavari district,	Abstract Accepted and Presented during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025

S.No.	Author/Authors	Topic of Paper	National Journal
		Andhra Pradesh	
24	Resma S. Pillai, Swati P. Dhenkula, Swaroop Kalyan Sahoo, Ravi Kumar Gumma , P.Raghavender, T. Madhav & G.Krishnamurthy	Assessment of Aquifer Characteristics in Guntur District Using Vertical Electrical Sounding (VES): Insights on Weathered Thickness, Fractures, Water Levels and Salinity	Abstract Accepted during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025
25	T. Madhav, Resma S. Pillai, P Yadaiah, T Rajababu & G.Krishnamurthy	Assessment of Aquifer Characteristics in Sri Satya Sai District Using Vertical Electrical Sounding (VES), Well Inventory and Exploration data: Insights on Weathered Thickness, Fractures, Water Levels and Water Quality	Abstract Accepted during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025
26	Monika, Bijay Ketan Mohanta, Atharv S. Pawar, A. Rama Devi, G.Krishnamurthy	Aquifer Characterization in NTR District: A VES and TEM-Based Study on Weathered Zones, Fractures, and Ground Water Quality Trends	Abstract Accepted during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025
27	M. S. Goutham, P. Kiran Kumar, T. Venkatagiri, Susanta Ratha, G. Krishnamurthy	Unraveling Aquifer Dynamics: VES, TEM, and Recharge Strategies in Alluri Sitarama Raju District	Abstract Accepted during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025
28	L.N. Damodara, Md. Sarif Khan, Ravi Kumar Gumma, Dr D Anantha Rao, Nilima Patra, G. Krishnamurthy	A COMPREHENSIVE HYDROGEOLOGICAL STUDY IN WATER STRESSED AREAS-WESTERN PRAKASAM DSITRICT, ANDHRA PRADESH, INDIA	Abstract Accepted during the AGSEEN National Seminar organized by Dept of Geophysics, Andhra University during 17th -19th February 2025
29	S.S.Vittala, Bijay Ketan Mohanta & Krishnamurthy, G	Ground Water Sustainability through Aquifer Mapping and Management Plans for Sangareddy District of Telangana State	National Seminar on Earth Science Perspective on Groundwater Sustainability - Recent Advances, 1 - 2, March 2025 at Jaipur, Rajasthan, India organised by Indian National Chapter of the International Association of Hydrogeologists (INC-IAH) (Paper Presentation)
30	P. Kiran Kumar, M.	Climate Variability and	National Seminar on Earth

S.No.	Author/Authors	Topic of Paper	National Journal
	S. Goutham, K. Rambabu, & G. Krishnamurthy	Groundwater Recharge : The Role of Rainfall Distribution in Telangana State	Science Perspective on Groundwater Sustainability - Recent Advances, 1 - 2, March 2025 at Jaipur, Rajasthan, India organised by Indian National Chapter of the International Association of Hydrogeologists (INC-IAH) (Paper Presentation)
31	Resma S. Pillai, Sugan Manimaram, Basant Maheshwari, et al..	Community-Driven Step-well Rejuvenation for Sustainable Groundwater Management: A Case Study of Bundi District, Rajasthan	National Seminar on Earth Science Perspective on Groundwater Sustainability - Recent Advances, 1 - 2, March 2025 at Jaipur, Rajasthan, India organised by Indian National Chapter of the International Association of Hydrogeologists (INC-IAH) (Paper Presentation)
32	Dr.S.S.Vittala, G.Praveen Kumar &G.Krishnamurthy	Aquifer Mapping and Management Plans for Sustainable Ground Water Resources in Kamareddy District, Telangana State	8th India Water Week 2024, New Delhi, India
33	Dr.S.S.Vittala & G.Krishnamurthy	Sustainable Development of Ground Water Resources through Aquifer Mapping and Management Plans for Nizamabad District of Telangana State	International Conference on Water and Agricultural Sustainability under Changing Climate (WASCC) which is organized by Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu and Association of Global Groundwater Scientist, India, scheduled on 7th - 9th October 2024.
34	Dr.S.S.Vittala, Bijay Ketan Mohanta & G.Krishnamurthy	Ground Water Sustainability through Aquifer Mapping and Management Plans for Sangareddy District of Telangana State	
35	Md Sarif Khan*, L.N. Damodara, Ravi Kumar Gumma, D Anantha Rao, Nilima Patra, G. Krishnamurthy	Groundwater Dynamics and Management Strategies in Dr. B. R. Ambedkar Konaseema District, Andhra Pradesh.	National Seminar on "Advances in Geophysics for a Sustainable Future: Energy, Environment, and Natural Resources (AGSEEN) held from 17th to 19th February 2025 at the Department of

S.No.	Author/Authors	Topic of Paper	National Journal
			Geophysics, Andhra University, Visakhapatnam, Andhra Pradesh.
36	Rinku R Das, T Chakraborty	Groundwater occurrence: Aquifer disposition and Irrigation water quality assessment in Barak Valley, Assam, North East India.	Seminar proceedings of 8th India Water Week 2024, New Delhi, India
37	Rinku R Das, T Chakraborty	Disposition of Aquifer its Potentiality and Irrigation water quality assessment in Cachar, Assam, North East India.	Seminar proceedings of National Seminar on Advances in Irrigation Technologies and Management (NSAITM)
38	Rinku R Das, T Chakraborty, K Radhapyari	Role of geology and geomorphology in arsenic distribution in groundwater of the Cachar fold belt, Barak Valley Assam	Conference proceeding of INTERNATIONAL GROUNDWATER CONFERENCE, Groundwater Vision 2047: Towards Water Security Under Changing Climate March 05-07, 2025, NIH Roorkee
39	Shasinlo Kent, Anenuo Pienyu, T Chakraborty	Springs of Northeast India: Hydrogeological characteristics, implications, and irrigation potential	National Seminar on Advances in Irrigation Technologies and Management (NSAITM) (Paper Presentation)
40	Chandreyee De, Anjali Kushwaha	Trends in Ground Water Quality and Heavy Metal pollution in Haridwar district, Uttarakhand: A call for sustainable management	Conference proceeding of INTERNATIONAL GROUNDWATER CONFERENCE, Groundwater Vision 2047: Towards Water Security Under Changing Climate March 05-07, 2025, NIH Roorkee , Pg 263
41	Debasish Bagchi, Suparna Datta, Naseema Jamal, Anadi Gayen, Aditya Sarkar, Saikat Majumder	Impact of Industrialization on the Salinized Aquifer System in Parts of Bengal Basin with Focus on the Lower Ganga Deltaic Plain	"Sustain Istanbul 2024: Towards Sustainable Groundwater Use Under Changing Climate", Book of Abstracts, Ed. Nadim K. Copty, Irem Daloglu Cetinkaya, and Cevza Melek Kazekeyilmaz Alhan, Bogazici University, Turkey, ISBN: 978-625-00-2417-1, p. 66 (2024)
42	Debasish Bagchi, Suparna Datta, Anadi Gayen	Management of diverse aquifer systems towards achieving Sustainable Development Goals: A case	Souvenir Cum Abstract Volume on "Earth Science Perspective on Groundwater Sustainability: Recent Advances", National

S.No.	Author/Authors	Topic of Paper	National Journal
		study	Seminar of the Indian National Chapter of International Association of Hydrogeologists (INC-IAH), Amity University Jaipur, India, p. 93 (2025).
43	Debasish Bagchi, Hrushikesh Behera, Anadi Gayen	Exploration of deep aquifers: Global scenario and regional perspective from West Bengal, India	Newsletter of Indian National Chapter of International Association of Hydrogeologists (INC of IAH), 12(1), pp. 35-38. (2025)
44	Borgohain A, Kundu SS, **Barman N**, Agrawal SP (2025)	17-Year Trend of Tropospheric Columnar NO2 Over India's Northeast Region Observed by OMI: Investigating Probable Anthropogenic and Natural Sources	Mausam Journal
45	Shaista Khan, Chirashree Mohanty, Suparna Datta, Subrata Das, Anadi Gayen	Hydrogeochemical Profiles of Coastal Confined Tertiary and Quaternary Aquifers System and Its Management Intervention- A case Study from Haldia Industrial Belt, Purba Medinipur, West Bengal, India	8th India Water Week 2024, New Delhi, India

3.14 (B) Technical Papers - International

Table 3.26: List of Published International Papers during 2024-25

S.No.	Author/Authors	Topic of Paper	Name of International Journal	Link on Web
1	Naved Alam, Mohd Amir Husain, Rakesh Singh*, Padam Kumar Jain, Elisabeth Eiche, Harald Neidhardt, Michael Marks, Manoj Kumar, Ashis Biswas	Geochemistry of fluoride mobilisation in hard rock aquifer of central India: Implication of fluoride safe drinking water supply	Applied Geochemistry, 171	https://doi.org/10.1016/j.apgeochem.2024.106106
2	Tiwari S., Sinha P., Khare M.,Srivastava R.K. & Biswal A.K.	A Critical Resource in a Changing Climate. Mitigation and Adaptation Strategies in Natural Systems Environmental Earth Sciences book series,	Springer Nature publication(under publication)	-
3	S. Tiwari, A.K. Biswal & G. Ramteke	Hydro-meteorological Research Study in Madhya Pradesh, Central India: A Literature Review. Pure	Pure Appl. Geophys. 181, 2935–2948	https://doi.org/10.1007/s00024-024-03553-6
4	S.N. Dwivedi, I. Roy, R. R. Shukla, Fakhre Alam, Suresh Kumar, Pankaj Kumar & Rakesh Singh	Controlling Arsenic Cross-contamination in Multi-aquifer System of Gangetic Flood Plain Through Optimal Well Spacing Based on Aquifer Response Modelling—A Case Study from Buxar District and Neighbouring Areas, Bihar, India	Water Conservation Science and Engineering Volume 9, Article no. 48,	https://doi.org/10.1007/s41101-024-00280-w,
5	Roy, I.	Assessing Groundwater Potential of Crystalline Basement in Greater Guwahati Area, Assam, North Eastern India, through integration of Lineament Mapping and Exploratory Drilling	Hydrogeology Journal, 23, 1-22	https://doi.org/10.1007/s10040-024-02852-5
6	Roy, I., Sarkar, S., Kumar, A., Mukherjee, M., Nandi, R., Chowdhuri, A. N., Barman, N, &	Hydrogeology of southeastern coastal parts of Great Nicobar Island, India: development of an integrated conceptual model for a hitherto unexplored	Quarterly Journal of Engineering Geology and Hydrogeology, 57(4), qjegh2024-063.	https://doi.org/10.1144/qjegh2024-063

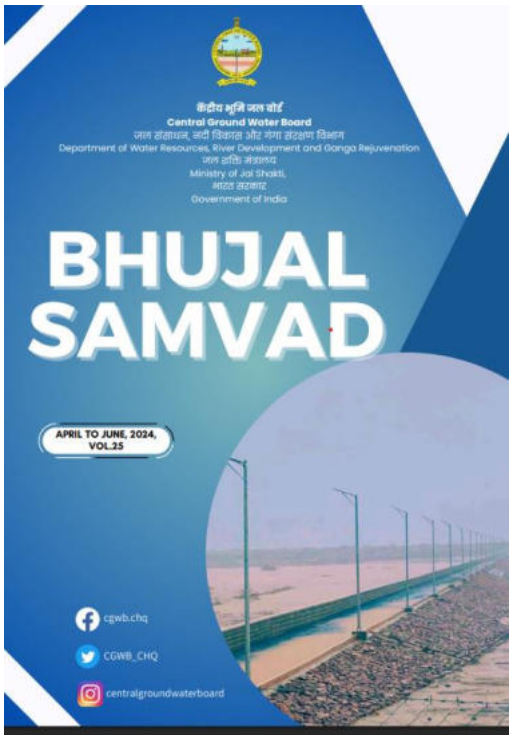
S.No.	Author/Authors	Topic of Paper	Name of International Journal	Link on Web
	Gayen, A.	island in the Indian Ocean		
7	**Barman Nilamoni**, Roy, I., & Gayen, A.	Long-term variation of hydro-meteorological parameters and their impact on groundwater of unconfined aquifer	Arabian Journal of Geosciences, 17(6), 1-17	https://doi.org/10.1007/s12517-024-11991-8
8	Roy, I., Dwivedi, S. N., Oraon, B. K., Raghavender, P., Singh, S. K., Ram, S., Alam, F., & Kumar, S.	Aquifer characterization of Rajgir thermal spring field, Bihar, Eastern India, in the face of climate change and rapid urbanization	Acta Geophysica, Vol. 72, pp. 4091–4110	https://doi.org/10.1007/s11600-024-01311-9
9	**Barman Nilamoni**, Roy, I., & Gayen, A	Investigation of land–atmosphere interaction at a high altitude (5330 m asl) snow covered station of East Himalaya.	Meteorology and Atmospheric Physics, 136(6), 49	https://doi.org/10.1007/s00703-024-01045-9
10	Fakhre Alam, S.K.Swaroop, S.G.Bhartariya	Assessing Aquifer Vulnerability Towards contamination of Groundwater using a new model (DRASTIC-LU) in parts of Ramganga Basin,India	Water Conservation Science and Engineering	https://doi.org/10.1007/s41101-025-00355-2
11	**Barman Nilamoni,** Kundu SS, Borgohain A (2024)	Observation and model simulation of aerosol optical properties and size distribution over the hilly terrain of Northeast India	*Aerosol Science and Engineering, Springer Nature*	Observation and Model Simulation of Aerosol Optical Properties and Size Distribution Over the Hilly Terrain of Northeast India | Aerosol Science and Engineering
12	Gogoi M, Borgohain A, Kundu SS, Sharma S, NVP Kiran Kumar, Roy R, Saha B, Bhuya PK, **Barman N,** Agrawal SP (2024)	Investigation of the Atmospheric Boundary Layer Characteristics over a High Altitude Station in North East India using Measurements and Reanalysis Datasets	*Theoretical and Applied Climatology, Springer Nature*	Investigation of the atmospheric boundary layer characteristics over a high altitude

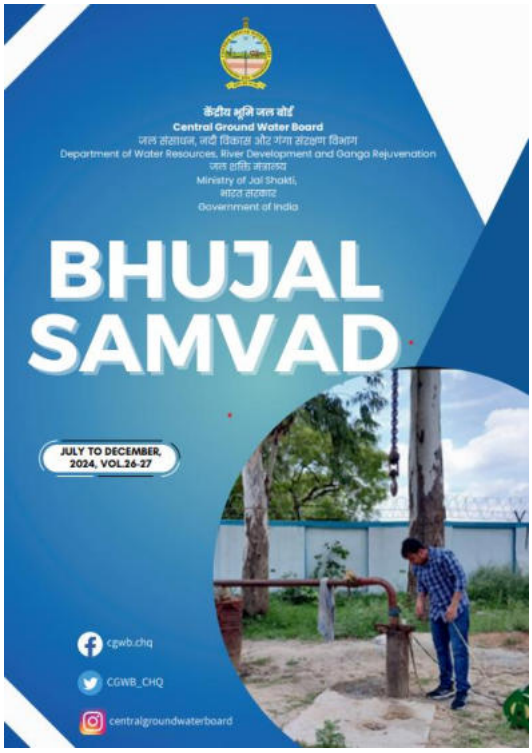
S.No.	Author/Authors	Topic of Paper	Name of International Journal	Link on Web
				station in North East India using measurements and reanalysis datasets | Theoretical and Applied Climatology
13	Anirudh Singh, Birendra Pratap Singh, Tejas.Y.Mankikar and Shahi Kant Singh	Hydrogeological Resolution using Composite Geophysical methods: A case study of Inland salinity in part of Agra, Uttar Pradesh, India	Journal of the Geological Society of India	https://doi.org/10.17491/jgsi/2024/173996
14	N Narsimha Naidu, Veena R. Achuta, Mayank Dixit	Demarcation and Assessment of Inland Salinity/Groundwater quality in parts of Western Dharwad Craton from Transient Electromagnetic Soundings	Conference proceeding of INTERNATIONAL GROUNDWATER CONFERENCE, Groundwater Vision 2047: Towards Water Security Under Changing Climate March 05-07, 2025, NIH Roorkee	-
15	Mayank Dixit, Mohit Arora, H.P. Jayaprakash, Veena R. Achutha	Achievements and significance of national hydrograph stations network and 55 years of continuous groundwater level monitoring in India	Groundwater for Sustainable Development	https://doi.org/10.1016/j.gsd.2024.101313
16	Mayank Dixit, A.R. Bansal, M. Ravi Kumar, Rajat Pasricha	Dynamically Triggered Events in Mining and Monsoon Induced Regions of Northwestern Deccan Volcanic Province of India	Natural Hazards	https://doi.org/10.1007/s11069-024-06787-8
17	Mayank Dixit, A.R. Bansal, M. Ravi Kumar, Rajat Pasricha	Mining and monsoons may increase risk of earthquakes in northwest India	Nature	https://www.nature.com/articles/d44151-024-00129-0
18	Debasish Bagchi, Anadi Gayen (2025)	Source Sustainability of Fractured Anorthositic Aquifer System in the Rarh Bengal and Socio-Cultural Implication	Journal of Hydrogeology and Hydrologic Engineering, 14:1.	DOI: 10.4172/2325-9647.1000361.

S.No.	Author/Authors	Topic of Paper	Name of International Journal	Link on Web
19	Shaista Khan, Suparna Datta, Debasish Bagchi, Anadi Gayen, Aditya Sarkar	Hydro-geochemical characterization and mapping of coastal aquifer system in the Lower Ganga Deltaic Plains (LGDP), West Bengal, India	Conference proceeding of INTERNATIONAL ASSOCIATION OF HYDROGEOLOGISTS, Davos, Switzerland	Abstract ID: 107410
20	Shaista Khan, Ghazala Siraj , Anadi Gayen	Impact of anthropogenic activities on groundwater quality: A case study from Murshidabad district, West Bengal, India	Conference proceeding of INTERNATIONAL ASSOCIATION OF HYDROGEOLOGISTS, Davos, Switzerland	Abstract ID: 106588
21	Shaista Khan, Ghazala Siraj , Anadi Gayen	Hydrochemical characterization and groundwater quality assessment under natural and anthropogenically influenced areas of lower Gangetic plains, West Bengal	SCI Journal of Water Resources	Article id:WatRes2560090Khan (under publication)

3.15 Bhujal Samvad

1. Bhujal Samvad is a quarterly magazine of the Central Ground Water Board (CGWB), launched in January 2018. As of December 2024, 27 issues have been published, all of which are available on the CGWB's official website.
2. The magazine includes several prominent sections such as Cover Story, In Focus, Reports, Pathshala, and Social Media Highlights and Collectables. The Cover Story highlights significant initiatives and key contributions of the Board, while In Focus provides a quarterly overview of major activities and achievements. The Reports and Pathshala sections aim to enhance awareness and understanding of various hydrogeological concepts among readers.
3. In line with its dynamic nature, Bhujal Samvad has recently introduced a new feature—the Bhujal Samvad Talk Series—which captures key takeaways from thought-provoking talks delivered by experts from within and outside the organization.





3.16 Public Interaction Programmes (PIP)

Public Interaction Programme being implemented by CGWB are one day programme for dissemination of outputs of NAQUIM and other related studies and create awareness. The programme also involves area specific discussions on ground water issues and possible solutions. The Public Interaction Programme have been dovetailed with the celebration of Azadi Ka Amrit Mahotsav and Mission Life Campaign. to promote the concept of LiFE (Life Style for Environment), The PIP activities were introduced in 2018. The details of activities carried out in all the Regional offices are as below in **Table 3.24**.

Table 3.24: List of PIPs conducted during the year (2024-25)

S.No.	State /UT	Region	Annual Target (2024-25)	PIPs done upto 31.03.2024	No of Total Participants
1	Madhya Pradesh	NCR	**4**	3	309
2	Himachal Pradesh	NHR	**1**	1	120
3	Jammu and Kashmir & Ladakh	NWHR	**4**	4	510
4	Haryana	NWR	**2**	2	226
5	Punjab & Chandigarh	NWR	**2**	2	192
6	Gujarat	WCR	**5**	5	571
7	Rajasthan	WR	**4**	4	389

8	Uttar Pradesh	NR	**4**	4	470
9	Delhi	SUO	**2**	2	160
10	Uttarakhand	UR	**1**	1	75
11	West Bengal	ER	**4**	4	401
12	Bihar	MER	**3**	3	392
13	Jharkhand	MER	**1**	1	133
14	Chhattisgarh	NCCR	**5**	3	483
15	NER	NER	**5**	3	546
16	Odisha	SER	**4**	3	246
17	Maharashtra	CR	**3**	3	421
18	Kerala	KR	**3**	3	235
19	Tamil Nadu	SECR	**4**	4	552
20	Andhra Pradesh	SR	**2**	3	188
21	Telangana	SR	**1**	1	120
22	Karnataka	SWR	**4**	4	355
	Total		**68**	**63**	**7094**

4. Public Investment Board (PIB) Project for Strengthening NAQUIM's Data Generation Infrastructure

4.1 Introduction of Public Investment Board (PIB)

The activities proposed under the National Aquifer Mapping and Management Project were originally proposed as part of the Central Sector Scheme "Ground Water Management and Regulation" under Component I (Aquifer Mapping and Management Programme, NAQUIM) of the Scheme for implementation during the period 2021-26. As per the recommendations of the EFC meeting of GWMR scheme held on 31st August 2021, proposals for three sets of activities of time bound nature were prepared and presented before the Public Investment Board (PIB) Committee and approved as NAQUIM(PIB) Project under the GWMR scheme. The NAQUIM(PIB) project has three components, namely: (i) Automation of infrastructure for water level monitoring through construction of Piezometers and installation of Digital Water Level Recorders (DWLRs) in the country. (ii) High resolution aquifer mapping through Heli-borne geophysical surveys and (iii) Data generation for aquifer mapping through well construction with an approved outlay of Rs.805 crore.

4.2 Creation of Infrastructure for high density automated monitoring of water levels

It is proposed to create infrastructure for automated monitoring of water levels to generate high frequency water level data in areas falling under assessment units that have been categorised under Over-exploited (OE), Critical and Semi-Critical category and other priority areas like large urban agglomerations, industrial clusters, mining areas, coastal areas, seismic zones IV and V and other such assessment units where adequate monitoring networks are not available. It is proposed to install Piezometers with Digital Water Level Recorders and telemetry in such areas for generation of high frequency water level data. The exercise would be taken up in convergence with the National Hydrology Project (NHP) and Atal Bhujal Yojana. Considering existing piezometers with CGWB and State Governments and those proposed under NHP and Atal Bhujal Yojana, additional requirement of piezometers has been estimated as 9,000. The actual requirement may vary depending upon field conditions like availability of suitable sites, security issues etc. Out of the 9000 Piezometers, 2000 piezometers are planned to be constructed through in-house resources of CGWB under the Ground Water Management and Regulation Scheme. Construction of remaining 7000 piezometers and installation of 7000 DWLR with telemetry is envisaged to be implemented through outsourcing under this project at an estimated expenditure of Rs 529 Cr. High frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas besides providing inputs in planning for

source sustainability under Jal Jeevan Mission and participatory ground water management under Atal Bhujal Yojana.

4.3 High resolution Mapping of aquifers through heli-borne geophysical survey in nearly 3 lakh sq. km area in arid parts of NW India

It is proposed to take up high resolution mapping of aquifers on 1:10,000 scale in ~3 lakh sq km area covering the arid parts of the States of Rajasthan, Gujarat, Haryana, Punjab and Himachal Pradesh using heliborne geophysical studies. The heli-borne study is aimed at providing 3D Geophysical model of the area, geophysical thematic maps at horizontal and vertical planes, Aquifer Geometry in 2D and 3D, Aquifer system with fresh and saline zones and suitable sites for artificial recharge. Spatial distribution of paleochannel network, if any, and its linkage with aquifer system would also be studied. The high resolution maps would help in formulating site specific plans for improving groundwater situation in the water stressed regions. An outlay of Rs 151 crores is envisaged towards conducting heliborne survey along with associated ground geophysical surveys covering 3 lakh sq km during 2022-25.

4.4 Data Generation for Aquifer Mapping through construction of 1135 wells in parts of 11 States of the country

National Aquifer mapping and Management Programme was initiated under the Ground Water Management and Regulation Scheme during the XII Plan (2012-17) period. Under the programme, an area of nearly 24.8 lakh sq km of the country has been identified for mapping out of which an area of 21.15 lakh sq km has already been covered upto March 2022. To meet the data generation requirement for completing aquifer mapping on 1:50,000 scale by March 2023, drilling of 1135 wells through outsourcing has been envisaged in parts of 11 States of the country. This activity is proposed to be taken up through outsourcing. The data generated through outsourcing will be used for preparation of aquifer maps and management plans. An outlay of Rs 125 Crore has been envisaged towards this activity.

4.5 Project Activities and Outputs

The outputs of this project will serve as inputs for preparing of aquifer maps and management plans under the ground water management and regulation scheme. This information, in turn will be shared with state government departments for planning sustainable ground water management. Activity wise major outputs are given in the **Table 4.1** below.

Table 4.1: Activity wise major outputs of PIB

S.No.	Item	Measurable Outcomes
1	Construction of 7000 Piezometers and installation of DWLRs with telemetry in OE, Critical, Semi-Critical assessment units and other priority areas	Creation of infrastructure for automated monitoring of water levels in water stressed assessment units. High frequency data would also provide scientific inputs for driving corrective action by State/Central agencies based on ground water level monitoring system. High density and high frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas and in improved assessment of groundwater resources.
2	Heli-borne surveys for high resolution mapping in ~3 lakh sq km area in arid parts of NW India	High Resolution mapping of aquifers on 1:10,000 scale in arid parts of the country will help formulate better ground water management plans. Will help provide Pin-pointed sites and depth zones for taking up appropriate management interventions for water harvesting and recharge in water stressed areas.
3	Data Generation for Aquifer Mapping through construction of Wells in 11 States	Construction of 1135 wells for data generation for completing remaining works of Aquifer Mapping on 1:50,000 scale. Improved information about aquifer disposition, quantification of available ground water resources and assessment of groundwater quality.

4.6 Estimated Cost

Total estimated cost of the project is Rs 805 Cr (Rupees Eight Hundred and five crores) for the entire project period i. 2022-23 to 2025-26. Year wise and activity wise proposed outlay for the project are given in the **Table 4.2** below.

Table 4.2: Activity wise estimated cost of PIB project

Activity	Total Physical targets	Financial outlay (Rs in Cr)				
		2022-23	2023-24	2024-25	2025-26	Total
Construction of Piezometers	7000 no of piezometers	25	150	150	59	**384**
Installation of DWLRs with telemetry	7,000 no of DWLRs	-	21	62	62	**145**
Heli-borne surveys for high resolution mapping in 3 lakh sq km area in arid parts of NW India	~3 lakh sq km	45	45	61	-	**151**
Data Generation for Aquifer Mapping through construction of Wells in parts of 11 States	1135 number of wells	55	70	-	-	**125**
	Total	**125**	**286**	**273**	**121**	**805**

4.7 Progress of various activities under PIB Project as on 31.03.2025.

4.7.1 Construction of 7000 Piezometers and installation of DWLRs

As per the AAP 2024–25 (as on 31st March 2024), a total of 2,826 Piezometers have been constructed across various states, along with the installation of 298 DWLRs. The highest progress was recorded under Package 1, with Tamil Nadu (276 Pzs., 90 DWLR), Andhra Pradesh (206 Pzs., 95 DWLR), Telangana (165 Pzs., 113 DWLR), and Kerala (18 Pzs.). The state-wise details are provided in **Table 4.3** below. Additionally, an amount of Rs.80.03 crore was released under the RA Bill during the same period.

Table 4.3 Details of Piezometers and DWLRs

S.No.	Package No.	State	Construction of Pzs. (Nos.) (upto Compressor Development)	Installation of DWLR
1	1	Andhra Pradesh	206	95
		Telangana	165	113
		Tamil Nadu	276	90
		Kerala	18	0
2	2	Gujarat	46	0
		Maharashtra	307	0
3	3A	Madhya Pradesh	0	0
4	3B	Madhya Pradesh	131	0
5	3C	Madhya Pradesh	35	0
6	3D	Chhattisgarh	175	0
7	4	Delhi	0	0
		Punjab	0	0
		Haryana	0	0
		Himachal	0	0
		Uttarakhand	0	0
8	5A	Rajasthan	282	0
9	5B	Rajasthan	297	0
10	5C	Rajasthan	478	0
11	6	Uttar Pradesh	260	0
		Bihar	27	0
		Jharkhand	53	0
		West Bengal	64	0
		Odisha	6	0
12	7A	Assam	0	0
		Meghalaya	0	0
		Arunachal Pradesh	0	0

S.No.	Package No.	State	Construction of Pzs. (Nos.) (upto Compressor Development)	Installation of DWLR
13	**8A**	**Jammu**	0	0
14	**8B**	**Kashmir**	0	0
			2826	**298**

4.7.2 Construction of EW/OW

A total of 567 Exploratory Wells (EWs) and Observation Wells (OWs) have been constructed across various states under different packages, as detailed in **Table 4.4** below. Additionally, an amount of Rs.26.49 crore of RA Bill was released under the AAP 2024–25.

Table 4.4 Details of various EW/OW constructed

S.No.	Package No.	State	No. of EW Constructed	No. of OW Constructed	Total No. of EW/OWs constructed
1	**I**	**Assam**	2	2	4
		Bihar	3	3	6
		Odisha	38	8	46
		West Bengal	60	19	79
2	**II**	**Andhra Pradesh**	45	11	56
		Karnataka	14	3	17
3	**III**	**Madhya Pradesh**	14	5	19
		Chhattisgarh	19	7	26
4	**III-R**	**Madhya Pradesh**	0	0	
5	**IV**	**Rajasthan**	51	9	60
		Gujarat	108	23	131
6	**V**	**Uttar Pradesh**	71	52	123
			425	**142**	**567**

4.7.3 Heli-borne survey

The tender document for high-resolution mapping of aquifers through heli-borne geophysical survey in the arid parts of northwest India, covering an area of 2.7 lakh sq.km, was prepared and floated on GeM for bidding.

5. Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) - non Scheme

5.1 Introduction

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI)is the training wing of CGWB**,** under the aegis of Department of Water Resources, River Development & Ganga Rejuvenation, Ministry of Jal Shakti, Govt. of India with the vision as a "Centre of Excellence for Training and Research for Sustainable Development of Ground Water Resources of the Country"

Since its Inception in 1997, as India's first exclusive Training Institute in Groundwater studies to improve the capacity building of groundwater professionals for sustainable groundwater development & management with participatory approach, a total of 1,16,626 participants, including ground water professionals as well as stakeholders at the grassroots level, participated in the 1,753 programs conducted by the Institute.

The institute mainly conducts training courses for Officers of Central Ground Water Board (CGWB). The subject matters in the training courses cover various stages of their professional careers starting from the Induction, Mid-career and Senior Management levels. In addition to this, officers from other Central and State Government Organizations, Non-Governmental Organizations (NGO) as well as University Faculty Members and Students associated with the realm of ground water are also beneficiaries of RGNGWTRI training programs.

5.2 Three-tiered Training Programme

Since 2012 (XII Plan) RGNGWTRI has embarked upon a three-tier training system (Tier-I: National Level, Tier-II: State/District Level & Tier-III: Block Level) to improve the capacity building of groundwater professionals for sustainable groundwater development & management involving participatory approach to create a pool of trained resource persons who would be working towards sustainability of ground water resources across the country with community participation. A

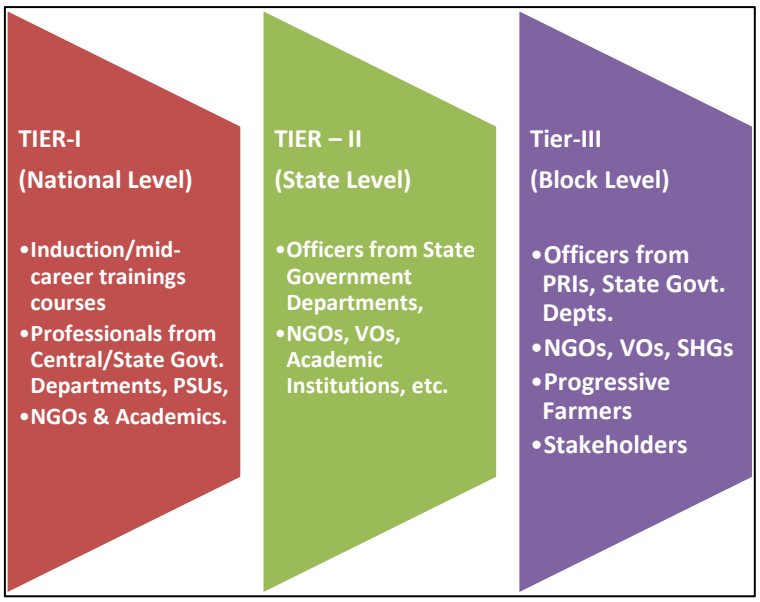


Fig. 5.1: Three-tiered Training Program structure at RGNGWTRI

glimpse of the three-tiered training program structure at RGNGWTRI is being displayed in **Figure 5.1**.

The Tier-I trainings are conducted at National Level and cover three verticals of capacity building *viz*., domain specific trainings, behavioral and functional competency improving training programs. During 2024-25, 61 Tier-I trainings (Domain: 25 Nos; Functional: 28 Nos & Behavioural: 8 Nos) were organized by RGNGWTRI, against the target of 60, with a total of 1489 participation (including 426 female participants). The Tier-I training also included:

- 23 mandatory trainings for Scientific, Ministerial and Engineering cadres of CGWB,
- 2 Immersion Trainings for the Officers and Officials of DoWR, RD & GR, Ministry of Jal Shakti, Govt. of India
- 3 demand driven trainings for Implementing Agencies and state government agencies,
- 2 International Trainings in Collaboration with AARDO
- 3 Trainings under National Hydrology Project,
- 1 Water Auditor Certification training and
- 2 trainings (1 each for NWA & NERIWALM) under synergy of the three Training institutes, RGNGWTRI, NERIWALM & NWA of Ministry of Jal Shakti.

RGNGWTRI as a part of Tier-I training progammes organizes Induction Level Training Courses (ILTC) for newly recruited scientific officers of CGWB. In the year 2024-25, the training for the first batch of this newly designed 1 year Induction Level Training Course was successfully completed in November 2024 as shown in **Figure 5.2**. In August 2024, training for the Batch 2 and Batch 3 was started (**Figure 5.3)** and in November 2024, the training for Batch 4 started.

Fig. 5.2: Dr S. K. Ambast, Chairman, CGWB along with the Trainee Officers of ILTC Batch 1 and Faculty Members of RGNGWTRI in the Valedictory Session Group Photograph

Fig. 5.3: Dr S. K. Ambast, Chairman, CGWB, Shri T. B. N Singh, Member Secretary (CGWA) along with the Trainee Officers of ILTC Batch 2 & 3 of RGNGWTRI in the Inaugural Session

Tier-II trainings programs are conducted at the State level for capacity building. These trainings are organized through CGWB Regional Offices. During the AAP 2024-2025, the Tier-II State Level training programs were held on the theme "Local Ground Water Management Issues". A total of 20 training courses against the target of 20 were organized in the entire country from April 2024 to March 2025, in which 545 trainees have participated including 216 women participants.

During the AAP 2024-2025, 52 Block level (Tier-III) trainings on "Local Ground Water-Related Issues and Participatory Ground Water Management" were organized to build capacity at the village/block level. All 52 trainings were organized in physical/offline mode, in which 6420 trainees have participated including 2315 women participants. Summary of the Targets and Achievements are as given below in **Table 5.1**. **Figures 5.4** and **5.5** represent the total Tier-I, Tier-II, and Tier-III training programs and the male, female, and total participants for each tier, for the period April 2024 to March 2025, respectively.

Table 5.1: Summary of Targets & Achievements of training courses in AAP 2024-2025

Training Courses	Training Target (Nos.)	Training Achievement (Nos.)	No. of Participants		
			TOTAL	Male	Female
TIER–I (National Level) at RGNGWTRI	**60**	61	1489	1063	426
TIER–II (State Level) *	**20**	20	545	329	216
TIER–III (Block Level) *	**50**	52	6420	4105	2315
TOTAL	**130**	**133**	**8454**	**5497**	**2957**

*Tier–II and Tier III training courses are conducted through all the Regional Offices of CGWB.

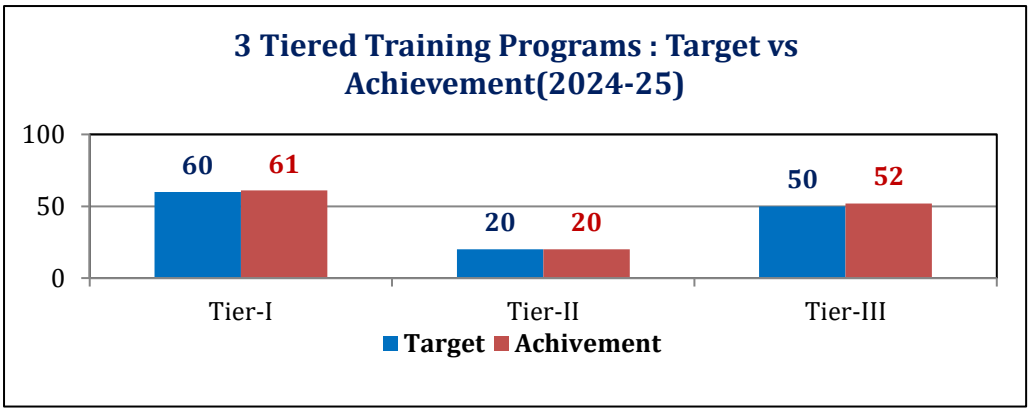
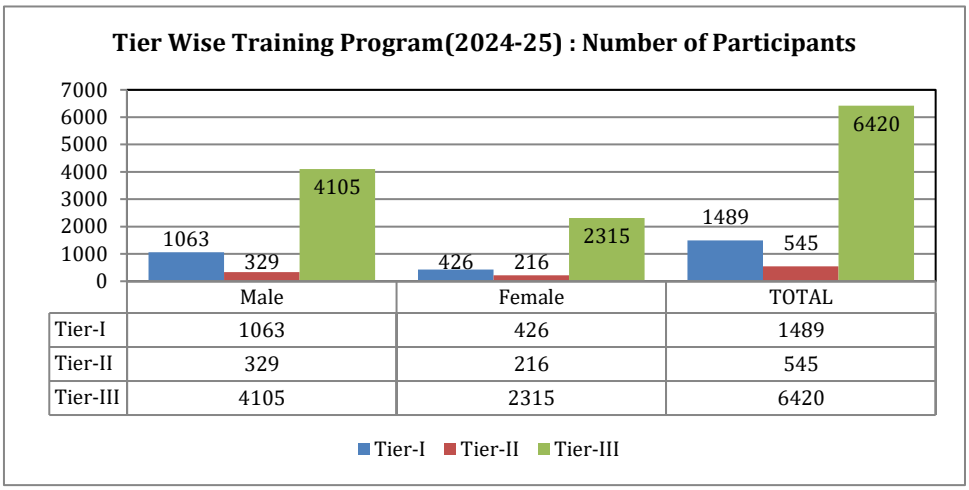


Fig. 5.4: Training Targets vis-à-vis Achievements in 2024-25



	Male	Female	TOTAL
Tier-I	1063	426	1489
Tier-II	329	216	545
Tier-III	4105	2315	6420

Fig. 5.5: Tier wise trainee Participation in Training programs of 2024-25

5.3 BHUJAL News

Bhujal News is a quarterly research Journal of CGWB which is being published by Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), Raipur from the year 2020.

The works in CGWB are mostly field-based and the outputs are measured by the reports based on these field studies. Bhujal News serves as a medium to highlight CGWB's contribution in the field of ground water.

RGNGWTRI, Raipur has published NAQUIM Special Volume - showcasing the outcomes of the NAQUIM 1.0 in the Bhujal News, Volume 33 (2023-24, No 1-4) in the year 2024. It includes 25 papers that covers Aquifer mapping in various hydrogeological environments, along with Geophysical and Chemical aspects of ground water in different aquifer systems. The compilation of volume 34 is in progress and 7 papers were received in the AAP of 2024-25. **Figure 5.6** features the cover page of Bhujal News, Volume 33 (NAQUIM Special Volume).

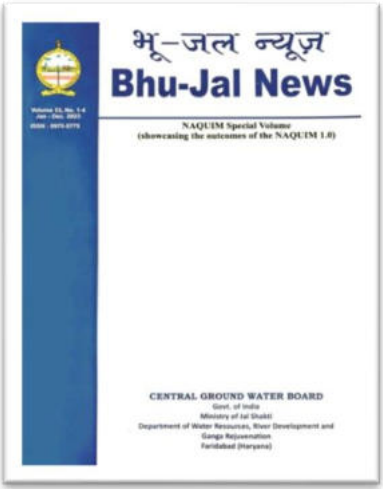


Fig. 5.6: Cover Page of Bhujal News Volume 33 (NAQUIM Special Volume)

5.4 Accreditation of Ground water Professionals

Accreditation is a process of instilling assurance and confidence amongst all stakeholders to submit reports to acceptable standards. As per the guidelines of Central Ground Water Authority (CGWA), applicants who seek permission for Ground Water extraction hire the services of accredited individuals or institutions for conducting hydrogeological survey and prepare a report on the groundwater scenario and the impact of the proposed extraction on the ground water regime. RGNGWTRI is entrusted with the screening and scrutiny of applications received from individuals/Institutions for accreditation, conducting examinations, and issuance of certificates. RGNGWTRI has conducted five rounds of Accreditation till 2024-25 (7 Individuals accredited in the 5th Round of Accreditation held in 2024-25). The summary of the Accredited individuals/Institutes as of 31.03.2025 are as below:

- Total number of Groundwater professionals accredited are: 73
- Total number of Institutions accredited are: 30

5.5 Training Programme for Water Auditors Certification

RGNGWTRI has completed one batch of training on Water Auditors Certification culminating into 3 Water Audit Certification Courses. A total of 13 water auditor certification has been awarded during 2024-25 totaling it to 46 Certified Water Auditors. The trainee participants were from former Scientists and Engineers from organizations in water sector and aspiring water audit professionals. Glimpses of Water Auditor Certification training 3rd batch are shown in **Figure 5.7**.

Fig. 5.7: Glimpses of Water Auditor Certification training 3rd batch

5.6 Other Important Trainings programs

- **International Training program for AARDO Member Countries:** During the current Year RGNGWTRI conducted two Training in association with African Asian Rural Development Organization (AARDO) which were organized during 22nd - 25th July 2024 (Online Mode: About 168 ground water professionals from 27 member countries participated) as shown in **Figure 5.8** and during 23rd September - 4th October 2024 (Physical Mode: 5 ground water professionals from 4 member countries participated) as shown in **Figure 5.9**.

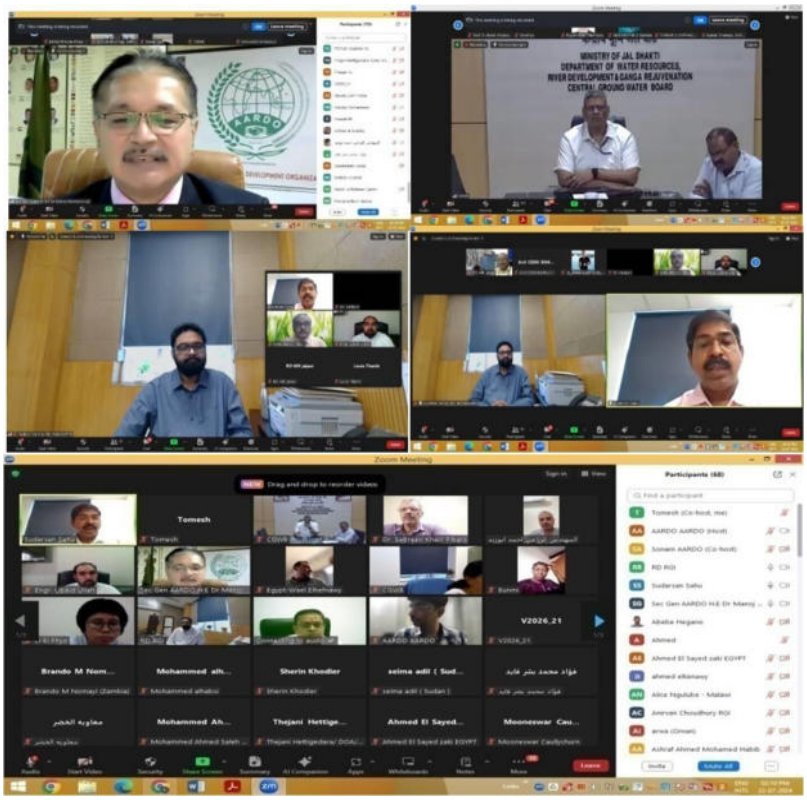


Fig. 5.8: Glimpses of the Online Training Programme for GW Professionals of AARDO Member Countries (22 – 25.07.2024)

Fig. 5.9: Glimpses of the Physical Training Program for GW Professionals of AARDO Member Countries (23.09.2024 – 04.10.2024)

- **Training programme on AI & ML:** For the first time, RGNGWTRI conducted a three days Training programme on Artificial Intelligence and Machine Learning and its application in ground water during 4 – 6th June 2024 for the Officers of CGWB, in collaboration with the Wadhwani Foundation, at RGNGWTRI. A glimpse od the training program is shown in **Figure 5.10**.

Fig. 5.10: Glimpse of the Inaugural Session of Training Program on AI & ML & its application to GW at RGNGWTRI

- **Training Program on role of Rajbhasa Adhikari of CGWB:** For the first time one excusive Tier- I Training Programme on Rajbhasa conducted on "Role of Rajbhasa Adhikari" during March 2025 as shown in **Figure 5.11**.

Fig. 5.11: Glimpses of the Training Program on Role of Rajbhasa Adhikari of CGWB at RGNGWTRI

- **Youth Exchange Program – Awareness Training:** RGNGWTRI successfully organized awareness cum training for 150 Youths from different Districts of J & K under 16th Kashmir Youth Exchange Programme in collaboration with Nehru Yuva Sangathan in January 2025, and a few glimpses of which are shown in **Figure 5.12**.

Fig. 5.12: Glimpses of the One Day Awareness Training Program on Yuva Bharat - Kashmir Youth Exchange Program

- **Yuva Bharat – Mera Bharat:** Under Mera Bharat Yuva Bharat one day training programme on Ground Water Development & Management was organized in March 2025 at RGNGWTRI for the Students of IIIT, Naya Raipur, as shown in **Figure 5.13**.

Fig. 5.13: Glimpses of the One Day Training Program on Mera Bharat – Yuva Bharat

5.7 Other Achievements

- **MoU With NIT Raipur:** On 25th Nov 2024, RGNGWTRI and NIT Raipur signed an MoU for collaborative R&D, internships, and dissertations for NIT students, and a few glimpses of which are shown in **Figure 5.14**. The partnership aims to enhance groundwater research, with outcomes featured in Bhujal News.

Fig. 5.14: MoU Signing between RGNGWTRI & NIT, Raipur

- **Workshop on Smart India hackathon:** One workshop on the software developed during Smart India Hackathon was organized on 11.03.2025 with participation from ILTC Trainee Officers, Students & Research Scholars of Academic Institutes and Officers from State Govt Departments, as shown in **Figure 5.15**.

Fig. 5.15: Inaugural Session of one day workshop on Smart India Hackathon at RGNGWTRI

- **First Prize in TOLIC:** RGNGWTRI as a part of Rajbhasha initiative – regularly organizes quarterly Sanjyukt Rajbhasa Workshop in association with the other Central Govt. Organizations. In the 2nd Half Yearly Meeting of TOLIC, RGNGWTRI was awarded the First Prize in recognition of the activities carried out by RGNGWTRI in propagating Rajbhasa in daily Official work, as shown in **Figure 5.16**.

Fig. 5.16: Regional Director, RGNGWTRI receiving the First Prize in the 2nd Hal Yearly meeting of TOLIC

- **Prasikshan Bharti:** The first publication (Volume-1) in Rajbhasa named प्रशिक्षण भारती was published in the AAP 2024-25 at RGNGWTRI.

Fig. 5.17: Cover Page of the Volume -1 of प्रशिक्षण भारती

6. National Hydrology Project (NHP)

6.1 National Hydrology Project (NHP)

National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of the Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with a total outlay of Rs. 3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs. 39.7674 Crore for the establishment of the National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources Data. The project spans over a period of 08 years starting from 2016-17 to 2023-24, which has been extended for a period of one-and-half years from April 2024 and up to Sep 2025. Out of the total outlay of Rs. 3679.7674 Crore, 50% of the amount has to be repaid by Central Government to the Word Bank (WB) and the remaining 50% would be Central Assistance. The project will be implemented by 48 agencies involving 9 Central and 34 States agencies, 02 UTs and 03 River Basins. Central Ground Water Board (CGWB) is one of the Central implementing agencies under the National Hydrology Project.

6.2 Objective of NHP

To improve the extent, quality, and accessibility of water resources information, decision support system for floods and basin level resource assessment/ planning and to strengthen the capacity of targeted water resources professionals and management institutions in India.

6.3 Project Components:

A. Water Resources Monitoring Systems:

This component supports the establishment/modernization of new and existing hydro met monitoring systems primarily for real time data transmission pertaining to meteorology, stream flow, ground water, water quality and water storage measurements, and establishment of hydro-informatics centers. This component will be implemented by states/UTs with the support of core central agencies. The major activities include:

1. Establishment of real time data acquisition hydro met networks;
2. Establishment of Supervisory Control and Data Acquisition (SCADA) systems for water infrastructure; and
3. Establishment/upgradation of Water Quality Laboratories.

B. Water Resources Information Systems:

Component B supports the strengthening of national and sub-national water information centers with web-enabled WRISs through standardization of databases and sources/departments and make comprehensive, timely and integrated water resources information available to decision makers for effective planning, decision making and operations.

The sources of data/information include the real-time data acquisition networks and centers under Component A remote sensing data, and topographical maps and knowledge products developed under Component C.

Emphasis will be on improving quality of and access to water information and on expanding public access beyond data to analytical results (trends, water balance, and so on) as well as to contribute to evidence based operational and investment plans. The project will support development or strengthening of centers for web-based WRIS at the central, regional, river basin, and state/UT levels. Some of the key activities under this component are:

1. Strengthening of National Water Informatics Centre (NWIC) including Water Information Management System (WIMS) and Water Resources Information System (WRIS).
2. Regional/State Water Resources Information System including formation of State Water Informatics Center (SWIC).

C. Water Resources Operations and Planning Systems:

This component supports the development of interactive analytical tools and decision supports platform that would integrate database, models and scenario manager for hydrological flood forecasting, integrated reservoir operations, and water resources accounting for improved operation, planning and management of both surface water and ground water, based on basin approach. The component will provide interactive systems to analyze the impacts of alternative management scenarios and generate knowledge products using real –time data under Component B. Component C has three subcomponents.

1. Development of analytical tools and decision-support platform (river basin modelling, stream flow forecasting and reservoir operation systems, inundation flood forecasts etc.
2. Purpose driven support.
3. Piloting innovative knowledge.

D. Institutional Capacity Enhancement:

Component D aims to build capacity for knowledge-based water resources management. It will support subcomponents in the establishment of (i) water resources knowledge centers, (ii) professional development, (iii) project management, and (iv) operational support.

The project will develop partnerships with national and international institutes; establish communities of practice, internships and visiting expert programs, customized training and workshops for knowledge exchange and profession networking. Outreach and awareness programme are an integral part of the project and will showcase the NHP to a broad audience, both for specific target audiences such as local communities in irrigated or flood prone areas and for the public at large.

6.4 Beneficiaries

The project has two groups of direct beneficiaries:

1. Central and state implementing agencies (IAs) responsible for surface water and /or ground water planning and management, including river basin organizations (RBOs)

2. Users of the WRIS across various sectors and around the world. The ultimate beneficiaries will be the selected farm communities which benefited from pilot projects for water management; rural and urban water and power users; populations affected by floods and droughts, especially poor rural people, and farm families who may benefit from improved irrigation water supply and management; stakeholders across the energy, inland waterways, environment, and agriculture ministries; research and educational institutions; students and researchers; and non-governmental organizations, civil society organizations, and the private sector.

6.5 Role of CGWB in NHP

CGWB being the Nodal Agency for Ground Water in the country, NHP cell in all the Regional Offices have been created for providing Technical Guidance & Support to States in the implementation of NHP.

6.6 The proposed major activities of CGWB under NHP

- **Real time monitoring Water Quality in coastal aquifers of Tamil Nadu & UT of Puducherry** through construction of 60 piezometers & installation of 60 DWLRs.
- **Establishment of Center of Excellence for groundwater modeling** for providing technical support to the Central/ State Agencies in Ground Water Flow Model Studies.
- **Real time monitoring of Water Level through Installation of 5200 additional DWLRs (Phase-1 & 2) in the existing Piezometers** located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Capacity Building** of Officers from CGWB, State / other Central implementing agencies through National and International level trainings every year.
- **Awareness Campaigns** are conducted at six places every year on state specific groundwater issues.
- Organizing **National Level Workshops** on need based Topics.

6.7 The Status of various activities of CGWB under NHP along with photographs are given below

6.7.1 Visit of NHP, NPMU and World Bank Team to CHQ Faridabad

On 22nd May 2025, the Senior Joint Commissioner-3 from the National Project Management Unit (NPMU) under the National Hydrology Project (NHP), accompanied by their team and experts from the World Bank, visited the Central Headquarters (CHQ) of the Central Ground Water Board (CGWB) in Faridabad. During the visit, a comprehensive presentation was delivered on the topic **"Visualization and Analysis of Geospatial Data using ArcGIS Node.".** The session aimed to showcase the advanced capabilities of geospatial technologies in groundwater data management, analysis, and visualization to support informed decision-making. The demonstration highlighted practical applications

of ArcGIS Node in real-time monitoring, mapping, and data integration for enhanced hydrological assessments.

The meeting was chaired by the **Chairman of CGWB**, and was attended by **Members of the Board** along with several **senior officers from CHQ Faridabad**, reflecting the importance of adopting innovative tools in the field of water resources management.

Meeting on Innovative Tools for Water Resources Management at CGWB Headquarters

6.7.2 Workshop on Best Practices and Achievements under NHP

Dr. Sunil Kumar Ambast, Chairman, CGWB, has participated in the inaugural session of a two-day workshop on "Best Practices and Achievements under the National Hydrology Project (NHP)" on February 11, 2025, at Kaushal Bhavan, New Moti Bagh, New Delhi. Dr. A. Ashokan, Member (HQ), has Chaired multiple sessions on 11th & 12th, Feb 2025, focusing on the "Real-Time Data Acquisition System for Ground Water," implemented by various State and Central Government Departments. Er. N. Varadaraj, Member (East) along with 25 Senior Officers from CGWB, have attended the workshop on 11th & 12th, Feb 2025. Mr. Nidhish Verma, Regional Director, RGI Raipur, has delivered a presentation highlighting the achievements and future way forward of 18 No. NHP training programs conducted by RGI Raipur from 2016–17 to 2024–25, aimed at building the capacity of State and Central Government officers. Additionally, Dr. M. Gobinath, Scientist-C, CHQ

Faridabad, presented an overview of the Real-Time Data Acquisition System for Ground Water, implemented across India by CGWB, on February 12, 2025.

Workshop on Best Practices and Achievements under the NHP

6.7.3 DWLR Training

As per the contractual agreement, DWLR vendors were required to provide hands-on training to CGWB officers covering the fundamentals of DWLR systems, field-based data calibration, data retrieval, and integration of data with the CGWB's RoDC server as well as the NWIC's WIMS server.

Accordingly, a total of **34 one-day training programs** were conducted by vendor across 13 **states**, namely Telangana, Maharashtra, Karnataka, Kerala, Tamil Nadu, West Bengal, Odisha, Uttar Pradesh, Madhya Pradesh, Gujarat, Bihar, Uttar Pradesh and Chandigarh. These training sessions witnessed active participation from **over 200 officers and officials** of CGWB.

The programs included both **classroom sessions** and **field demonstrations**, providing participants with practical exposure to DWLR operations and data management workflows. A visual overview of these sessions are presented in the pictures below.

Hands-on training to CGWB officers on the fundamentals of DWLR systems

6.7.4 Site inspection DWLRs stations

On 1st and 2nd July 2025, Member (East) conducted a comprehensive review of the DWLR data utilization, DWLR dashboard, and field inspection of DWLR sites to assess the quality of installation and ensure seamless data transmission to the NWR Chandigarh RoDC servers and the WIMS server of NWIC, within the jurisdiction of NWR, Chandigarh.

During the visit, detailed discussions were held regarding data flow and system performance. Representatives from M/s Encardio Rites and M/s Aaxis Nano Technologies were present and actively participated in the review and inspection process.

Glimpse of DWLR site Inspection in State Punjab and Chandigarh

Glimpse of DWLR site Inspection in State Punjab and Chandigarh

6.7.5 International training on Groundwater resources management from 30th September 2024 to 4th October 2024 in Denmark

A delegation comprising **nine senior-level officers from the Central Ground Water Board (CGWB)** undertook an international visit to **Denmark** to participate in an **advanced training program on Groundwater Resources Management.** This high-level professional development initiative was specifically designed to enhance participants' technical expertise, practical competencies, and strategic policy understanding in the domain of **sustainable groundwater management.** The program provided a unique platform for in-depth learning through expert-led sessions, technical workshops, and site visits, with a focus on Denmark's globally recognized best practices in integrated water resources management. Key topics included groundwater monitoring and modeling, aquifer recharge techniques, water quality management, and climate-resilient strategies for groundwater sustainability.

This international exposure aimed to empower CGWB officers with cutting-edge knowledge and innovative tools, enabling them to strengthen India's groundwater governance framework and apply globally informed approaches to local and regional water challenges.

International training on Groundwater resources management

6.7.6 Cohort (II) - India Young Water Professional Programme (YWP)

Government of India signed the Memorandum of Understanding (MoU) with the Government of Australia to cooperate in the field of Water Resources Management on 10th November, 2009 and subsequently renewed on 5th September, 2014. The said MoU was superseded with the signing of a new MoU on 20th May, 2020. The effectiveness of the MoU is for a period of 5 years with effect from the date of signing. The Para 3 of signed MoU stipulates 'forms of cooperation' in the identified areas related to Water Resources Management. One of the identified forms of cooperation is capacity building, training, study tours, education and knowledge sharing. Under this form of cooperation, Australia Water Partnership (AWP), during the year 2019-20 had proposed to initiate India Young Water Professional Programme (YWP) in collaboration with National Hydrology Project (NHP).

The specific objectives of this training program are:

- To create a group of Young Water Professional in India who are transdisciplinary in their approach and can provide effective leadership in implementing water projects.
- Develop leadership skills and management competencies of YWPs in India.
- Improve practices and management approaches in sustainable water management.
- Widen knowledge base related to the water sector reforms based on Australia's experience in this area.
- Build a strong alumni network of YWPs for seamless and continuous professional developmentand networking.

Under the Co Cohort (II) - India Young Water Professional Programme (YWP) 06 officers from CGWB have undergone 03 weeks Physical Training in Sydney, Australia.

6.7.7 Workshop on High Frequency Hydraulic Head Data Validation, Analytics, And Decision-Making Techniques for Aquifer Management

NPMU, NHP has requested CGWB to Organize 2-days review cum workshop on data validation, analytics and decision making with respect to high frequency time series hydraulic head data under NHP at 03 Locations and the details are given below.

S.No.	Regional Office In-charge	Date	Location	Participating Offices
1	KR, Trivandrum	03rd – 04th March, 2025	Trivandrum, Kerala	SECR, SWR, SR, KR CR
2	SER, Bhubaneshwar	17th – 18th March, 2025	Bhubaneshwar, Odisha	SER, ER, MER, NER, NCR, NCCR, NR
3	WR, Jaipur	20th – 21st March, 2025	Jaipur, Rajasthan	NWR, WCR, WR, UR, NHR, NWHR, SUO-Delhi

The objective of the work includes

- At least 20 time-series hydraulic head of the region to be presented for review and discussion.
- At least 3 aquifer basins to be showcased with at least 7 nos. of Hydraulic head time series graph representing at least 2 in recharge, 2 in intermediate and 2 in discharge regions.
- RTDAS-GW stations with faulty and erroneous data.
- Hydraulic Head Data validation
- Hydraulic Head Data Analytics
- Application of results of Data Analytics in response to Hydrologic stress: a) Monsoon Period b) Non Monsoon period c) Drought period d) Monthly and Annual forecast e) Mitigation measures based on Hydraulic Head Data Analytics.
- Hydrogeological Tool to address the above

Glimpse of the Workshops organized at Trivandrum, Bhubaneshwar and Jaipur are given below:

6.7.8 Domain Specific Training

03 Nos Domain Specific Training has been organized through RGI Raipur for Officers from State and Central Government departments under NHP.

Date	Name of the Training	Learning Outcomes	No. of participants
14-10-2024 To 18-10-2024	Source Sustainability: MAR/AR to Groundwater	Aquifer storage, transfer and recovery, Different Artificial Recharge Structures, Conjunctive use	17
04-11-2024 To 08-11-2024	Statistical tools for groundwater data management and High Frequency Data Analysis	Data Validation Identifying the outliers, Obtain, clean/process, and transform data, Data Analysis and interpret	17
17-03-2025 To 21-03-2025	Application of Remote Sensing & Geographic Information Systems in Ground Water Studies	Able to interpret the remote sensing image that can be subsequently applied in groundwater exploration in different hydrogeological terrain including the inaccessible area. Use of RS & GIS in groundwater in Studies	12

Domain Specific Training

7. Pradhan Mantri Krishi Sinchai Yojana - Har Khet Ko Pani - Ground Water (PMKSY-HKKP-GW)

Pradhan Mantri Krishi Sinchai Yojana- Har Khet Ko Paani- Ground Water scheme, launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, envisages to provide irrigation facility for small and marginal farmers in areas having sufficient potential for future development of ground water.

The scheme provides financial assistance to States for assured ground water irrigation to small and marginal farmers with priority to SC/ST and women farmers. The funding pattern is in the ratio of 90:10 (C:S) in case of NE/Hilly areas and 60:40 (C:S) in case of other areas. The scheme is applicable only in areas having stage of ground water development less than 60%, average rainfall more than 750 mm rainfall and having shallow ground water levels (less than 15 m below ground level).

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Arunachal Pradesh, Assam, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, and Uttarakhand. All the 13 projects have been successfully completed. Under these thirteen projects, construction of 29779 wells was targeted with the creation of 88679 Ha of command area benefitting 67930 small & marginal farmers. Total 29779 wells have been constructed, 88547 ha command area have been created benefitting 67909 small & marginal farmers.

The total cost of these 13 projects is Rs. 978.00 Crore and Central Assistance is Rs.827.15 Crore of which **764.89** crore has already been released by the Ministry of Jal Shakti. All 13 projects have been completed by March 2024.

8. Jal Shakti Abhiyaan (JSA)

Jal Shakti Abhiyan (JSA) was launched by Ministry of Jal Shakti in the year 2019 in the 256 water stressed districts of the country. Jal Shakti Abhiyan: Catch the Rain (JSA: CTR) has become an annual feature. In continuation to this, JSA: CTR 2023 was implemented across the country with the theme of "Source Sustainability for Drinking Water" focusing on 150 districts of the country. In 2024, 151 officers from CGWB were nominated as Technical Officers (TO) in the Central Team of JSA for various districts across the country along with Central Nodal Officer (CNO).

Glimpse of Jal Shakti Abhiyaan 2024, Faridabad District

ਦੇਰਾਨ ਨਾਇਬ ਤਹਸੀਲਦਾਰ ਦੀ ਭੂਮਿਕਾ ਵੀ ਪੜਤਾਲੀ ਜਾਵਗੀ।

ਰੀਚਾਰਜ਼ ਕੀਤੇ ਪਾਣੀ ਦੀ ਗੁਣਵੱਤਾ ਦੀ ਕੀ
ਜਾਂਚ : ਨੋਡਲ ਅਫ਼ਸਰ ਜਲ ਸ਼ਕਤੀ ਅ
ਟੀਮ ਨੇ ਪਿੰਡ ਧਰਦਿਓ ਦਾ ਕੀਤਾ ਦ

ਅੰਮ੍ਰਿਤਸਰ 1 ਅਗਸਤ (ਹਰਪਾਲ ਸਿੰਘ ਭੰਗੂ, ਮਿੰਟੂ ਅਰੋੜਾ)- ਰਾਸ਼ਟਰੀ ਜਲ ਮਿਸ਼ਨ
ਜਲ ਸ਼ਕਤੀ ਅਭਿਆਨ ਭਾਰਤ ਸਰਕਾਰ ਦੇ ਮੁੱਖ ਨੋਡਲ ਅਫ਼ਸਰ ਸ੍ਰੀ ਭਰਤ ਭੂਸ਼ਨ ਵਰਮਾ
ਅਧਿਕਾਰੀਆਂ ਨਾਲ ਜਲ ਸ਼ਕਤੀ ਅਭਿਆਨ ਸਬੰਧੀ ਮੀਟਿੰਗ ਕਰਦਿਆਂ ਕਿਹਾ ਕਿ ਰੀਚ
ਪਾਣੀ ਦੀ ਗੁਣਵੱਤਾ ਦੀ ਜਾਂਚ ਜ਼ਰੂਰ ਕੀਤੀ ਜਾਵੇ ਤਾਂ ਜੋ ਪਤਾ ਲਗ ਸਕੇ ਕਿ
ਰੀਚਾਰਜ਼ ਕੀਤਾ ਗਿਆ ਪਾਣੀ ਕਿੰਨੀ ਮਾਤਰਾ ਤੱਕ ਸਹੀ ਹੈ। ਉਨਾਂ ਕਿਹਾ ਕਿ
ਜਲ ਸ਼ਕਤੀ ਅਭਿਆਨ ਦਾ ਮੁੱਖ ਟੀਚਾ ਪਾਣੀ ਬਚਾਓ, ਪਾਣੀ ਦੀ ਬਚਤ,
ਨਵੀਆਂ ਸਿੰਚਾਈ ਵਿਧੀਆਂ, ਵਾਟਰ ਹਾਰਵੈਸਟਿੰਗ ਅਤੇ ਪਾਣੀ ਦੀ ਦੁਰਵਰਤੋਂ
ਹੋਣ ਤੋਂ ਬਚਾਉਣ ਸਬੰਧੀ ਜਾਗਰੂਕ ਕਰਨਾ ਹੈ। ਉਨਾਂ ਕਿਹਾ ਕਿ ਪਾਣੀ ਦੇ ਡਿੱਗਦੇ
ਪੱਧਰ ਨੂੰ ਬਚਾਉਣ ਲਈ ਬਾਰਿਸ਼ ਦੇ ਪਾਣੀ ਨੂੰ ਰੀਚਾਰਜ਼ ਕਰਨਾ ਬਹੁਤ ਜ਼ਰੂਰੀ ਹੈ ਤਾਂ
ਜੋ ਪਾਣੀ ਦੇ ਡਿੱਗਦੇ ਹੋਏ ਪੱਧਰ ਨੂੰ ਰੋਕਿਆ ਜਾ ਸਕੇ। ਮੀਟਿੰਗ ਨੂੰ ਸੰਬੋਧਨ ਕਰਦੇ ਹੋਏ ਸ੍ਰੀਮਤੀ ਰਚਨਾ
ਕੇਂਦਰ ਗਰਾਊਂਡ ਜਲ ਬੋਰਡ ਨਵੀਂ ਦਿੱਲੀ ਨੇ ਕਿਹਾ ਕਿ ਰੀਚਾਰਜ਼ ਕੀਤੇ ਗਏ ਪਾਣੀ ਨੂੰ ਖੇਤੀ ਲ
ਸਕਦਾ ਹੈ, ਜਿਸ ਨਾਲ ਪਾਣੀ ਦੇ ਡਿੱਗਦੇ ਪੱਧਰ ਨੂੰ ਵੀ ਰੋਕਿਆ ਜਾ ਸਕੇਗਾ ਅਤੇ ਜਮੀਨੀ ਪੱਧਰ ਤੇ
ਘੱਟ ਨਿਰਭਰ ਹੋਵਾਂਗੇ। ਉਨਾਂ ਕਿਹਾ ਕਿ ਜਲ ਸ਼ਕਤੀ ਅਭਿਆਨ ਵਿੱਚ ਯੁਵਕਾਂ ਦੀ ਭਾਗੀਦਾਰੀ ਵ
ਸੰਭਵ ਯਤਨ ਕੀਤੇ ਜਾਣਗੇ ਤਾਂ ਜੋ ਦੇਸ਼ ਦਾ ਯੂਵਾ ਬੱਚੇ ਇਸ ਮੁਹਿਮ ਤੋਂ ਪ੍ਰੇਰਿਤ ਹੋ ਕੇ ਪਾਣੀ ਬਚਾ
ਲੈਣ। ਇਸ ਮੌਕੇ ਮੀਟਿੰਗ ਦੇ ਕਨਵੀਨਰ ਭੂਮੀ ਅਤੇ ਜਲ ਸੰਭਾਲ ਵਿਭਾਗ ਦੇ ਮੰਡਲ ਭੂਮੀ ਰੱਖਿਆ ਅ
ਸਿੰਘ ਨੇ ਦੱਸਿਆ ਕਿ ਜਿਲ੍ਹੇ ਵਿੱਚ ਪਾਣੀ ਦੀ ਸਾਂਭ ਸੰਭਾਲ ਲਈ ਅਨੇਕਾਂ ਯਤਨ ਕੀਤੇ ਜਾ ਰਹੇ
ਕੀਤੇ ਗਏ ਕੰਮਾਂ ਸਬੰਧੀ ਜਾਣਕਾਰੀ ਵੀ ਦਿੱਤੀ। ਇਸ ਉਪਰੰਤ ਜਲ ਸ਼ਕਤੀ ਅਭਿਆਨ ਦੀ ਟੀਮ ਵ
ਦਾ ਦੌਰਾ ਕਰਕੇ ਉਥੇ ਚੱਲ ਰਹੇ ਕੰਮਾਂ ਦਾ ਜਾਇਜਾ ਵੀ ਲਿਆ ਅਤੇ ਪਾਣੀ ਬਚਾਉਣ ਦੇ ਢੰਗ ਵੀ ਦ
ਰਹੇ ਸੋਲਰ ਦੀ ਮਦਦ ਨਾਲ ਛੱਪੜ ਦੀ ਸਿੰਚਾਈ ਕਰਨ ਦੇ ਪ੍ਰੋਜੈਕਟ ਦਾ ਜਾਇਜਾ ਵੀ ਲਿਆ। ਇ
ਪਿੰਡ ਧਰਦਿਓ ਦੇ ਸਕੂਲ ਵਿੱਖੇ ਵਾਤਾਵਰਨ ਦੀ ਸਵੱਛਤਾ ਨੂੰ ਬਰਕਾਰ ਰੱਖਣ ਲਈ ਪੌਦਾ ਵੀ ਲ
ਇਸ ਮੀਟਿੰਗ ਵਿੱਚ ਵਧੀਕ ਡਿਪਟੀ ਕਮਿਸ਼ਨਰ ਪੇਂਡੂ ਵਿਕਾਸ ਮੈਡਮ ਪਰਮਜੀਤ ਕੌਰ, ਜਿਲ੍ਹਾ ਵਿ
ਅਫ਼ਸਰ ਸ੍ਰੀ ਸੰਦੀਪ ਮਲਹੋਤਰਾ, ਜਿਲ੍ਹਾ ਜੰਗਲਾਤ ਅਫ਼ਸਰ ਸ੍ਰੀ ਅਮਨੀਤ ਸਿੰਘ, ਐਸ.ਡੀ.ਓ ਪੁੱਡਾ
ਐ.ਸ.ਡੀ.ਓ. ਮਜੀਠਾ ਸ: ਧਰਵਿੰਦਰ ਸਿੰਘ, ਮੁੱਖ ਖੇਤੀਬਾੜੀ ਅਫ਼ਸਰ ਸ: ਤਜਿੰਦਰ ਸਿੰਘ , ਐਸ.
ਰਾਜ ਸ: ਗੁਰਿੰਦਰ ਸਿੰਘ ਤੋਂ ਇਲਾਵਾ ਹੋਰ ਅਧਿਕਾਰੀ ਵੀ ਹਾਜ਼ਰ ਸਨ।

Glimpse of Jal Shakti Abhiyaan 2024, Amritsar District

9. Rajbhasha

राजभाषा हिन्दी का प्रचार, प्रसार एवं कार्यान्वयन

- केंद्रीय भूमिजल बोर्ड और इसके सभी अधीनस्थ क्षेत्रीय, प्रभागीय और राज्य एकक कार्यालयों में वर्ष 2024-25 के दौरान राजभाषा हिन्दी के कार्यान्वयन और प्रचार–प्रसार के सक्रिय प्रयास जारी रखे गए। केंद्रीय भूमि जल बोर्ड राजभाषा हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिबद्ध है। राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धारित लक्ष्य को प्राप्त करने के लिए बोर्ड सतत प्रयत्नशील है। राजभाषा विभाग, गृह मंत्रालय, जल शक्ति मंत्रालय और नगर राजभाषा कार्यान्वयन समिति से समय–समय पर प्राप्त अनुदेशों और दिशानिर्देशों के पूर्ण अनुपालन सुनिश्चित करने तथा संघ की राजभाषा नीति के प्रावधानों के कार्यान्वयन और वार्षिक कार्यक्रम में निर्धारित लक्ष्यों को प्राप्त करने के उद्देश्य से सभी अधीनस्थ कार्यालयों और मुख्यालय के सभी अनुभागों को इससे अवगत कराया गया।

- सरकारी काम काज में हिंदी के प्रयोग को बढ़ाने के लिए नियमित रूप से विभिन्न परिपत्र, आदेश और अपील जारी किए गए । कार्यालय में हिन्दी में टिप्पण और आलेखन को बढ़ावा देने के उद्देश्य से राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम 2024-25 की प्रति, द्विभाषी मानक मसौदे, प्रशासनिक शब्दावली आदि सभी अधिकारियों और कर्मचारियों को उपलब्ध कराये गए। सभी कम्प्यूटरों पर हिन्दी में काम करने की सुविधा प्रदान करने हेतु यूनिकोड सुविधा उपलब्ध कराई गयी ।

- राजभाषा हिन्दी के प्रभावी प्रचार-प्रसार और कुशल कार्यान्वयन के लिए राजभाषा विभाग, गृह मंत्रालय द्वारा जारी समस्त प्रोत्साहन योजनाओं यथा मूल रूप से हिन्दी में टिप्पण–आलेखन, हिन्दी में टंकण आदि योजनाओं को कार्यालय में लागू किया गया है तथा अधिक से अधिक अधिकारी और कर्मचारी उत्साहपूर्वक इन योजनाओं में भाग ले रहे हैं।

- राजभाषा हिन्दी के प्रभावी कार्यान्वयन और इसकी प्रगति की मॉनिटरिंग के लिए बोर्ड और इसके समस्त अधीनस्थ कार्यालयों में राजभाषा कार्यान्वयन समिति का गठन किया गया है। इस समिति की बैठकें नियमित रूप से प्रत्येक तिमाही में आयोजित की जाती हैं। इन बैठकों में राजभाषा विभाग, गृह मंत्रालय द्वारा जारी वार्षिक कार्यक्रम में निर्धारित लक्ष्यों की प्राप्ति हेतु विभिन्न मुद्दों पर विचार-विमर्श द्वारा कार्यनीति तैयार की जाती है। इन बैठकों में लिए गए निर्णयों पर प्राथमिकता के आधार पर अनुवर्ती कार्रवाई की जाती है।

- गृह मंत्रालय द्वरा जारी दिशा निर्देश के अनुसार सितंबर माह में दिनांक 17.09.2024 से 30.09.2024 के दौरान बोर्ड और इसके समस्त अधीनस्थ कार्यालयों में हिन्दी दिवस, हिन्दी सप्ताह, हिन्दी पखवाड़ा का आयोजन किया गया। इस दौरान अधिकारियों और कर्मचारियों के लिए कई प्रतियोगिताओं जैसे हिन्दी निबंध प्रतियोगिता, टिप्पण-आलेखन, टंकण, वाद-विवाद , प्रश्न मंच आदि का आयोजन किया गया। सभी अधिकारियों और कर्मचारियों ने इन प्रतियोगिताओं में बढ़-चढ़कर भाग लिया। अध्यक्ष, केन्द्रीय भूमि जल बोर्ड की गरिमामयी उपस्थिति में हिन्दी पखवाड़े का समापन एवं पुरस्कार वितरण समारोह का आयोजन किया गया। अध्यक्ष महोदय को राजभाषा हिन्दी के कार्यान्वयन में बहुमूल्य मार्गदर्शन के लिए सदस्य एवं राजभाषा प्रभारी महोदया द्वारा सम्मानित किया गया । इसके पश्चात सभी विजेताओं को नगद पुरस्कार के साथ साथ प्रमाण पत्र भी प्रदान किया गया।

- भारत सरकार की राजभाषा नीति के अनुसरण में राजभाषा अधिनियम, 1963 की धारा 3(3) के तहत सभी दस्तावेज हिंदी और अंग्रेजी दोनों में तैयार किए जाते हैं। राजभाषा नियम, 1976 के नियम 5 के तहत हिन्दी में प्राप्त सभी पत्रों के उत्तर अनिवार्य रूप से हिन्दी में ही दिये जा रहे हैं। वार्षिक कार्यक्रम में निर्धारित लक्ष्यों को ध्यान में रखते हुए कार्यालय में जांच बिन्दु स्थापित किए गए हैं, ताकि इस दिशा में किए जा रहे कार्यों की मॉनिटरिंग की जा सके।

- राजभाषा विभाग, गृह मंत्रालय द्वारा समय-समय पर जारी दिशानिर्देशों के अनुपालन में केंद्रीय भूमिजल बोर्ड, मुख्यालय और इसके अधीनस्थ कार्यालयों में नियमित रूप से प्रत्येक तिमाही मे हिन्दी कार्यशालाओं का आयोजन किया जा रहा है।

- माननीय संसदीय राजभाषा समिति की दूसरी उपसमिति द्वारा केंद्रीय भूमि जल बोर्ड के विभिन्न अधीनस्थ क्षेत्रीय / प्रभागीय / राज्य एकक कार्यालयों का राजभाषायी निरीक्षण किया गया। समिति द्वारा कार्यालयों में राजभाषा के प्रचार-प्रसार और इसके कार्यान्वयन की दिशा में किए जा रहे कार्यों पर संतोष प्रकट किया गया।

- केंद्रीय भूमि जल बोर्ड हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिबद्ध है । राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धारित लक्ष्य को प्राप्त करने के लिए बोर्ड सतत प्रयत्नशील है

10. Vigilance and RTI

10.1 Vigilance

During the year 2024-25, out of total 10 disciplinary cases, 4 cases are with Ministry and 2 cases with CGWB for preliminary investigation. 1 case has been disposed off and 3 cases are cases are under investigation by the CBI.

10.2 RTI

सूचना का अधिकार

केन्द्रीय भूमिजल बोर्ड, मुख्यालय फ़रीदाबाद में सूचना का अधिकार अधिनियम 2005 के अंतर्गत **वर्ष 2024-2025** के दौरान प्राप्त ऑनलाइन एवं ऑफलाइन प्राप्त आवेदनों, निपटान किए आवेदनों, प्राप्त किए गए अपील आवेदनों तथा निपटान किए गए मामलों का विवरण निम्नलिखित तालिका में दिया गया है :

सूचना हेतु प्राप्त आवेदनों की संख्या	निपटान किए गए आवेदनों की संख्या	प्राप्त अपील आवेदनों की संख्या	निपटान किए गए अपील आवेदनों की संख्या	लंबित आवेदनों की संख्या	केंद्रीय सूचना आयोग में द्वितीय अपील निपटान की संख्या
1	2	3	4	5	6
267	**267**	**47**	**47**	**शून्य**	**09**

11. Budget and expenditure

Table 11.1: Budget Details for the financial year 2024-25

Name of the Scheme	Revenue			Capital			Total		
	BE	RE	Expenditure	BE	RE	Expenditure	BE	RE	Expenditure
GWM&R	9395	8387	7784.26	20605	13588	13432.16	30000.00	21975.00	21216.42
TSP	800	725	678.84	1700	1300	1284.12	2500.00	2025.00	1962.96
Total	**10195**	**9112**	**8463.10**	**22305**	**14888**	**14716.28**	**32500.00**	**24000.00**	**23179.38**
HQ-Establishment	29650	31140	29077.71	1350	160	18.58	31000.00	31300.00	29096.29
RGI	385	294	226.07	15	6	2.97	400.00	300.00	229.04
External Support (NHP)	150	171.40	21.67	500	915	779	650.00	1086.40	800.70
Domestic Support (NHP)	150	170.65	16.45	500	940	786	650.00	1110.65	802.78
Total (NHP)	**300**	**342.05**	**38.12**	**1000**	**1855**	**1565**	**1300.00**	**2197.05**	**1603.48**

***Figures in Lakhs**

12. Photo - Gallery

Photo Gallery of the various activities held during the year 2024-25

High yielding EW at Umri, Saoner Block, Nagpur District (Q=17.90 lps)

High yielding EW at Sindhi Vihiri, Karanja (Gh.), Wardha district (Q=7.73 lps)

High yielding EW at Film City, Goregaon, Mumbai Suburban district (Q=12.19 lps)

High yielding EW at Manjri, Sangola Block, Soalpur district (Q=17.92 lps)

Inaugural Programme of PIB project at Mojhari site, Tiwsa taluka, Amravati district

Meeting of Stake holder Advisory committee (SAC) attended by Hon'ble Chairman, CGWB at NEERI, Nagpur

Activities undertaken by CGWB Officers

1

Central Ground Water Board (CGWB), Faridabad, celebrated the 76th Republic Day with enthusiasm. Chairman, CGWB hoisted the national flag, followed by activities and prize distribution for children of officials who participated in various events.

The Sixth Bhujal Samvad Talk Series has been conducted on 16th January 2025, at CGWB, Bhujal Bhawan Faridabad, on Hybrid Mode. The talk was delivered by Dr R S Chatterjee, Scientist G, Indian Institute of Remote Sensing (IIRS), Dehradun on "Satellite Gravity Observation and Interferometric SAR for Ground Water Storage Change and Aquifer-system Compaction Studies".

2

On 31st Dec 2024, Hon'ble Minister of Jal Shakti, Shri C.R. Patil, released two key reports by CGWB: National Compilation of Dynamic Ground Water Resources of India, 2024 and Annual Ground Water Quality Report, 2024. The

Shri N. Jyothi Kumar, RD, SWR, met Shri D.K. Shivakumar, Deputy CM of Karnataka, who confirmed chairing a session on "Water

	event was attended by senior officials including Smt. Debashree	Vision @ 2047: The Way Forward" at the All India State Ministers' Conference in Udaipur, Feb 18-19, 2025. He also gifted a Kannada book on groundwater.
3	 Regional Directors' Meeting of CGWB was successfully organised at CGWB, SWR, Bengaluru on 21st February 2025. The meeting was chaired by Dr Sunil Kumar Ambast, Chairman CGWB. Dr. Ram Prasath Manohar, IAS, Chairman, BWSSB was the Chief guest & Sh. K. Raghavan, Secretary MI, Govt of Karnataka was the Guest of Honour in the occasion	 As per the MoA between CGWB, BWSSB, and IISc, a meeting was held on 18.01.2025 to assess Bangalore's groundwater for summer. IISc presented a GW outlook, and Sh. Rahul R. Shende highlighted vulnerable wards and mitigation measures based on CGWB data.
4	 CGWB, NWR, Chandigarh organized a Tier-II Training on "Aquifer Management Through Participatory Approach" at CCSHAU, Hisar. VC Sh. V. R. Kamboj inaugurated the event, highlighting groundwater	 On 21st January 2025, CGWB, Eastern Region, Kolkata organized a Public Interaction Programme at Chalsa Gayanath Vidyapith, Jalpaiguri, under Dr. Anadi

	importance. The 3-day program, concluding on 12.02.2025, included lectures, hands-on sessions, and field demos.	Gayen's supervision. Students participated in a painting and quiz competition, raising awareness on groundwater sustainability and environmental protection.
5	 Honourable Union Minister of Jal Shakti, Sh. C. R. Paatil, accompanied by the Honourable Chief Minister of Chhattisgarh, Sh. Vishnu Deo Sai, visited Rajnandgaon district. During the visit, women from the Self-Help Groups (SHGs) in Barga village shared the benefits of the percolation tank with a recharge shaft. The Union Minister praised the efforts of the Gram Panchayat and the district administration in promoting water conservation and recommended constructing multiple recharge shafts in each percolation tank.	 CGWB took part in the 8th India Water Week 2024 in New Delhi. Ms. Debashree Mukherjee, Secretary (DoWR, RD&GR), and Mr. Subodh Yadav, Additional Secretary, visited the CGWB exhibition stall.
6		 Officers of CGWB, Smt. Rachna Bhatti, Dr.

	A Step Drawdown Test (SDT) was conducted in the Exploratory Well at Joypur PHED Campus, Joypur block, Purulia district, West Bengal. During the SDT, the step-wise discharge was 48 lpm, 54 lpm and 60 lpm, respectively.	Gobinath Manivel, and Shri Vipin Kumar, attended a technical meeting at the Indian Embassy in Copenhagen, Denmark. They discussed and provided technical inputs on ongoing and potential collaborations in the groundwater sector under the India-Denmark Collaboration.
7	 CGWB Eastern Region organized a Public Interaction Programme in Rangat, Andaman & Nicobar Islands, focusing on "Adopting Lifestyle Changes for Groundwater Sustainability." Activities included painting, quizzes, and a tree-planting initiative, engaging 80 students.	

सोलापूर
प्रखर राष्ट्रीय विचारांचे दैनिक
तरुण भारत
न मे कर्मफले स्पृहा
संभाजीनगर, सातारा, सोलापूर, अहमदनगर, धाराशिव, लातूर, नांदेड, परभणी, हिंगोली, जालना व बीड येथून प्रकाशित
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सांगोला प्रतिनिधी-दिपक भाकरे
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स्थानिक भूजल समस्या, व्यवस्थापनावर विद्यामंदिरमध्ये प्रशिक्षण

तभा वृत्तसेवा,
सांगोला, दि. २० डिसेंबर-

भारत सरकार जलशक्ती मंत्रालय, जलसंपदा नदी विकास व गंगा पुनरूज्जीवन विभाग केंद्रीय भूजल मंडळ, नागपूर यांचेकडून माझा युवा भारत कार्यक्रम अंतर्गत एक दिवस प्रशिक्षण कार्यक्रमात स्थानिक भूजल समस्या व भूजल व्यवस्थापन याविषयी सांगोला विद्यामंदिर प्रशाला व ज्यु. कॉलेज येथे विद्यार्थ्यांना मार्गदर्शन करण्यात आले.

प्रशिक्षण उद्घाटन प्रसंगी सां. ता. शि. प्र. मंडळाचे अध्यक्ष प्रा. प्रबुद्धचंद्र झपके, भूजल शास्त्रज्ञ जी. एस. लोंढे, भू भौतिक शास्त्रज्ञ एस. एस. पुरती, प्राचार्य अमोल गायकवाड, उपमुख्याध्यापिका शहिदा सय्यद, उपप्राचार्य प्रकाश म्हमाणे, प्रशिक्षण व्याख्याते भूजल शास्त्रज्ञ अमरनाथ, उज्ज्वलकुमार, दिलीपकुमार मीना उपस्थित होते.

यावेळी प्रा. प्रबुद्धचंद्र झपके यांनी आपण सर्व शेती संस्कृतीशी निगडित आहे. त्यामुळे प्रत्येकाचा जिव्हाळ्याचा विषय पाणी यासंदर्भात जलशक्ती मंत्रालय भारत सरकारकडून सांगोला विद्यामंदिरची निवड करून यासंदर्भात भूजल समस्या व भूजल व्यवस्थापन याविषयी प्रशिक्षणामध्ये मिळणारे अनमोल मार्गदर्शन महत्त्वपूर्ण आहे असे सांगत प्रशिक्षण विषयाच्या संदर्भात विद्यार्थ्यांमध्ये सजगता निर्माण केली व आयोजकांचे कौतुक केले. अमोल गायकवाड यांनी पाणी प्रश्न महत्त्वाचा असून या प्रशिक्षणातून यासंबंधीचे विचार विद्यार्थी आत्मसात करून आपल्या कुटुंबीयांमध्ये व गावामध्ये मांडतील व त्यातून या प्रशिक्षणाचा हेतू साध्य होईल असा विश्वास व्यक्त केला.

प्रशिक्षणामध्ये उज्ज्वलकुमार यांनी बेसीक हायड्रोजियोलॉजी, अमरनाथ यांनी सांगोला तालुक्यातील जल उपलब्धता, व्यवस्थापन आणि संवर्धनाची जलविज्ञान स्थिती व महाराष्ट्र जलव्यवस्थापन मॉनिटरिंग हायड्रोलॉजी व दिलीपकुमार यांनी एक्सप्लोरेशनमधील जिओफिजिकल सर्वेक्षण या संदर्भात अनमोल मार्गदर्शन केले. सूत्रसंचालन प्रा. धनाजी चव्हाण यांनी केले.

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